

Agenda Consistency in the German Bundestag

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Introduction

Elections are built on promises, promises parties make in a time of intense competition and fierce rivalry. The clearest form these take is the party manifesto, the document in which a party sets out what it promises, what it prioritises and what it stands for. Against it stands a suspicion as old as party democracy, that what is announced before an election bears little relation to what follows it, and that such announcements serve to soothe an expectant public rather than to bind the party that makes them. Where the suspicion can be checked against the record, the evidence runs the other way: parties that reach office keep considerably more of their pledges, at least in part, than their reputation allows, although how many depends on whether they govern alone or share power with others (Thomson et al., 2017). That record is read off legislation and policy outputs. But what about the arena in between, the plenary chamber? This thesis asks whether parties practise in parliament what they announce in their programmes. At issue is not the fate of single pledges, but whether the emphasis a party sets out on paper returns in what it chooses to talk about on the floor. To our knowledge, that comparison has not been made systematically for Germany.

In the literature the consensus is that the attention profiles in the political arena are a reputational asset. By prioritising issues the individual party forms a recognisable profile, which motivates it to keep emphasising its preferred topics (Petrocik, 1996). This would mean that manifestos should be recognisable in a party's shared parliamentary agenda. But another angle, also well documented in the literature, is that there is a party-system agenda in parliament to which all parties are subject. While each party takes an active part in its formation, that agenda is formed jointly and, in the German system, negotiated by consensus, so it binds all parties to a jointly negotiated order of business (Green-Pedersen and Mortensen, 2010; Sieberer, 2006). This means that party speech agendas as represented by their MPs' speeches taken together must be subject to this party-system agenda and cannot be fully individually adjusted. We therefore read the divergence between the two channels against two quantities the parties themselves supply: the distance between their programmes as the upper reference, since it is the largest distance parties deliberately create, and the distance between their speech agendas as the lower reference, since a shared order of business keeps any party from bringing its speeches fully into line with its programme. Neither is a bound the design derives; both are scales that make an otherwise uninterpretable distance readable. How far apart, then, do the announced and the

practised agenda of the same party lie? This thesis quantifies that divergence and asks where it falls across the topic buckets.

This thesis measures the agendas of both channels in the same category space by using the manifesto categories of the MARPOR coding scheme (Lehmann et al., 2025) to classify speeches thematically. This is implemented by using a classifier model, ManifestoBERTa (Burst et al., 2024), fine-tuned on hand-coded quasi-sentences from the Manifesto Corpus, to classify plenary speeches from GermaParl2 (Blätte and Leonhardt, 2025) sentence by sentence, while the manifesto side enters directly from the project's main dataset. In total the plenary speech corpus classified in this way holds more than six million sentences. The analysis runs on 38 party-period cells over eight legislative periods from 1994 onwards, each of which aggregates between 41,562 and 279,947 sentences once the two filters of Chapter 4 have been applied. The measure we use to quantify the divergence is the Jensen–Shannon divergence between the two distributions over the nine topic buckets into which we aggregate the 56 MARPOR categories, so what is measured is a distance in emphasis and not in position. As a JSD value carries no absolute reading of its own, the values are read against a double reference, the mean divergence between parties' speech agendas as its lower reference point and the mean divergence between their manifestos as its upper one, and against a second benchmark, whether the nearest programme to a party's speech agenda is its own. To further differentiate directional signals, we implement a second directional aggregation scheme separating economy, welfare, migration and European integration into sub-buckets. The analysis using this aggregation is read with more caution throughout, for reasons the validation establishes. Transferring a manifesto-trained classifier to parliamentary speech produces two structural failure modes, both of which Chapter 4 diagnoses and corrects before any result is read. The observation window also holds an opportunity, since a Eurosceptic challenger, the AfD, entered the Bundestag within it in 2017; because the system quantities are recomputed without that party, the part of any change that rests on the newcomer's own position and weight can be separated from what remains among the parties already present. Chapter 2 derives three expectations from this literature — that the two agendas are systematically linked, that the divergence which remains is shaped by the institution rather than spread evenly, and that polarisation moves with the arrival of the challenger — and Section 2.5 states them as hypotheses.

We find that the answer has two sides. Parties do adhere to the priorities they put forth in their manifestos, but this adherence is bounded by the agenda they share in the chamber. To put this in perspective, parties' speeches diverge on average roughly four times as much from their

manifestos as their speech agendas do from one another, and only 0.57 times as much as the manifestos of different parties diverge from each other. The congruence is therefore real and at the same time capped by the institution. Concretely, a party's own manifesto is the nearest to its speech agenda in 42.1% of cells, against a chance baseline of 21.1%. The rate differs strongly between parties. The FDP is nearest to its own manifesto in all seven of its cells and the AfD in both of its two, the Linke in none of its five, the CDU/CSU in two of eight and the SPD in one of eight, while the Grüne reach it in four of eight — a spread that does not sort along ideological lines.

The part of a party's agenda that does not carry into the chamber has a shape of its own: it is not noise but structure. In the speech channel, content on governance and on foreign policy and defence is systematically over-weighted. On average, speech agendas raise their share of Democracy & Political System content from about 15% to about 20% relative to manifestos, and Foreign Policy content from about 7% to about 12%. This increase holds in every one of the six parties, and these are the only two topics on which all of them move the same way. Other topics are under-weighted, and in many cases it is the topic a party emphasises most in its manifesto that is given less room in its speech agenda, so that wide programmatic spreads narrow into a common band. This is the institutional filter levelling attention rather than positions. Furthermore, under the directional aggregation, speech positions follow manifesto positions with slopes well below one, so the differences between parties carry over only in part and the parties end up closer to one another. The broad ordering largely survives — manifesto position still predicts speech position, with R^2 between 0.53 and 0.78 — but it is not preserved pair by pair. This decrease in dispersion holds in all four directional domains. Migration alone adds a level shift to this compression, moving from a liberal manifesto mean to a speech mean on the restrictive side. Part of this may be an artefact of the classifier, one the validation identifies (Section 5.3.2). Taken together, these shifts leave the chamber with an agenda profile of its own, one in which governance and the economy carry more weight than welfare and migration do. No single programme resembles that profile as closely as the FDP's, which comes nearest in 18 of the 38 cells. This is a claim about which topics get emphasis, not about the positions taken on them.

Read at the system level, as the party-strength-weighted spread of the position scores, the trends differ strongly across domains, described here rather than tested. While the economy shows a rising trend in polarisation on the speech side over the observation period, welfare becomes less polarised in the manifestos over time, and both movements survive a refit of the trends that

leaves the AfD out. Europe shows a mechanism of its own. In 1994 the programmes hardly differ on Europe at all, so the axis of conflict forms inside the observation window, and over its course polarisation rises in both channels, with most of the movement falling in the final two periods. Refitting without the AfD, however, reverses the manifesto trend from $+0.0769$ to -0.0079 and cuts the speech trend to roughly a quarter of its size. The rise is therefore carried above all by the arrival of a pole: in the manifesto channel the established parties do not move apart at all but converge sharply on one another, while in the speech channel a reduced trend survives the refit. On migration, polarisation rises slowly on the speech side, and it is not the AfD story one might have expected, since it sets in before the party exists and only part of it drops out under that refit, the rest being a fanning out among the parties already present. A rising polarisation curve alone cannot tell a new pole from parties drifting apart, which is why the recomputation without the entrant was fixed in advance as part of the design and is reported alongside the trends themselves rather than as an afterthought (arm 6 of the pre-committed battery, Section 4.9).

The mandate literature this thesis joins has concentrated on the far end of the chain, following programmes into laws, executive programmes and parliamentary questions, and for Germany the closest study traces manifesto priorities into the legislative agenda across three decades (Guinaudeau and Guinaudeau, 2023). The designs that come nearest to the comparison made here each supply part of what it needs. Louwerse (2011, 2012) treats mandate fulfilment as manifesto-to-speech congruence, but positionally and for one election per decade. Osnabrügge, Ash and Morelli (2023) supply the transfer template, a manifesto-trained classifier carried over to plenary speech and validated against human coding, and name the comparison of priorities across a party's channels as the programme their method opens. Ivanusch (2025) makes such a comparison for press releases, social media and parliamentary speech, in a cross-section built to compare channels rather than to follow parties over a longer horizon. What the question asked here needs is all three at once: one category space on both sides, the party and the legislative period as the unit of observation, and a window wide enough to take in the arrival of a challenger. Supplying that combination is the first contribution of this thesis.

The second contribution is a pair of instruments that outlast the case. The double reference answers a question every text-distance measure runs into, namely when a distance counts as large, and it answers it with quantities the parties themselves provide rather than with a threshold set from outside. The recomputation of a system polarisation series without the entering party does the same service for polarisation series: it isolates the part of a trend that

rests on the newcomer's own position and weight, though not what its entry set in motion among the parties already present. The third contribution is diagnostic. A classifier trained on manifestos and turned loose on parliamentary language fails in two systematic ways. It is forced to assign a code because its label space holds no null class, and one category absorbs far more of the corpus than the reference coding warrants — in part because it catches those forced codes, in part independently of them, which is why each mode needs a correction of its own. Both are named, quantified and corrected before any agenda is compared, and as named and quantified failure modes neither has, to our knowledge, been documented in political science applications of manifesto-trained classifiers, even though the absence of a null class is anticipated by design where a classifier is trained for the purpose (Section 4.5.3). The first of the two follows from the training design rather than from the particular model, so it is to be expected wherever a manifesto-trained classifier without a null class meets parliamentary language; which category absorbs the forced codes, and how far it is over-assigned beyond them, has to be established corpus by corpus, as Chapter 4 sets out.

Chapter 2 develops the two strands of literature whose opposing pulls keep the question open and condenses them into the hypotheses, two that carry the argument and four that deepen it. Chapter 3 introduces the two corpora and the analysis sample. Chapter 4 builds the measurement chain from the raw protocols to the cell distributions, validates the transfer of the classifier to parliamentary language, diagnoses and corrects its two failure modes, and fixes every specification before any result is read. Chapter 5 reports the findings along the hypotheses and puts them through a battery of robustness checks. Chapter 6 draws the results together, reads them against the literature they came from, states the limits of what the design can carry and sketches where the question goes next.

Literature Review

This chapter locates the thesis in the literature for which the congruence between a party's announced and practised agenda is a theoretically meaningful quantity. Section 2.1 introduces the theory of issue competition, Section 2.2 develops the institutional asymmetry of the two arenas we compare, and Section 2.3 reviews the direct evidence linking programmes to parliaments. Section 2.4 turns to the dynamics of challenger entry and the politicisation of Europe. Section 2.5 condenses the argument into three substantive expectations and hands over to the research design.

2.1 Issue Competition, Saliency and Ownership

One of the early models that established the basis for the modern study of party competition is the spatial model of parties and the electorate. Formulating a purely theoretical "economic theory of democracy", Downs (1957) develops a model in which parties as rational actors compete for votes by taking positions on a shared policy dimension and voters maximise their utility by choosing the platform closest to their own preferences, which in two-party competition drives both parties towards the median voter. Stokes (1963) objected that this model quietly rests on several assumptions that do not hold empirically. One of these is the assumption of a conflict space that is stable and shared across actors and over time, against which he argued that the dimensions salient to voters change over time. As a second objection, he introduced the concept of valence issues, topics on which there is effectively only one position a party can take: voters broadly agree on the desirable outcome, and the conflict between parties is over who takes this position more convincingly. What is needed, Stokes concluded, is "language that would express the fact that different weights should be given different dimensions at different times" (Stokes, 1963, p. 372). Saliency theory turned this objection into a model of inter-party competition in its own right. Parties are less prone to confronting each other than to selectively emphasising the topics they are positively associated with, keeping quiet on other topics (Robertson, 1976; Budge and Farlie, 1983; for a conceptual overview, Budge, 2015). The real competition then lies not in the confrontation with other parties but in control over which topics are on the agenda in the first place. This theoretical approach supplied the intellectual foundation of the Comparative Manifesto Project and its successor MARPOR, whose category scheme we use as the common measurement language of both channels. If competition takes place on the agenda-setting dimension, then the composition of that agenda, and the congruence between the agenda a party announces and the agenda it practises, become measurements with theoretical importance.

Issue ownership explains why these attention profiles are stable enough to be measured. Petrocik (1996) defines ownership as a reputation for handling an issue, produced by a party's history of attention and initiative on it. Later work separates this competence dimension from an associative one, the spontaneous association of a party with an issue irrespective of perceived competence, and shows that the two are conceptually and empirically distinct (Walgrave, Tresch and Lefevere, 2015). Ownership so understood is a long-term asset: comparative evidence finds it stable over time and similar across countries rather than fluid and context-bound (Seeberg, 2017). It also disciplines behaviour. Parties prioritise their owned issues in

campaigns and in office (Egan, 2013), and how strongly a party leans on its best issues, as opposed to riding the wave of current concerns, varies systematically with its organisation and resources (Wagner and Meyer, 2014). But the saliency prediction is not a law. In a direct test with independent measures of salience and positions, parties do disproportionately emphasise the issues they own, yet they engage one another on the same issues far more often than saliency theory allows (Dolezal et al., 2014). Ownership also reaches beyond the campaign: issue reputations shape executive and legislative agendas once parties govern (Green and Jennings, 2019). A party's profile of attention is thus to a considerable degree a reputational asset it carries across contexts, not a choice it makes anew in each arena.

Stable profiles do not, however, add up to segregated agendas. In United States presidential campaigns between 1960 and 2000, a high degree of similarity in the issue emphases of the two sides was the norm rather than the exception (Sigelman and Buell, 2004). This holds even though the experimental evidence gives candidates no reward for synchronising their messages with the news agenda, and rewards them instead for advertising on the issues they own (Ansolabehere and Iyengar, 1994).

European party research reaches the same conclusion from a different direction. Issue competition, the contest over what the party-political agenda should be about, has become a defining feature of West European party competition (Green-Pedersen, 2007), and parties engage with the issues of their competitors far more than a pure avoidance logic predicts, in patterns structured by party family and party size (Green-Pedersen and Mortensen, 2015). The reason is that no party competes in a vacuum. Drawing on the agenda-setting tradition (Baumgartner and Jones, 1993), Green-Pedersen and Mortensen (2010) model a party-system agenda, a common stock of salient issues that no party can ignore. In twenty-five years of Danish data they find that opposition parties remain freer to stay on the issues that favour them, while governing parties are the ones repeatedly forced to respond. Party issue profiles are thus sticky enough to be recognisable, yet observed agendas overlap far more than a pure ownership logic predicts. The congruence between a party's announced and its practised agenda is therefore an empirical quantity, not settled in advance.

2.2 The Manifesto and the Plenary Arena

The two channels we compare differ in who controls their agenda. The manifesto side is the clearest case: the manifesto is the one major channel whose agenda a party authors alone. Manifestos are programmatic documents that a party authorises as a collective actor, even though the drafting process itself varies with party organisation and government status (Dolezal

et al., 2012). Their functions reach beyond advertising: candidates treat the manifesto as the compendium of valid party positions, as a device that streamlines the campaign, and as campaign material (Eder, Jenny and Müller, 2017). The production and purposes of manifestos have become objects of comparative study in their own right (Harmel, 2018). In the mandate tradition, the manifesto counts as the most authoritative expression of what a party asks to be elected for (Klingemann, Hofferbert and Budge, 1994). We return to this role in Section 2.3. For present purposes one property matters most: on the manifesto side the agenda is essentially unconstrained. External events and rival parties shape what a party considers worth addressing, but the weight it ultimately gives each issue is its own choice.

Plenary speech follows a different logic. In the comparative theory of legislative debate, floor time is a scarce party resource: where electoral success runs through the party label, as in Germany, party leaderships control access to the floor and allocate it in the service of the party's message (Proksch and Slapin, 2012). Debates are staged as position-taking before an audience (Proksch and Slapin, 2015), in ways that vary systematically across parliaments (Bäck and Debus, 2016). What is spoken about is constrained before any speaker rises: the agenda of the Bundestag is set consensually, negotiated in the Council of Elders with unanimity as the working rule, and government control over it is weak by design (Sieberer, 2006). Governing adds pressures of its own, since coalition parties balance programmatic differentiation against the demonstration of unity across the electoral cycle (Sagarzazu and Klüver, 2017). Governments must also attend to problems as they arise rather than only to their announced priorities, a point we develop in Section 2.3.

Direct channel comparisons make the resulting asymmetry visible. Across parliamentary speeches, press releases and social media in Germany, Austria and Switzerland, channel characteristics moderate how much of its ideal agenda, the agenda of its manifesto, a party can communicate. That share is smallest in channels that operate under a pre-given agenda, of which plenary speech is the model case (Ivanusch, 2025). Within a single chamber, parties still put considerable emphasis on their issue preferences in speeches, but how much depends on their agenda-setting power, as evidence from the Austrian Nationalrat shows (Ivanusch, 2024). Positions, by contrast, travel across arenas and measurement sources more robustly than salience profiles do (Helbling and Tresch, 2011; Gessler and Hutter, 2026), which marks the agenda as the channel-sensitive layer of party communication. In the Bundestag, floor time is a party resource allocated under a pre-set agenda. Plenary speech therefore measures a party agenda after the institution has filtered it.

Together, the two arguments spelled out above bound what congruence between the channels can look like. Section 2.1 established that programmatic differentiation is real and strategically intended, with parties cultivating distinctive profiles of attention and association. Based on this, the distance between programmes can be taken as the outer bound of the divergence we measure. Furthermore, Section 2.2 showed the restrictions on differentiation in the plenary arena, where each party's speech agenda is strongly bound by the shared parliamentary agenda. While each party can exert some influence on that shared agenda, precisely its shared character limits each party's ability to align its speech agenda fully with its own programme. Programmatic differentiation supplies the upper reference for the divergence a channel switch could produce, and the shared parliamentary agenda the lower one, which is why we later read manifesto-to-speech distances against both references rather than against an absolute standard (Section 4.8).

2.3 From Programmes to Parliamentary Speech

That programmes matter beyond the campaign is one of the older findings of comparative party research: across ten democracies, the emphases of election programmes systematically prefigure the policy priorities of the governments that follow (Klingemann, Hofferbert and Budge, 1994). The dominant strand of the mandate literature has since become the pledge approach, which codes specific, testable commitments and traces their enactment. The approach goes back to Royed (1996), who traced individual pledges under Reagan and Thatcher and found fulfilment markedly higher in Britain — an institutional difference rather than a partisan one. In the largest comparative study, spanning thousands of pledges in twelve countries, governing parties fulfil far more of their promises than their public reputation suggests, conditional on how power is shared (Thomson et al., 2017; Naurin, Royed and Thomson, 2019). We ask a related but distinct question: not whether individual promises are kept, but whether the allocation of attention announced in the programme recurs in a party's communication.

This attention-based reading of the mandate has a literature of its own. In Britain, the legislative programme of governments follows the priorities of the governing party's platform, but competes with an anticipatory agenda that responds to public concern and to unforeseen problems (Froio, Bevan and Jennings, 2017). The sovereign debt crisis raised the weight of current voter priorities in Southern European parliaments without displacing the party mandate (Borghetto and Russo, 2018), and comparatively, electoral priorities shape policy agendas across Western Europe without amounting to partisan mandates in the conventional sense (Grossman and Guinaudeau, 2022). For Germany, the closest study finds the legislative agenda

from 1983 to 2016 markedly responsive to manifesto priorities, though again conditional on government capacity. A one per cent rise in a topic's manifesto salience goes with some 2.8 per cent more legislation on the topic in a given month (Guinaudeau and Guinaudeau, 2023). Nearly all of this evidence ends at policy outputs: laws, questions, executive programmes. Between the programme and the statute sits plenary speech, the arena in which parties actually spend their words. On this arena, the mandate literature is thinnest.

Four designs come closest to the one adopted here. Louwerse (2011, 2012) reads collective mandate fulfilment as congruence between manifestos and parliamentary speeches, for one election per decade between 1950 and 2010 in the Netherlands and the United Kingdom. He finds no difference in mandate fulfilment between the consensus and the majoritarian democracy. His congruence, however, is positional: within issue areas, texts are scaled into party positions, so what is compared are spatial configurations rather than distributions of attention in one shared coding scheme. Osnabrügge, Ash and Morelli (2023) supply the transfer template. They train a classifier on coded manifestos, apply it to the plenary speeches of the New Zealand Parliament from 1987 to 2002, and validate the transfer against human coding. They also explicitly name the systematic comparison of priorities across manifestos, speeches and other party documents as a research programme their method opens. Ivanusch (2025) carries out a first such comparison at scale, classifying speeches, press releases and tweets from Germany, Austria and Switzerland with a manifesto-trained model, and shows that channel characteristics moderate how much of its ideal agenda a party communicates. His design is a cross-section of the years 2019 to 2021, built to compare channels, not to measure congruence per party and legislative period over a long horizon. For the Bundestag itself, Rauh (2015) demonstrates for European Union affairs that issue salience in the plenary is measurable and can be explained institutionally. We return to this study in Section 2.4. Each of these studies supplies one element our design needs: a shared category space, a manifesto-to-speech comparison, or a long horizon. None supplies all three at once. Table 2.1 gives an overview of the comparison with neighbouring studies.

What is missing is a design that measures manifestos and plenary speech in one categorical space, at the level of the party and the legislative period, across a horizon long enough to contain the entry of a challenger party. We provide that design for Germany, over eight legislative periods from 1994 onwards. To our knowledge, it is the first systematic measurement of agenda congruence between programmes and plenary speech for the German party system. We extend the comparison by a directional reading of the same texts and by the polarisation dynamics

motivated in Section 2.4. Transferring a manifesto-trained classifier to parliamentary language does require corrections of its own, which Chapter 4 sets out.

Table 2.1: Direct predecessors: manifesto-to-parliament agenda linkage

Study	Country and period	Channels linked	Category space	Congruence concept	What remains open
Louwerse (2011, 2012)	NL and UK, one election per decade, 1950 to 2010	Manifestos and plenary speeches	Issue areas with positions scaled by Wordfish	Positional congruence of electoral and parliamentary competition	Salience distributions; one shared coding scheme on both sides
Froio, Bevan and Jennings (2017)	UK, 1983 to 2008	Manifestos and the government's legislative programme	CAP topics	Attention responsiveness (promissory versus anticipatory agenda)	The speech channel; congruence at the party level
Brouard et al. (2018)	France, 1981 to 2012	Manifestos and legislation	CAP topics	Conditional programme-to-policy attention linkage	The speech channel
Borghetto and Russo (2018)	Italy, Portugal, Spain, 2003 to 2014	Manifestos and parliamentary oral questions	CAP topics	Issue attention balancing mandate and current voter priorities	Plenary speech; distribution-level congruence
Guinaudeau and Guinaudeau (2023)	Germany, 1983 to 2016	Manifestos and legislation	CAP topics	Mandate responsiveness, conditional on government capacity	The speech channel; congruence per party and legislative period
Osnabrügge, Ash and Morelli (2023)	New Zealand, speeches 1987 to 2002	Manifesto-trained classifier applied to plenary speeches	Manifesto-derived topics (44 and 8 categories)	Method precedent; cross-channel priority comparison named as future research	Germany; a long panel; congruence as the quantity of interest; a directional layer
Ivanusch (2025)	Germany, Austria, Switzerland, 2019 to 2021	Manifesto-trained model applied to speeches, press releases, tweets	Manifesto-derived issue categories (20)	Channel moderation of the ideal agenda	A long horizon; congruence per party and legislative period; challenger entry

Note: The final column records design differences relative to the present study, not a quality assessment of the works listed.

2.4 Party System Change and the Politicisation of Europe

How party systems change when a new competitor arrives is a question the literature has answered mostly for the electoral arena. Niche and challenger parties compete by politicising issues the mainstream neglects or consciously suppresses, and their fate depends on how the established parties respond. Mainstream strategies of ignoring, accommodating or attacking the newcomer's issue shift that issue's salience and ownership, and with them the challenger's

electoral fortunes (Meguid, 2005, 2008). Challengers act, in this reading, as issue entrepreneurs who challenge the established order by mobilising around an underrepresented topic. Parties that lose on the dominant dimension of competition have the strongest incentive to mobilise new issues, most prominently European integration (Hobolt and de Vries, 2015; De Vries and Hobolt, 2020).

The responses of the established side are real but heterogeneous. Radical right success causes mainstream parties on the left and the right to shift their positions on the cultural dimension, as a regression discontinuity design around electoral thresholds shows (Abou-Chadi and Krause, 2020). The effect differs by challenger type: radical right success raises anti-immigrant emphasis, while green success leads established parties to de-emphasise the environment (Abou-Chadi, 2016). Anti-immigration parties also exert a contagion effect on immigration positions across entire party systems, with opposition parties more vulnerable than governing ones (van Spanje, 2010). Two features of this literature matter for our design: it is predominantly about positions, and it is predominantly about electoral competition. How the entry of a challenger reshapes agendas and speech inside parliament has been studied far less. Challenger parties thus change party systems in two ways: through what they say themselves, and through what they make everyone else say.

Of the four domains we read directionally, Europe is the one whose place needs an argument. Economy, welfare and migration are standing axes of German party competition, whereas the conflict structure of the European domain forms within the observation window itself, which is what makes it theoretically interesting here. Integration was long explained by the functional pressures of interdependence and settled among elites, while publics looked on with indifference. The postfunctionalist account, which breaks with that functional logic, holds that this permissive consensus is gone: integration is now contested in elections and referendums, and the contestation runs on national identity rather than on economic efficiency. Governments must therefore anticipate opposition at home before conceding further competences, a condition Hooghe and Marks (2009) call a constraining dissensus. The reaction against European and transnational integration has since been read as a new transnational cleavage, which transforms party systems through rising parties rather than through the conversion of programmatically inflexible established ones (Hooghe and Marks, 2018).

The empirical politicisation research qualifies this picture. Politicisation of Europe is episodic and crisis-driven rather than secular, varies strongly across countries, and is carried above all by populist challengers of the mainstream. Mainstream parties themselves, bound by coalition

considerations and internal divisions, often have incentives to keep the issue off the agenda (Hutter, Grande and Kriesi, 2016; Dolezal and Hellström, 2016; Hutter and Kriesi, 2019). Across Europe the issue has nonetheless remained a giant fast asleep, rarely politicised because mainstream parties had little incentive to raise it (Green-Pedersen, 2012). Germany fits that pattern. In the Bundestag between 1991 and 2013, attention to European Union affairs grew with the transfers of authority to the European level and was pushed by the mainstream and particularly the governing parties, not by challengers (Rauh, 2015). Until the euro crisis, Europe in German politics was a mainstream project with little partisan contestation, which is why the arrival of a Eurosceptic challenger is the sharpest available test of how entry reshapes a domain. That challenger arrived in two steps. The Alternative for Germany (AfD) was founded in 2013 as an anti-euro party. On the evidence of its early programmatic material, it was positioned at the far right of the German party system but neither populist nor radical right, and is best classified as soft Eurosceptic (Arzheimer, 2015). By its parliamentary entry in 2017, the party had transformed into a radical right party, with general right-wing views and negative attitudes towards immigration as the main motivation of its electorate (Arzheimer and Berning, 2019). What populist challengers do once inside parliament is documented mostly beyond Germany: in the Netherlands, populist opposition parties favour scrutiny of the government over participation in policy-making (Louwerse and Otjes, 2019). For the Bundestag itself, the closest study scales plenary debates around the entries of the Greens, the PDS, the Left Party and the AfD with Wordfish, a positional method. It finds that new parties often use populist language, but that the arrival of new contesters does not necessarily increase polarisation on their core issues: the established parties distance themselves from the newcomer's framing rather than converge on it (Lewandowsky et al., 2022). That finding is positional and rests on a different measurement logic than an agenda-based design, but it sharpens the question we put to the data. Whether the entry of a challenger polarises a parliament is not settled by its arrival alone. It depends on whether the established parties move.

Before polarisation can be traced, it needs a concept. At the level of the party system, polarisation is understood as the dispersion of party positions around the system's centre of gravity, weighted by party strength (Dalton, 2008). Dispersion of this kind can also be measured in parliamentary speech and interpreted over time. For the United Kingdom, the separability of the two main parties' speech tracks polarisation across 78 years of House of Commons debates and places today's Westminster at roughly the level of the mid-1960s (Peterson and Spirling, 2018). In the econometric literature on language divergence between groups, the partisanship

of United States congressional speech is far higher today than through most of the preceding century and has risen sharply from the early 1990s (Gentzkow, Shapiro and Taddy, 2019). One identification problem is central for what follows. When a new party enters and system polarisation rises, the rise can have two observationally equivalent sources at the trend level. The first is contagion, in which the established parties move apart, the mechanism of the accommodation literature above. The second is composition, in which a new pole is added while the established parties stand still. Separating the two requires recomputing the system quantity without the entering party, and that recomputation is therefore built into our design rather than added as an afterthought. A polarisation trend that disappears once the entering party is excluded identifies composition, one that persists identifies contagion, that is, movement among the established parties. The exclusion separates the arrival of a pole from movement among the parties already present, but not the direction of that movement, for which the weighted mean position has to be read alongside the dispersion.

2.5 Expectations and Hypotheses

From the literature we discussed above, we derive three distinct substantive expectations, stated below as six hypotheses. Two of them carry the main argument, H1a on the topics parties raise and H2a on where the two agendas come apart, and four deepen it on the directional layer, on which the validation of Section 4.5 sets a weaker floor.

First, we expect manifesto and speech agendas of the same party to be systematically linked: close relative to how strongly programmes differ across parties, and recognisable among the other parties' manifestos at a rate above chance, if not uniformly so, yet never so congruent that the institution in between would leave no mark. This follows from the mandate evidence of Section 2.3 read through the ownership logic of Section 2.1: if a party's attention profile is a reputational asset built for association and the manifesto its authoritative announcement, then speeches should resemble the announced agenda closely enough to be recognised, but, as discussed in Section 2.2, only within the limits the institution sets. To the extent that direction can be recovered from text, the same logic extends to the directional composition of what parties say within a policy domain.

H1a: Parties speak in parliament about the topics their election programmes emphasise. A party's speech agenda stands closer to its own programme than party programmes stand to one another, and is identifiable among the other parties' programmes at a rate well above chance.

H1b: The same holds for the side a party takes. Within a policy domain, the balance a party strikes between opposing positions in parliament resembles the balance struck in its own programme.

Second, if the institution drives the remaining divergence, then that divergence should look much the same for every party. It should hold content attributable to the parliamentary setting itself, such as the workings of the political system, the conduct of government and reactions to external events, rather than spreading evenly across the respective party agendas, and it should point in the same direction for every party. Where directional content can be recovered, speeches should show attenuated positions, moderated by the shared agenda towards a negotiable middle position. And across parties, speech positions should lie closer together than programme positions do, showing convergence across the channel within a period, not convergence over time.

H2a: Where speech and programme diverge, the divergence is concentrated on a few topics rather than spread evenly across the agenda, and it is largest on the Democracy & Political System bucket. It also points the same way for every party.

H2b: Parties argue less one-sidedly in parliament than in their programmes. A party that argues strongly for one side of an issue in its programme strikes a more even balance in plenary speech.

H2c: Party positions lie closer together in parliamentary speech than in party programmes, within the same period rather than as a development over time.

H2b and H2c follow from the same mechanism but are not the same claim, since a flatter mapping of manifesto onto speech positions compresses the individual party while the spread across parties can stay where it is, and a shift that moves every party alike changes the mapping without changing that spread at all.

Third, we expect polarisation in the European domain to rise in both channels once the AfD becomes part of them, which happens with its 2013 programme on the manifesto side and with its entry into the Bundestag in 2017 on the speech side, and the same logic extends to the migration domain, on which the AfD's radicalisation centred. Whether such a rise reflects only the entrant or also movement among the established parties is the fork discussed in Section 2.4: the arrival of a polarising challenger does not by itself settle how the established parties react. The design therefore separates the two sources by recomputing the system quantity without the AfD. On the speech side, this expectation inherits the conditionality of the directional layer, holding to the extent that direction can be recovered from text.

H3: Over the observation window the positions of the parties move further apart, in particular on Europe and migration, and around the AfD's entry into parliament in 2017.

The measurement decisions behind every term used here follow in Chapter 4, and Section 4.8 restates each hypothesis in the form in which it is taken to the data.

Data

3.1 GermaParl2

The speech side draws on GermaParl2, a linguistically and structurally annotated corpus of Bundestag plenary protocols covering debates since 1949, designed for the quantitative study of parliamentary speech (Blätte and Blessing, 2018); we use release v2.3.0-rc1 (Blätte and Leonhardt, 2025). The corpus is distributed in Corpus Workbench (CWB) format and accessed through polmineR, whose structural attributes (legislative period, date, speaker role, party) drive the extraction described below. Our window covers LPs 13 to 20, from 1994 to the present, as a design choice on comparability grounds: three decades give the time dimension enough variation, while reaching further back would import a party system and political discourse increasingly remote from the periods under study. Among available alternatives, notably ParlSpeech V2 (Rauh and Schwalbach, 2020), GermaParl2 offers the annotation depth and CWB structure this pipeline requires.

We extract only plenary contributions tagged as speech and made by an MP or member of the federal government, which excludes chair and presidium interventions. Interjections or Annotations like applause enter the corpus as parenetical annotations, we strip them from the speech text. We then count the words and drop all speeches below a word count of 100 after clearing. We restrict attention to plenary speech, the party-controlled, fully documented, and cross-party comparable arena of parliamentary communication (Proksch and Slapin, 2015).

3.2 Party Manifestos: MARPOR

The Manifesto Project (MARPOR; Lehmann et al., 2025) provides human-coded content analyses of party manifestos since 1945. Trained coders split each manifesto into quasi-sentences, i.e. sentence sub-parts carrying one distinct political statement, and assign each quasi-sentence to one of the 56 policy categories of the project's coding scheme (Handbook 4; Werner et al., 2011). The scheme is constructed on salience-theoretic grounds, which makes it a natural measurement language for an agenda comparison (Budge et al., 2001; Klingemann et

al., 2006). For every party-LP pair this yields an observed distribution over the 56 categories directly: on the manifesto side, no classification step is required.

The speech side carries no such annotation and Sections 4.3 to 4.6 spell out how we close this asymmetry. We use the project's main dataset of aggregated category shares rather than its annotated corpus, whose quasi-sentence texts cover only a subset of our party-LP cells. The permutation procedure originally specified on pooled sentences was dropped with this choice. Category shares are renormalised to sum to one per manifesto. Two properties of the manifesto side matter later: MARPOR codes the CDU and CSU jointly as one programmatic actor, which matches their treatment as one parliamentary party on the speech side, and only Handbook 4 three-digit categories are available, without directional sub-codes, as have been introduced in Handbook 5, a constraint Section 4.7.1 returns to. Human coding is itself imperfect at the fine-code level (Mikhaylov et al., 2012). This is the accepted field standard, and the bucket aggregation of Section 4.7.1 smooths fine-code noise.

3.3 Linking and the Analysis Sample

Each legislative period is paired with the manifesto of the election from which it emerged: the programme states the announcement, the speeches that follow are read against it. Pairing with the following election's manifesto would invert this logic. Which of the party-LP combinations enter the analysis, and why, is declared once in Section 4.8; the resulting sample comprises 38 valid cells.

Methods

The essential conception of the thesis is to measure the congruence between the proposed agenda of a given party represented by the manifesto of the election from which a legislative period (LP) emerges and the practised agenda in the form of plenary speeches in that LP. This congruence is operationalised as the distance between two topic distributions and measured by the Jensen–Shannon divergence (Section 4.7.3). To this end we work with a small set of politically distinct topic buckets. We also include a sub-aggregation to some of these categories allowing a directional analysis (Section 4.7.1). The analysis is conducted at the party-period cell level. The chain runs from speech to sentence to MARPOR category to topic bucket to distribution to divergence.

The primary specification and the acceptance criteria were fixed before the analysis (Section 4.2). The two data sources were introduced in Chapter 3. This chapter proceeds along the

measurement chain: the research design (Sections 4.1 and 4.2), speech-side preprocessing (Section 4.3), classification (Section 4.4), validation (Section 4.5), corrections (Section 4.6), aggregation and measurement (Section 4.7), the empirical strategy (Section 4.8) and, finally, the robustness design (Section 4.9).

4.1 Unit of Analysis and Comparison Logic

The unit of analysis is the party-period cell, one party observed over one legislative period. For each cell we derive two probability distributions over the same topic buckets (Section 4.7.1), one from the party's manifesto (*m*) and one from its parliamentary speeches (*s*). Congruence is then read as the closeness of *s* to *m*, measured by the Jensen–Shannon divergence (Section 4.7.3). The divergence summarises in a single number how differently the two distributions spread their attention, and runs from zero, where they are identical, to one, where they place their mass on entirely separate buckets.

Speaking time in parliament is budgeted: attention devoted to one issue is unavailable to another, so the scarce resource a party allocates is relative attention. The full distribution of that attention across topic buckets, rather than any single statement or position, is therefore the level at which an announced and a practised agenda can be compared (Chapter 2). What the comparison measures is accordingly salience and not position. The divergence responds to how much attention each bucket receives and not to what is said inside it, so a manifesto and a speech that give migration the same share of attention agree on that bucket even when one argues for restriction and the other against it. The directional sub-aggregation of Section 4.7.1 recovers part of that positional content, but it rests on the weaker validation floor of Section 4.5 and is read with corresponding caution throughout.

4.2 Pre-Commitment Architecture

If analytic choices are made after seeing the data, nominal inference loses its meaning even without intentional p-hacking (Gelman and Loken, 2014). The accepted answer in automated text analysis is to fix design decisions before results are inspected (Barberá et al., 2021; Grimmer, Roberts and Stewart, 2022). We follow this approach without a formal external registry: the decision rules were fixed in code and project documents before we inspected the primary results. Five choices carry the design: the primary specification and the aggregation scheme (Section 4.7.2), the procedural-sentence typology, decided on formal grounds before its effects were inspected (Section 4.3.2), the acceptance criteria of the robustness battery (Section 4.9), the thresholds of the redundancy verdict on the two corrections (Section 4.6.2),

and the frozen gold-standard sample (Section 4.5.1). The sections of this chapter document each of these choices in the order of the measurement chain.

4.3 Speech-Side Preprocessing

4.3.1 Sentence Segmentation (SoMaJo)

The speech side is classified sentence by sentence. The classifier used (ManifestoBERTa) takes a single sentence as its input unit. Thus, the sentence is the atomic classification unit on the speech side. Manifestos are classified by quasi-sentences, i.e. partial sentences that carry independent coding-relevant statements, as determined by the human coders. Because quasi-sentence boundaries reflect subjective human-coder decisions, this segmentation cannot be reproduced on the speech side, and we are restricted to the sentence as unit of measurement. As discussed in Section 4.1, the unit of analysis is the party–LP cell, and in that the probabilities are aggregated over all speeches in the given LP by a given party. Therefore, the comparison is never made at the unit (sentence to quasi-sentence) level but done as a distributional average over the whole cell mass. Although the quasi-sentence coding allows a more granular analysis, the granularity difference cannot be assumed to be uncorrelated with topic. The soft aggregation of Section 4.7.2 spreads each sentence's posterior across categories, so a sentence touching several topics does register on more than one of them, but its total weight stays at one under the inclusive specification, and this is more a representation of classifier uncertainty than the classifier assigning more than one topic. A manifesto sentence that a coder splits into three quasi-sentences contributes three coded units instead. Content that typically enters as a subordinate statement may therefore carry less relative mass on the speech side than the same content carries in a manifesto. The classifier is also trained on quasi-sentences each carrying a single code, so the spread of its posterior expresses uncertainty about which one code applies rather than a mixture of topics. We treat this as a limitation of the transfer design rather than as a difference that averages out, and three considerations bound its weight. The validation of the speech corpus is conducted on the same sentence basis, in both the gold- and the silver-standard step, so the reliability figures are measured under the same unit constraint the analysis operates under. Däubler et al. (2012) question quasi-sentence unitisation on the grounds that human identification of quasi-sentence boundaries is itself unreliable, and argue for natural sentences as an exogenous, reproducible coding unit. And the threshold arm of Section 4.9 replaces the soft aggregation with argmax assignment, and on the salience layer the cell rankings survive that replacement up to a level shift, so the salience results do not turn on how the posterior is read. The essential discrepancy is nevertheless acknowledged, which is why inference is

restricted to cell-level distributions. Because the sentence is the measurement unit, segmentation quality is itself measurement-relevant and non-trivial for German texts. The period is an overloaded character: phrases like "1. Oktober" and "Art. 5 Abs. 1" make a naive splitting of sentences on punctuation insufficiently accurate. We account for this by utilising SoMaJo (Proisl and Uhrig, 2016) for corpus segmentation, which is a rule-based, deterministic tokeniser and sentence splitter for German, run under its German EmpiriST tokenisation guidelines ("de_CMC").¹ It relies on hand-crafted rules and curated abbreviation lists rather than a statistical language model, making its output deterministic and, at a pinned version, fully reproducible, with no learned parameters or random seed.² SoMaJo tokenises the text and sets sentence boundaries; each sentence surface is then reconstructed from the token stream via the `space_after` annotation, which reattaches the punctuation that the CWB representation stores as detached tokens. Sentences that hold only non-alphabetic content after splitting — bare numbers, punctuation and the like — are removed from the corpus. The criterion is alphabetic content rather than length, so short but genuine sentences ('Ja.', 'Nein.') are retained; genre-specific but linguistic material such as salutations remain in place and are handled by the procedural filter (Section 4.3.2). Overall, the segmentation of sentences utilising SoMaJo yields a total of 6,139,035 sentences. Segments containing no alphabetic character, such as isolated numbers or punctuation, are discarded at this step and are not part of that count. Proisl and Uhrig report a tokenisation F1 of 99.57 for SoMaJo, averaged across the web and CMC subcorpora of the EmpiriST 2015 shared task; the figure therefore speaks to tokenisation of written online German, not to sentence splitting of parliamentary speech. We therefore validate segmentation quality internally. The silver annotation scheme included a dedicated mis-segmentation category, so each sentence was explicitly checked for mis-segmentation. The mis-segmentation rate under SoMaJo is 0.895% across the two silver samples (Section 4.5.2), which is small enough not to shift the cell-level distributions on which inference is built. We acknowledge that the silver samples are coded by an AI model (Sonnet 4.6) and cover only a fraction of the full corpus; they are nonetheless large enough, at $n = 20,000$ sentences in total, to support these claims. Implementation details of the segmentation are documented in [Appendix A](#).

¹ Contrary to the default behaviour, `split_camel_case` is set to false, since abbreviations such as GroKo must not be split.

² The version of SoMaJo used here is 2.4.3.

4.3.2 Procedural-Sentence Filter

In contrast to manifestos, parliamentary speeches carry additional genre-specific language, such as procedural formulae and opening and closing remarks (Proksch and Slapin, 2015). This difference creates a significant problem for cross-domain topic classification as the classifier trained on manifestos only simply has no category for this type of language but is designed to assign a category to every sentence. Hence, it systematically misclassifies sentences of this type. To remove some of these sentences we built a dictionary of procedural sentence forms. We compiled a corpus-wide frequency list of normalised sentence forms and hand-coded the 500 most frequent, flagging 228 of them across five types: salutations, thanks and closing formulas, structuring markers, interjections and the short exchanges that accompany an intervention.³ Because normalisation masks names, one entry covers all of its surface variants, so 'herr <name>' matches 'Herr Präsident!' as well as any address to a named member, while 'vielen dank' covers the closing formula in its most frequent form. A sentence is flagged as procedural when its normalised form exactly matches a dictionary entry. The primary specification then keeps only the non-flagged sentences, while the full corpus is retained for robustness checks. Because the entire corpus is already classified, flagging rather than deleting lets both the filtered specification and the unfiltered robustness arm derive from the same basis, requiring no re-classification. This procedural sentence filter in total excludes 354,364 sentences, 5.77% of the full corpus after sentence segmentation. The procedural filter and the null-class filter of Section 4.6.1 are complementary rather than redundant, as the 000-filter also catches procedural sentences outside of the most common sentences forms but still without political information and would be misclassified, while the surface filter catches procedural sentences to which the classifier assigns a spurious substantive label, which would thus survive the 000-filter. We have nevertheless to acknowledge that by coding only the 500 most common normalised sentence forms of the corpus the long tail of such sentences is not caught and remains in the corpus; the residual mass is small, partly absorbed by the 000-filter and the unfiltered robustness arm confirms the results are stable with or without the filter. The full hand-coded typology is reproduced in [Appendix B](#).

³ Normalisation masks named appellations to a placeholder ('Herr Kubicki' → 'herr <name>'), then lowercases, collapses whitespace, and strips leading/trailing punctuation; the identical routine builds the candidate list and matches the corpus, so a single entry ('herzlichen dank, herr <name>') captures all name variants.

4.4 Automated Classification (ManifestoBERTa)

Classifying 6,139,035 sentences by topic is only feasible through automated classification. Transfer learning with models of the BERT family has been established as a standard approach for such tasks in political science (Laurer et al., 2024). We decided on using a transformer encoder model based on the XLM-RoBERTa-large architecture, a multilingual model pretrained by Facebook AI (Conneau et al., 2020).⁴

We use an adaptation of this model called ManifestoBERTa (Burst et al., 2024), which has been fine-tuned in a supervised fashion on 1,691,707 hand-coded quasi-sentences of the Manifesto Corpus, version 2024a (Merz et al., 2016), covering manifestos in 38 languages. Each quasi-sentence is coded into exactly one of the 56 three-digit Handbook-4 codes of the MARPOR scheme, as discussed in detail in Section 3.2.⁵

There are two model types of ManifestoBERTa, a context and a sentence-only one. The version used is 56topics-context-2024-1-1. In the context variant, the input is a sentence pair: the sentence to be classified, and a context block in which the sentence is embedded in its surrounding text. During training, this context was greedily filled up to 200 tokens. In the authors' held-out evaluation, the context variant achieves an accuracy of 0.64, compared to 0.57 for the sentence-only variant.⁶ Our implementation, however, differs from the training convention: we construct the context as the immediately preceding and following sentence only, rather than filling the 200-token budget, a mild train–inference discrepancy; the model card provides for a context that falls short of the budget, since it instructs that a sentence without available context be passed as its own context, and we return to the point in the limitations. Context is constructed strictly within speech boundaries. The model is used as released on

⁴Pretrained by Facebook AI on 2.5 TB of filtered CommonCrawl data in 100 languages, with about 550M parameters (Conneau et al., 2020).

⁵The standard classification head was modified by removing the tanh activation and the intermediate linear layer, reducing it to a direct linear projection from the sentence representation onto the 56 logits (Burst et al., 2024).

⁶Accuracy figures as reported on the two Hugging Face model cards (accessed 10 July 2026).

Hugging Face with frozen weights which are not changed at any point in the process.⁷ All corrections applied in this thesis operate downstream of the frozen model.

The output we save from the classification run is the full 56-dimensional softmax vector, from which the top-1 category and its probability are derived. The full vector of probabilities is retained because reducing each sentence to its argmax would discard information about certainty of the assignment. Two sentences with distributions of (.35, .33, .32, ...) and (.95, .03, .02, ...) are indistinguishable using an argmax decision, whilst differing substantially in certainty of prediction. This allows the inference to be built on soft aggregation (Section 4.7.2), robustness checks such as temperature scaling and hard assignment under confidence thresholds (Section 4.9), and the confidence diagnostics of Section 4.5.3.

To make a meaningful comparison between manifestos and Bundestag speeches, we need to measure both channels in the same categorical space. Because the ManifestoBERTa model is trained to predict into the very scheme on which manifestos are coded, both distributions live in the same 56-category space. Without using the same scheme for both channels a crosswalk has to be made between two schemes. Any such mapping introduces an additional, unvalidated measurement assumption: because the two schemes are not matching, part of the divergence one measures could originate in the mapping rather than in the texts themselves. This makes the shared coding inside the MARPOR scheme not a convenience but a design requirement. Osnabrügge et al. (2023) establish a precedent for this approach, applying a manifesto-trained classifier to parliamentary speech and validating it against expert coding. Their finding that accuracy varies considerably by topic anticipates the category-specific diagnostics we undertake in Section 4.5.

While the approach of zero-shot domain transfer has practical merit and avoids the mapping problem set out above, it introduces two structural risks that follow directly from the difference between the two domains.

Risk one has its origin in the general difference in genres that are being compared. On the one hand there are manifestos, written documents aimed strictly at conveying a political program; on the other hand, Bundestag speeches, also vehicles of political messages but in a very different format, shaped by the spoken word and the debate culture of the institution. Examples

⁷The model is run in evaluation mode, which deactivates dropout; inference is thus deterministic in design (no stochastic components), although exact bitwise reproducibility across hardware configurations is, as usual for GPU inference, not guaranteed.

would be the voicing of subjective feelings on a given topic and direct attacks on political opponents. This means that the classifier, which has been trained on manifestos alone, has to classify speech texts and their unfamiliar style into categories learned from manifesto-style texts. The risk that follows is that the classifier has learned to look for specific signals in syntax and rhetoric in manifestos for a given topic which then leads to misclassifications in speeches as some signals simply do not exist or are typical of a different topic in this domain. All accuracy figures mentioned above are from in-domain tests, as the model card reports no numbers on cross-domain validity of any sort. Our setting is what Ramponi and Plank (2020) describe as unsupervised domain adaptation: application to a target domain for which no labelled data exists. Their survey documents domain shift as an established methodological problem in NLP. Evidence from related PLM-based (Pretrained language model) classifiers shows this concern to be more than theoretical: Roussinov and Sharoff (2023) report F1 drops of 20–30 percentage points for BERT- and RoBERTa-large under distributional shift between training and application. Their configuration is the inverse of ours. Genre is the target and topic varies, whereas here topic is the target and genre varies, but the underlying entanglement of topic and style is the same. Crucially, this degradation in accuracy is unlikely to be uniform across categories: both Osnabrügge et al. (2023) and Roussinov and Sharoff (2023) find that the size of the performance drop varies substantially across topics. An aggregate accuracy figure would thus conceal rather than reveal the problem, which is why the validation in Section 4.5 evaluates the classifier at the category level.

The second risk announced above is the absence of a residual category. Although manifestos in hand coding do have a 000 category for non-codable content (Werner et al., 2011), the classifier's label space comprises only the 56 political categories (Burst et al., 2024). The model card states that training used all quasi-sentences of the corpus yet does not document how 000-coded quasi-sentences were handled. In the speech domain there is a considerable part of text that has no inherent political meaning. These genre-specific elements range from procedural content, to voicing of disdain or disagreement, to greetings and thanks. The procedural filter introduced in Section 4.3.2 removes part of this material, but only on the procedural side. By construction, however, it cannot exhaust it: a rule-based surface filter only detects what it matches. Non-codable content that follows no fixed formula leaves no surface signature to match on, passes the filter, and reaches the classifier, which, lacking a residual category, must assign it one of the 56 codes. This is an instance of what the literature calls the closed-world assumption, the premise that every case at test time belongs to one of the classes known at training time (Scheirer et al., 2013). In the present setting that premise is enforced mechanically:

the softmax mass over the 56 categories sums to one even when the classified sentence fits none of them. This forced assignment does not announce itself as uncertainty. Neural classifiers of this type have been shown to be systematically overconfident, as it has been shown that their softmax probabilities overstate the empirical accuracy of their predictions (Guo et al., 2017). A code assigned in the absence of any fitting category will therefore appear in the output as a confident prediction. Because the misdirected probability mass is concentrated rather than dispersed, it does not average out across a cell: it shifts the topic distribution in a systematic direction. Consequently, the classifier's own confidence cannot serve as a signal for detecting non-codable sentences: identifying them requires a signal independent of ManifestoBERTa's posterior.

The two risks discussed above are logically distinct. The first concerns codable sentences assigned to the wrong category, which is a failure of the mapping from surface to category. The second risk concerns sentences being forced into a wrong category by the classifier, which is a failure of the output range. Neither is remedied by resolving the other: a perfectly genre-adapted classifier would still have no code for a procedural announcement, and a classifier equipped with a residual category would still misread the rhetoric of a hostile intervention. Whether these risks materialise, and how strongly, is an empirical question that Section 4.5 will address.

4.5 Validation

Because automated text-classification methods carry no context-free guarantee of accuracy, they require validation against the specific task to which they are applied (Grimmer and Stewart, 2013). To validate the results from the classification two complementary validation instruments are implemented. The first is a gold-standard sample, hand-coded by a human coder, which supports per-class reliability and a targeted precision estimate for code 305; it is deliberately not corpus-proportional. Second, a silver-standard validation sample, that has been blind coded by Anthropic Claude Sonnet 4.6, to allow us to measure corpus-weighted consistency based of two independent automatic coders. These two instruments together allow the assessing of the two risks discussed in Section 4.4. The purpose of this section is diagnosis, the solutions will be discussed in Section 4.6.

4.5.1 Gold-Standard Validation

The purpose of the gold-standard validation is per-class validation and precision estimation; the sample is therefore by design not proportional to the corpus. The gold sample comprises 259 sentences, stratified on the model's predicted bucket at the finest level the analysis distinguishes: the B-bucket where the predicted code carries one, the A-bucket otherwise. This

yields 16 strata, consisting of the nine B-buckets, the five non-directional A-buckets, and two residual strata for Economy and Welfare codes that carry no B-assignment. For each category the sample is drawn randomly but must contain at least 10 observations. Furthermore, early checks indicated that the DPS bucket (Democracy & Political System), and code 305 in particular, is over-assigned by the classifier, most likely on procedural content. The DPS bucket is therefore deliberately oversampled ($n = 56$). Readers seeking corpus-representative rates must turn to Section 4.5.2. The realised allocation of the gold sample across the strata is documented in [Appendix C](#).

The coding was done by a single human coder relying on MARPOR Handbook 4, identically to the coding scheme ManifestoBERTa is trained with. The labelling was done blind to classifier labels and metadata. The coder first assigned an A-bucket label and then, where that bucket is one of the four directional domains, a B-bucket label. Agreement is thus estimated at the levels the analysis uses, not at the 56-code level the pipeline never reports. This decision is supported by the literature, specifically by Mikhaylov, Laver and Benoit (2012), who report that MARPOR-trained coders reach only moderate reliability on the 56-category scheme. Validating at the 56-code level would therefore measure the granularity of the scheme rather than the performance of the classifier, which inherits that ceiling from its training labels. As the coding has been done by a single human coder, what is measured here is human-versus-model agreement with the human as reference, not inter-coder reliability.

Ten of the 259 coded sentences could not be matched to the final corpus version and are excluded, leaving 249, with two strata thereby falling to nine.⁸ The coder-vs-model agreement is measured between the A-bucket carrying the classifier's argmax code and the A-bucket label the coder assigned. The comparison is made using Cohen's κ . Cohen's κ reports agreement corrected for the agreement expected by chance alone, so a value of zero indicates performance at chance and a value of one perfect correspondence. The $\kappa(A)$ measured on the 249 sentences is 0.562 with raw agreement of 0.614. Since the classifier has no category in which it could ever express the human coder's judgement of 000, i.e. a non-codable sentence not fitting any of the 56 categories, every such sentence enters the comparison as a disagreement by construction, and agreement is therefore reported a second time with them removed. Restricted to the 226 sentences the human coder identified as codable, $\kappa(A)$ rises to 0.630 at a raw agreement of 0.677. Krippendorff's α returns 0.561 and 0.630 (codable sentences) respectively, so neither

⁸ Against the prediction stored at sampling time, $\kappa(A)$ over all 259 coded sentences is 0.570, against 0.562 over the 249 carried. The loss does not shift the estimate.

figure is an artefact of the κ statistic. Against the conventional interpretive bands (Landis and Koch, 1977), the results fall at the upper end of the moderate range and the lower end of the substantial range respectively, but those bounds are general convention rather than grounded in test theory.

The gap between the two figures is structural rather than noise. The 23 sentences (9.2 per cent) the human marked as not assignable cannot be matched by the classifier, because code 000 is not included among the 56 categories and every sentence must therefore receive one of them. $\kappa(A)$ thus contains a category that only one of the two coders can ever use. Section 4.5.3 returns to what follows from this. The codable-only figure is not a licence to disregard those sentences, as in the corpus they are not identifiable in advance, and the classifier assigns them a code regardless. Section 4.6.1 addresses this. $\kappa(A)$ scores the argmax label alone, whereas the analysis aggregates over the full posterior (Section 4.7.2). The figure reported here is therefore a conservative floor rather than a summary of the classifier's fitness for the use it is put to.

The gold sample was further designed to estimate precision for code 305, which is why the DPS bucket was oversampled. Because it is stratified on the classifier's predicted bucket rather than on the coder's judgement, precision is the quantity it delivers. The classifier assigns code 305 to 28 of the 249 sentences, of which 23 survive the procedural filter. Precision is reported on those 23, so that the estimate rests on the procedurally filtered corpus rather than on the raw classifier output (Section 4.3.2). This works against the exclusion argued in Section 4.6.2 rather than for it: not one of the five procedurally flagged sentences is confirmed by the coder, and their inclusion would lower precision from 43.5 to 35.7 per cent. The null-class filter of Section 4.6.1 is not applied here, since this validation is what motivates it. Because that filter removes precisely the kind of sentence the coder marks as non-codable, applying it would raise the reported precision further, so the figure also sits below what the primary specification would yield. Of the 23, the coder placed 10 (43.5 per cent) in the DPS bucket and 10 (43.5 per cent) in the non-codable category, the remaining three elsewhere. That as many of these sentences carry no codable content as carry content on democracy and the political system is the forced assignment of Section 4.4 surfacing in the gold data.

Two qualifications bound the figure. Confirmation here is bucket-level: the coder assigned the sentence to the DPS bucket, which contains nine codes, so a sentence genuinely belonging to code 301 counts as confirmed although the classifier's 305 is wrong. The 43.5 per cent is therefore an upper bound on precision for code 305 itself. And with $n = 23$ the estimate is correspondingly imprecise. Section 4.6.2 takes up what follows for the treatment of code 305.

Directional agreement is measured on the B-buckets, comparing the direction the coder assigned to the B-bucket carrying the classifier's argmax code. Since direction is defined only within the four directional domains, the comparison is restricted to the 98 sentences to which the coder assigned one; $\kappa(B)$ is 0.456 at a raw agreement of 0.500. Direction is thus recovered less reliably than domain, which is the reason for our decision to read findings on the B level with more caution.

4.5.2 Silver-Standard Validation

The gold-standard sample validation provides insights into precision but, being stratified on the classifier's predictions, it cannot support corpus-representative claims about distributions. That is why we designed a silver-standard validation, for which we used Anthropic's Claude Sonnet 4.6 to categorise a simple random sample of 10,000 sentences drawn from the full classified speech corpus, a design choice supported by evidence that LLM annotation of political text matches or exceeds crowd-coder quality (Gilardi et al., 2023).⁹ The coding is based on a prompt that has been modelled on the MARPOR Handbook 4 (Werner et al., 2011), the same scheme on which the classifier is trained.¹⁰ It differs in one essential respect, in that the model was explicitly instructed to assign all non-codable sentences to a non-codable class. This allows us to make an estimate of the corpus-wide share of non-codable sentences. The model received only the sentence in question together with the previous and following sentence, so here too the classification is done blind, as in Section 4.5.1, without metadata or classifier labels.

What the silver sample measures has to be stated precisely: it captures the corpus-weighted consistency of two independent automated coders, not validity in the strict sense. Claude Sonnet is no ground truth, and agreement between two models could in principle reflect shared error rather than shared accuracy. What lends it credibility as a reference in this comparison is its own anchoring in the gold sample: against the human coder, Sonnet reaches a $\kappa(A)$ of 0.710 across all 249 carried gold sentences and 0.693 restricted to the sentences the human coder identified as codable, and thereby exceeds the classifier's 0.562 and 0.630 on the same sentences (Section 4.5.1). One qualification belongs with that comparison. The gold sample is stratified on the classifier's own predicted bucket and oversamples the Democracy & Political System bucket, so the two coders are compared on ground chosen where the classifier is weakest, and the gap it reports is an upper bound on the gap that would obtain corpus-wide. What the comparison establishes is the ordering of the two coders, not its size, and the ordering is what

⁹ model identifier claude-sonnet-4-6, accessed via the batch API.

¹⁰ See Appendix D.

the argument uses. The silver sample therefore does not replace the human validation but extends it. The gold anchor establishes which of the two automated coders is the more reliable one; the silver sample then projects that comparison onto the corpus-representative basis that 249 gold sentences structurally cannot provide. Two limitations bound this construction. First, the annotation quality of LLMs varies strongly across tasks, and performance on one annotation task does not predict performance on another (Ziems et al., 2024; Pangakis et al., 2023). Benchmark results from other applications therefore carry no guarantee for MARPOR coding of parliamentary speech, which is why the instrument is validated against human labels for this specific application, as discussed above. Second, a proprietary model can be modified or withdrawn by its provider, so that the same request may not return the same output at a later point (Ollion et al., 2024). Full reproducibility can therefore not be guaranteed; what can be fixed is fixed: the exact model identifier and the coding instructions, documented verbatim in [Appendix D](#).

The results of the silver-standard validation are reported in the same way as the gold-standard results. On the domain level, $\kappa(A)$ is 0.429 across all 10,000 sentences and rises to 0.583 once the comparison is restricted to the 7,711 sentences Sonnet identifies as codable. Krippendorff's α is 0.408 on all and 0.579 on codable sentences, ruling out the possibility that these figures are artefacts of the κ statistic. The gap between the two figures is not noise but the first corpus-scale symptom of the missing null class anticipated in Section 4.4; Section 4.5.3 traces where those sentences go. On the directional level, the two automated coders reach a $\kappa(B)$ of 0.698 on the 2,576 sentences Sonnet assigns to a directional domain. Conditional on the domain placement, agreement on the direction is therefore higher than agreement on the domain itself.¹¹ This consistency does not revise the assessment of the B level in Section 4.5.1, since agreement between two automated coders cannot substitute for the thin human anchor on which that decision rests. On the fine-code level, the consistency of the two automated coders drops to a $\kappa(F)$ of 0.363 across all 56 categories, one reason the analysis proceeds on aggregated buckets rather than on raw categories (Section 4.7.1).

Beyond validation, the silver sample serves a second purpose: it provides the training labels for the text-based null-class classifier introduced in Section 4.6.1. For this purpose, we coded a second, disjointly drawn random sample of 10,000 sentences under the same specifications; all reliability figures reported in this section rest on the first sample alone. That second sample also

¹¹ The gold sample supports the same ordering at the directional level: against the human coder, Sonnet reaches a $\kappa(B)$ of 0.622 to the classifier's 0.490 ($n = 111$, strict direction handling, full gold basis, which is why $\kappa(B)$ here differs to the $\kappa(B)$ reported in Section 4.5.1).

provides an unplanned check on the reproducibility caveat above. Coded independently under the same instructions, it returns a null-class share of 23.0% against 22.9% in the first sample, a domain-level $\kappa(A)$ of 0.429 against 0.429, and 0.583 against 0.583 once the comparison is restricted to codable sentences. Of the figures reported here, the only one whose difference reaches conventional significance is Sonnet's own use of code 305, at 3.4% against 4.1% ($z = 2.38$, $p = 0.02$), which widens rather than narrows the discrepancy reported in Section 4.5.3. The figures of this section are therefore not an artefact of a single coding run.

4.5.3 Diagnosis

The validation results of the two preceding sections point to two distinct failure modes of the cross-domain transfer. One is the null-class gap, anticipated as the second risk in Section 4.4. Sonnet, whose label space includes the 000 category, assigns it to 22.9% of the corpus-representative sample, 2,289 of the 10,000 sentences. ManifestoBERTa assigns it to none of them. This is not a difference in judgement but a structural property: the category does not exist in the classifier's label space, so every sentence receives one of the 56 substantive codes regardless of whether it carries codable content. The silver sample shows where these forced codes go. On the code level, 54.2% of the sentences Sonnet deems non-codable (1,240 of 2,289) receive code 305, and on the bucket level 68.6% end up in the Democracy & Political System bucket. These misassignments do not announce themselves as uncertain. The classifier's median confidence on the very sentences that Sonnet codes as 000 and that the classifier codes as 305 is 0.746, a confident error rather than a hesitant one. A confidence threshold can therefore not close the gap: 9.4% of the non-codable sentences receive codes other than 305 at confidences above 0.7, so any cutoff strict enough to catch the forced codes would discard correct predictions wholesale. The gap is a property of the label space, not of calibration, which is why Section 4.6.1 introduces a text-based null-class classifier rather than a confidence filter.

The other, sink-category inflation, concerns code 305 itself. Across the silver sample, the classifier assigns 305 to 24.9% of the sentences, while Sonnet assigns it to 4.1% of the same sentences. Because both coders read the same speeches under the same scheme, the discrepancy cannot be a property of the genre or of the coding scheme and must lie with one of the two coders. Given the gold anchoring established in Section 4.5.2, it lies with the classifier. That accuracy varies strongly by topic in this kind of transfer is documented (Osnabrügge et al., 2023), but the direction does not carry over: in their English-language application political authority is one of the better-performing categories, which is why the inflation observed here has to be established on this corpus rather than assumed from theirs. Diagnosing the inflation

requires the silver sample. Although the gold design deliberately oversamples the DPS bucket (Section 4.5.1), code 305 enters the gold sample at only 11.2% against its 24.9% corpus rate, so the inflation rate is structurally invisible there. What the gold sample contributes is precision: only 43.5% of the classifier's 305 assignments are confirmed by the human coder ($n = 23$, Section 4.5.1), a figure computed over all of them, including those the coder marks as non-codable. The two failure modes are connected but not identical. The forced codes of the null-class gap feed the inflation, since 54.2% of them land on 305, yet 305 remains over-assigned even among the sentences both coders consider codable, which is the residual the null-class filter cannot reach and which Section 4.6.1 quantifies. Neither correction can therefore substitute for the other.

The two modes do not stand in one-to-one correspondence with the two risks of Section 4.4. The null-class risk materialises in both of them, whereas the genre risk has no mode of its own, surfacing here only as the part of the 305 inflation that survives among codable sentences, and otherwise in the category-level agreement figures rather than in any single displaced category. That is why it receives no correction but bounds how the results are read. A diffuse loss of accuracy cannot be subtracted, only measured, which is what the category-level design of Section 4.5 is for and why the directional layer is held to the weaker floor throughout.

Both failure modes are instances of the domain-shift problem discussed in Section 4.4 (Ramponi and Plank, 2020). As named and quantified failure modes, however, forced assignment through a missing null class and sink-category inflation have, to our knowledge, not been documented in political science applications of manifesto-trained classifiers. Osnabrügge et al. (2023) document topic-dependent accuracy and propose diagnostics for it, but neither mode is named or measured there. Their own label space does carry a no-topic category, so the gap arises where a pre-trained manifesto classifier without one is transferred, as here. The diagnosis is therefore a contribution in its own right, and the corrections in Section 4.6 rest directly on it.

Values of κ around 0.6 on the domain level carry aggregated distributional claims, which is what the research design requires. Every Jensen–Shannon divergence in the analysis compares thematic distributions in which each party–LP cell on the speech side aggregates between 41,562 and 279,947 sentences, so random classification noise averages out at this scale. Systematic error does not average out, and the two failure modes described above are systematic, which is why they receive corrections in Section 4.6 rather than caveats. What the figures do not carry are sentence-level claims, fine-grained statements about individual cells,

and the directional B level, for which Section 4.5.1 sets the weaker floor. The per-axis decomposition of the silver sample shows why this designation stands even though the pooled $\kappa(B)$ of 0.698 appears solid: agreement ranges from an F1 of 0.848 for welfare expansion and 0.800 for the Pro-EU direction down to 0.148 for the limitation of the welfare state, so the pooled figure conceals axes that differ by more than half the scale. The weakest of them is also the thinnest. It rests on 33 reference sentences, and the second silver sample returns 0.310 on 38, so the point estimate itself carries little precision. Both readings place the axis far below every other one, which is what the designation rests on. Landis and Koch (1977) locate the domain-level values in the moderate to substantial range. We cite the convention as orientation rather than as a licence, because what matters is not the label but what the estimand requires.

4.6 Corrections

The validation diagnosed two failure modes of the classification (Section 4.5.3). In the following, we propose and implement one correction for each.

4.6.1 Non-Codable Sentences Filter (gbert-000)

The goal of this instrument is to alleviate the null-class gap diagnosed in Section 4.5.3. The classifier has no null class, so sentences that carry no codable policy content are nonetheless forced into a substantive category. To address this, we trained a deepset/gbert-large model (Chan et al., 2020) to predict whether a sentence is codable or non-codable. Its training data are the 20,000 randomly drawn sentences of the two silver samples, and its label is the code Claude Sonnet assigned to each of them under the coding instructions shown in Appendix D. This allows a preselection of the corpus, so that results are computed on the sentences the filter labels codable. The use of LLM labels for training a student classifier follows established precedent (Pangakis and Wolken, 2024) and rests on the gold validation of the labelling model itself (Section 4.5.2).

The filter was evaluated out-of-fold by stratified five-fold cross-validation on the 19,821 usable sentences of the two silver samples, 4,414 of which Sonnet coded as 000. The production model applied to the corpus was subsequently retrained on the full labelled sample. Across all thresholds, the filter reaches a PR-AUC of 0.845 and a ROC-AUC of 0.931.¹² At the operating point of 0.5, precision is 0.801 and recall 0.716 (F1 = 0.756), which means that four in five flagged sentences carry the non-codable label and that roughly seven in ten of the sentences

¹² The ROC-AUC is the probability that a randomly drawn non-codable sentence receives a higher filter score than a randomly drawn codable one. The PR-AUC summarises precision and recall across all thresholds and is the stricter measure under class imbalance, since its chance baseline equals the positive share of 0.223.

carrying it are caught. The operating point was not fixed in advance but chosen on the cross-validated evidence, and we state that openly. It is justified on the quantity that matters. Under the filter, the DPS share in the classifier's coding of the evaluation sample falls from 40.0% to 32.2%, almost exactly the oracle value of 31.7% that a direct adoption of the Sonnet labels would yield. What this establishes is that the filter reproduces its training target closely at this threshold, not that the target is itself correct. That second claim rests on the gold anchoring of Section 4.5.2. Applied to all 6,139,035 classified sentences, the filter flags 19.34% (1,187,584 sentences) as non-codable. This is not a recovery of the corpus-representative silver rate of 22.9% reported in Section 4.5.3, nor should it be. At the cross-validated recall of 0.716 and precision of 0.801, that rate implies a flag rate of about 20.5%, which is what the corpus figure approaches. Nothing is removed at this stage. The flag takes effect as the third step of the filter axis at aggregation time (Section 4.7.2), which keeps the unfiltered robustness arm available without reclassification. The effect of the filter on the individual buckets is reported in [Appendix F.3](#) alongside the same diagnostic for the code-305 correction. Measured on the analysis sample, it removes a quarter of the Democracy & Political System bucket against an overall rate of 14.5%, and no other bucket lies above that overall rate.

The filter is not trained on human labels but on labels assigned by a model, so non-codability is defined operationally here. What the filter learns is Sonnet's notion of what counts as non-codable, and its anchoring in human judgement runs indirectly, through the gold validation of Sonnet itself (Section 4.5.2). We checked the filter against the gold sample directly, and report the result as it stands. Of the 23 sentences the human coder marked as non-codable across the whole gold sample, the filter catches 15, a recall of 0.652 against 0.716 in cross-validation, at a false-positive rate of 6.6%. The stratification of the gold sample lowers the share of non-codable sentences and with it the precision of the flags, which is why no precision figure is reported for the gold check. Recall and the false-positive rate are invariant to class shares, but the gold sample is stratified on the classifier's predicted bucket rather than on the class itself, and 68.6% of non-codable sentences are routed to the Democracy & Political System bucket (Section 4.5.3), so the 23 gold non-codables over-represent that route. With $n = 23$ the recall estimate spans roughly 0.45 to 0.81 at 95% confidence. We therefore read the check as a plausibility test that finds no contradiction, not as an independent estimate of corpus recall.¹³ What the filter cannot reach are the codable sentences themselves. Even with every non-codable

¹³ The interval is a Wilson score interval, obtained by asking which true recall values the observation would not reject rather than by placing a symmetric band around the point estimate. It is used in preference to the normal approximation, which understates the uncertainty at samples of this size because its actual coverage falls short of the nominal level.

sentence perfectly removed, the classifier's DPS share would stand at 31.7% against the 19.3% Sonnet assigns to the same sentences. This remaining gap is sink-category inflation at work among codable sentences, and it is the reason the filter cannot substitute for the correction of Section 4.6.2.

4.6.2 Exclusion of Code 305

Code 305, political authority, covers claims about a party's or a leader's competence to govern, including the lack of it in others, together with favourable mentions of strong and stable government. In scheme A it maps to the Democracy & Political System bucket ([Appendix E](#)). To alleviate sink-category inflation, we exclude code 305 from the per-cell distributions in the primary specification. As Section 4.5.3 established, this mode draws on two sources: the forced codes of the null-class gap, of which 54.2% land on 305, and the over-assignment of 305 that persists among the sentences both coders consider codable. The null-class filter reaches only the first of these. Three arguments underpin the exclusion, and they answer three different questions: what the filter leaves behind, whose behaviour that residue reflects, and how much of it is confirmed. First, the residue is substantial and general. Even after the non-codable sentences filter, the speech-side share of code 305 exceeds the manifesto share of 305-coded sentences in the 38 cells by 0.133 on average, and the excess is positive in every cell. The oracle comparison of Section 4.6.1 makes the same point from the other side, since even a perfect removal of non-codable sentences would leave the classifier's share of the Democracy & Political System bucket well above the share Sonnet assigns to the same sentences. Second, the residue is the classifier's. ManifestoBERTa assigns code 305 to 24.9% of the silver sentences where Sonnet assigns it to 4.1% of the same sentences, and the gold anchoring of Section 4.5.2 locates the discrepancy with the classifier rather than with the reference. Measured before the filter, this figure bounds the phenomenon as a whole rather than the residue alone. Third, little of the residue is confirmed. Of the 23 gold sentences ManifestoBERTa classified as 305, the human coder placed only 10 (43.5%) in the Democracy & Political System bucket, and because confirmation is bucket-level, no claim is possible about how many of those carry code 305 rather than another code of that bucket, so the figure is an upper bound. The bound gains force from the remaining sentences: the coder identified 10 of the 23 as non-codable, which is the artefact sink of Section 4.5.3 surfacing in the gold data. The measure is therefore applied on both the speech and the manifesto side, with the distributions renormalised afterwards. Part of the excess may well be genuine, as parliamentary speech plausibly carries more claims to political authority than manifestos do. The exclusion does not deny this: it removes a category

whose confirmed share is a minority and whose artefactual channel is identified, and the inclusive specification runs alongside as robustness (Section 4.9).

An obvious objection is that the two corrections might be redundant, since 54.2% of the forced codes land on 305 and the null-class filter removes many of them already. This question was anticipated, and two criteria were declared before the results were known (Section 4.2). The exclusion would have counted as redundant only if both had been met: an additional removal of less than 15% of DPS-driven divergence on top of what the null-class filter removes, and agreement of the new primary with the previously corrected primary to within 15% of the correction magnitude. Neither is met. On top of what the filter removes, the exclusion removes a further 115% of that amount, and the filter-only specification still stands 49.8% of the correction magnitude away from the previously corrected primary (Appendix F.1). The two corrections are therefore complements rather than substitutes. This is the quantitative counterpart of the gap left by the filter in Section 4.6.1. The scope of the exclusion has to be stated precisely. It defines the primary specification on the A level, where the salience distributions of H1a and H2a are compared. The B level runs inclusive, because code 305 maps to no directional sub-bucket. Its mass sits in the residual category and drops out when the directional buckets are renormalised, so the exclusion has nothing to act on, which is consistent with the weaker validation of that level (Section 4.5.1). A final alternative deserves an answer. Reclassifying the affected sentences instead of excluding the category would require a target label for every sentence, and no such label exists. The diagnosis establishes that the codes are wrong in the majority of cases, not what would be correct in their place, which is why exclusion with renormalisation is the conservative and transparent operation.

4.7 Aggregation and Measurement

4.7.1 Topic Buckets

The analysis proceeds on a level above the 56 MARPOR categories, which are aggregated into nine substantively defined buckets. We call this scheme Aggregation A. The nine buckets are Foreign Policy & Defence; European Integration; Democracy & Political System; Economy; Welfare & Social Policy; Environment; Law & Order and National Identity; Migration; and Social Groups (see Appendix E).

Two considerations motivate the aggregation. The first is the low reliability of the fine-grained codes. In the silver-standard validation, $\kappa(F)$ reaches only 0.363 (Section 4.5.2), which is insufficient for substantive claims, and Mikhaylov, Laver and Benoit (2012) show that fine-

code misclassification is substantial even among trained human coders. The second is that many of the codes encode directional positions, for example 108 (Pro-EU) against 110 (Contra-EU). A salience measure has to be blind to direction, so codes that differ only in the sign of the position they express belong in the same bucket.

The directional analysis rests on a second scheme, called Aggregation B, which redefines four domains as directional sub-buckets and assigns everything else to a residual category. Economy is divided into market liberalism (401, 402, 407, 414) and state intervention (403, 404, 405, 409, 412, 413), together with a third, non-directional bucket for general economic content (408, 410, 411), which remains of interest for the directional analysis even though it carries no ideological direction. Welfare is divided into expansion (503, 504, 506) and retrenchment (505, 507), migration into restrictive (601, 608) and liberal (602, 607), and Europe into pro-EU (108) and contra-EU (110). Aggregation B is not a refinement of Aggregation A. Three codes that Aggregation A places inside these domains carry no direction on the relevant axis and fall into the residual as well, namely 406, 415 and 502, so Aggregation B covers 24 of the 56 codes across nine directional sub-buckets and leaves the remaining 32 in the residual (see [Appendix E](#)).

One limitation of this directional scheme has to be stated openly. Handbook 4 provides no subcodes, so categories 601 and 602 bundle general appeals to the national way of life with genuine migration positions. The bundling is symmetric in design but not in effect, because 601 is considerably more frequent than 602, so the restrictive bucket absorbs the larger share of content that is not about migration at all. We therefore pre-committed a sensitivity arm that re-splits the migration axis to the clean multiculturalism pair alone, with 608 as the restrictive and 607 as the liberal pole. Its design is fixed here, and its results are reported as arm (5) of the robustness battery (Section 4.9) and in [Appendix F.8](#).

4.7.2 Speech-Side Aggregation and the Primary Specification

With the buckets fixed, the aggregation from sentence-level classifier output to per-cell distributions can now be stated. The unit of analysis is the cell, a party $\times$ legislative-period combination. Let p_s denote, as a column vector, the 56-dimensional posterior distribution ManifestoBERTa assigns to sentence s (Section 4.4), and let S_c be the set of sentences of cell c that pass the active filter, which under the primary specification are the sentences that are neither procedural (Section 4.3.2) nor non-codable (Section 4.6.1). Let M be the 56×9 indicator matrix of Aggregation A, each of whose rows carries a single one, in the column of the bucket

to which that code belongs. The speech-side agenda of cell c is then the bucket-mapped sentence mass of the cell, normalised to sum to one:

$$\pi_c = \frac{\sum_{s \in S_c} M^T p_s}{\mathbf{1}^T \sum_{s \in S_c} M^T p_s}$$

where $\mathbf{1}$ is the vector of ones, so that the denominator is the total mass of the numerator. Because M assigns every code to exactly one bucket, each mapped posterior is itself a distribution over the nine buckets. The denominator therefore equals the number of sentences in the cell, and π_c is the simple mean of the sentence-level bucket vectors. This holds in every arm in which no mass is removed in bucket space, both temperature arms and all three filter levels included. Under Aggregation B the same construction applies with a 56×10 matrix whose tenth column is the residual category of Section 4.7.1.

The normalisation acquires content only under the 305 exclusion, which subtracts each sentence's probability mass on code 305 from its Democracy & Political System entry, and under Aggregation B from the residual, before the cell-level normalisation (Section 4.6.2). Each sentence then enters with mass $1 - p_s(305)$ rather than one, so that π_c becomes the mean of the sentence-level bucket vectors weighted by their non-305 mass. A sentence that is almost entirely political-authority language contributes almost nothing, rather than contributing its remaining mass at full sentence weight. This is the intended reading of the exclusion, and it is the operation applied to the manifesto side as well. The buckets themselves are unchanged. π_c remains a point in the nine-bucket simplex, and only the Democracy & Political System bucket is now spanned by eight codes rather than nine.

The temperature arms rescale each posterior before the mapping by applying a softmax with temperature τ to its log probabilities. Because log probabilities and logits differ only by a row constant, this is exactly a tempering of the classifier's logits. The native $\tau = 1$ leaves the posteriors unchanged and defines the primary specification, while the sharpened ($\tau = 0.5$) and the flattened ($\tau = 2.0$) variants run as robustness arms (Section 4.9). Two properties of this arm have to be stated. Tempering acts on the 56 codes and does not commute with the bucket mapping, so it redistributes mass according to how widely a posterior is spread within a bucket. In the limit of full flattening every sentence approaches the distribution of codes across buckets, 26.8 per cent for the fifteen-code Economy bucket against 3.6 per cent for the two-code Environment bucket, rather than the uniform distribution over the buckets themselves. And because the manifesto side carries no posteriors, the arm perturbs only one side of the divergence. It is therefore a bounded sensitivity check on the confidence profile of the speech

measurement. Its limit as τ approaches zero is the argmax assignment, which places the sharpened arm on the same axis as the hard-assignment check of Section 4.9 rather than beside it as an independent one.

Aggregating full posteriors rather than argmax labels is the defining choice of this step. Its justification is aggregate rather than sentence-level. As Section 4.3.1 notes, the spread of a posterior expresses uncertainty about which single code applies and not a mixture of topics. The sum of posterior mass in a bucket is nevertheless the expected number of the cell's sentences belonging to it, so the soft sum estimates the cell's bucket composition directly. Argmax discards precisely the information that separates a sentence the classifier is certain about from one it is not, and across 56 classes it biases the resulting counts towards the categories the classifier favours a priori. What the soft sum carries into the measurement is therefore the expected category frequency and not the uncertainty itself, which is quantified separately in Section 4.7.4.

There are three corpus-wide specification decisions that together span the primary and the robustness specifications of the speech side. The first is the filter, which has three levels. The unfiltered corpus retains every sentence, the procedural filter removes procedural sentences (Section 4.3.2), and the full filter additionally removes non-codable ones (Section 4.6.1). The ladder isolates the marginal effect of each filter. The second is the treatment of code 305, which is either included or excluded (Section 4.6.2). We refer to the two levels as the inclusive and the exclusive specification. The third is the decoding temperature, at 0.5, 1 or 2. All combinations are computed in a single pass over the corpus, which yields eighteen specifications per cell and scheme. The primary specification, declared ex ante (Section 4.2), is the full filter with code 305 excluded at the native temperature for scheme A, and the full filter with code 305 included at the native temperature for scheme B, because code 305 maps to no directional sub-bucket and drops out when the directional buckets are renormalised.

The manifesto side requires none of these decisions except the treatment of code 305. Manifestos carry human-assigned MARPOR codes rather than classifier posteriors, so there is no temperature to apply. The non-codable share of a manifesto is already removed by the coding procedure itself, since MARPOR leaves uncoded quasi-sentences out of the per-category percentages and the remainder is renormalised to sum to one, so there is no filter axis either. The null-class filter of Section 4.6.1 is the speech-side counterpart of that step. The per-cell manifesto distribution is therefore the observed share of quasi-sentences per code, renormalised across the 56 codes and mapped into bucket space by the same indicator matrix as the speech

side. Which manifesto belongs to which cell is fixed by the pairing rule of Section 3.3. The treatment of code 305, however, applies to both sides identically. Under the exclusive specification, code 305 is removed from the manifesto distribution as well and both sides are renormalised, so that speech and manifesto distributions are formed on the same reduced code base (Section 4.6.2). This leaves two manifesto variants against the eighteen speech-side specifications.

4.7.3 The Jensen–Shannon Divergence

With the per-cell distributions defined, the comparison of speech and manifesto agendas requires a measure of distance between two distributions over the same buckets. Writing p for the speech-side distribution π_c of a cell and q for the manifesto distribution of the same cell, we use the Jensen–Shannon divergence:

$$\text{JSD}(p, q) = \frac{1}{2}\text{KL}(p \parallel m) + \frac{1}{2}\text{KL}(q \parallel m), \quad \text{with} \quad m = \frac{1}{2}(p + q)$$

Here KL denotes the Kullback–Leibler divergence, and both inputs are renormalised to sum to one before the computation, which matters under the exclusive specification (Section 4.7.2). Computed with the base-2 logarithm, the measure is bounded between 0 and 1 (Lin, 1991).

Three properties motivate this choice. First, the divergence is symmetric, so neither side is privileged as the true distribution, which fits a comparison in which speech and manifesto are equally measured quantities. Second, it remains finite when one side assigns zero mass to a bucket, a case in which the Kullback–Leibler divergence itself diverges. Under soft aggregation the speech side never assigns exactly zero, so the case arises on the manifesto side, and it is the rule rather than the exception on the directional layer, where the contra-EU bucket rests on a single code. Third, the bound makes values comparable across cells, parties and legislative periods measured on the same bucket set. The measure additionally decomposes into per-bucket contributions that are individually non-negative and sum exactly to the total, which is the instrument for H2a, because it makes visible which topics drive the divergence of a cell.

It has to be acknowledged that no canonical measure of agenda congruence between programmatic and parliamentary text has established itself in the field, which is why the choice is justified by these properties rather than by precedent. The closest established relative is the issue convergence index of Sigelman and Buell (2004), which subtracts half the summed absolute differences in issue shares from a fixed maximum. It runs in the opposite direction, high values marking convergence rather than distance, but it is, like the Jensen–Shannon divergence, bounded and additive across issues. The two differ in how they weight a given gap.

Their index counts an absolute difference equally wherever it occurs, whereas the divergence used here scales it to the mass of the bucket in which it occurs, so that a gap of two percentage points between shares of five and three per cent contributes roughly eight times what the same gap contributes between thirty-two and thirty per cent. Since the buckets differ by an order of magnitude in the mass they carry, this is the property the comparison needs.

A single divergence value carries no absolute interpretation. The scale is therefore provided empirically, by the distribution of divergences between different parties, which Section 4.8 introduces as the reference frame for H1a.

4.7.4 The Sentence Bootstrap

Each divergence value is a point estimate based on a finite set of sentences. We quantify the sampling uncertainty of the speech side by a nonparametric bootstrap within cells. For each of 1,000 draws, the sentences of a cell are resampled with replacement, the bucket distribution is re-aggregated as specified in Section 4.7.2, and the divergence against the fixed manifesto distribution is recomputed. The 95% confidence interval is the percentile interval of these draws, and a fixed seed makes the procedure deterministic. We report the interval of the primary specification. The intervals of the remaining filter and 305 combinations are reported in [Appendix F.3](#), which shows that neither correction inflates them.

The scope of these intervals has to be stated precisely. They capture the sentence sampling uncertainty of the speech side, conditional on the classification and on the manifesto. The manifesto side is treated as fixed, so the uncertainty of the human coding is not propagated ([Benoit, Laver and Mikhaylov, 2009](#)). The classification enters only through the posteriors of Section 4.7.2, which the bootstrap holds fixed, so classifier uncertainty is not resampled either. And the procedure treats sentences as independent draws, whereas sentences cluster within speeches and within speakers, so a block bootstrap at the speech level would widen the intervals further. All three omissions point in the same direction. The intervals are therefore lower bounds on the total uncertainty, and we read them as such.

With this, the measurement pipeline is complete. The final specification confronts the validation findings rather than evading them. Both failure modes diagnosed in Section 4.5.3 are structurally addressed, and the reliability evidence carries the aggregate level on which the analysis proceeds. Independent computation paths agree to within a numerical tolerance of 10^{-6} , and row-sum checks run over every cell and specification. The details are documented in [Appendix G](#). Section 4.8 states, for each hypothesis, the exact quantities and benchmarks that are computed.

4.8 Empirical Strategy

This section states, for each hypothesis, the objects computed, the benchmarks against which they are read, and the group to which each belongs. The base of every analysis is the set of party-period cells. Six parties over eight legislative periods (LPs 13 to 20) span 48 mechanical combinations, of which 42 carry speech data, as the AfD, founded in 2013, missed the five per cent threshold that year with 4.7% of the vote and entered the Bundestag only in 2017, which leaves its first six cells structurally empty. Two exclusions reduce this set further. The FDP was not represented in the 18th Bundestag, so its LP 18 cell is excluded and the few sentences attributed to it are treated as misattributions. The Linke cells of LPs 13 to 15 are excluded as a party-continuity decision rather than a data constraint. Coded PDS manifestos exist for 1994 to 2002, but we do not treat the PDS and Die Linke, which emerged from the merger of the PDS and the WASG (Decker, 2026), as the same programmatic actor, a break that the MARPOR party list itself draws with separate party ids and that the merger literature documents substantively (Olsen, 2007). This leaves $n = 38$ valid cells (CDU/CSU 8, SPD 8, Grüne 8, FDP 7, Linke 5, AfD 2). Under the full filter, the speech side of a valid cell comprises between 41,562 and 279,947 sentences (median 106,641), and cells enter the analysis only above a pre-specified floor of 1,000 sentences after the procedural filter, which the smallest cell clears by a factor of more than forty. At these counts the random component of the sentence-level error averages out, while the two systematic failure modes diagnosed in Section 4.5.3 do not, which is why they are corrected at source in Section 4.6 rather than absorbed by the aggregation. The analysis therefore proceeds on the aggregate level.

4.8.1 Hypothesis H1

H1a states that a party's speech agenda stands close to its own manifesto relative to how far party programmes stand apart, and that it is identifiable among the other parties' manifestos above chance (Section 2.5). Its first object is the per-cell JSD between the speech and manifesto distributions under the primary specification (Section 4.7.2), reported with the 95% percentile bootstrap interval of Section 4.7.4 (1,000 draws; the per-cell seeds are derived deterministically from the base seed 20260507). The full table of all 38 cells is given in Appendix F.2, and the intervals are read as lower bounds on the total uncertainty. We read these values against two benchmarks fixed in advance. The first is a nearest-manifesto test. For each cell, the speech distribution is compared against the manifestos of all parties that hold a valid cell in the same legislative period, and we record whether a party's own manifesto is the nearest and which rank it takes among the four to six candidates. Both baselines of this test are derived rather than

simulated. Under random assignment, a party's own manifesto is identified with probability 21.1%, and its expected rank is 2.95. The second benchmark is a double reference computed on the same specification and cell base as H1a itself. It comprises all unordered pairwise between-party JSDs, once on the speech side and once on the manifesto side ($n = 74$ pairs each). The speech-side reference is the lower reference point, the distance that remains between agendas under a shared parliamentary agenda. The manifesto-side reference is the upper one, the distance programmatic differentiation makes available. Neither is a bound: three cells exceed the upper reference (Section 5.2.1). The reference is a scale for what counts as a large agenda distance, not a second measurement of the same quantity. It compares speech with speech and manifesto with manifesto, while H1a compares speech with manifesto.

H1b extends the consistency claim of H1a to the directional layer. Within the four directional domains, the composition of parliamentary speech across directional sub-buckets is expected to resemble the composition of the same party's manifesto (Section 2.5). As diagnosed in Section 4.5.3, all analyses on scheme B rest on the weaker validation floor and are read with more caution throughout. The objects of H1b are the per-cell divergence across the nine directional sub-buckets, computed after the residual category is dropped and the remainder renormalised, and the within-domain composition of each of the four domains. The migration domain additionally carries a within-domain divergence, which is the quantity the re-split of arm (5) varies. H1b is read as a composition claim; no nearest-manifesto counterpart is computed for it. Code 305 requires no treatment on this layer, as its mass drops out under the within-domain renormalisation (Section 4.7.2). Migration values are defined only for cells that carry migration mass on both sides, which yields $n = 37$ for the baseline and $n = 36$ under the 608/607 re-split. The re-split was fixed as a sensitivity arm in Section 4.7.1 and appears as arm (5) in Section 4.9.

4.8.2 Hypothesis H2

H2a states that manifesto–speech divergence is not uniform across topics but concentrates in identifiable buckets, with the largest contributions expected in the institutionally driven Democracy & Political System bucket, and that it points in the same direction for every party (Section 2.5). Its first object is the per-bucket decomposition introduced in Section 4.7.3, computed per cell and aggregated across cells. Because the nine contributions sum exactly to the cell JSD, concentration can be read off directly. Part of the remaining contribution of the Democracy & Political System bucket is plausibly genuine after the corrections, as argued in Section 4.6.2. The second object records the direction of the divergence. The signed per-bucket

differences between speech and manifesto shares, in percentage points, state whether plenary speech over- or under-weights a bucket relative to the manifesto. The intervention ladder of Section 4.6, which crosses the filter with the code-305 treatment, remains diagnostic material for the corrections and is not an object of H2a; its results are reported in Appendices F.1 and F.3.

H2b states that parties moderate their directional positions in parliamentary speech relative to their manifestos (Section 2.5), and like every analysis on scheme B it carries the caution Section 4.5.1 sets out. H2b, H2c, and H3 all rely on the same polarisation score, defined here once. Per domain and side, the score is the share of the positive pole minus the share of the negative pole, divided by the sum of the two, so that it is signed and bounded in $[-1, 1]$, with positive values denoting the market-liberal, retrenchment, restrictive, and EU-sceptic poles. The general-economy bucket does not enter the denominator of the Economy score, and a cell without mass on either pole of a domain carries no score rather than a zero. H2b then fits an OLS of speech position on manifesto position per domain ($n = 38$; migration $n = 37$, Section 4.8.1) and reports slope and intercept. A slope below one against the 45-degree reference is the descriptive signature of compression, read as the between-party differences carrying over only in part, and the intercept states where the compressed positions sit. The two are read together, since compression towards a shifted level is not the same as compression towards a balanced position, and neither is read as a test.

H2c states that directional positions disperse less across parties in plenary speech than in their manifestos, the cross-channel convergence claim (Section 2.5), and it carries the same caution; measuring polarisation from what politicians say has established precedent ([Gentzkow, Shapiro and Taddy, 2019](#)). Its object is the Dalton-weighted standard deviation of position scores per legislative period, domain, and channel, defined in Section 4.8.3, compared between the two channels as levels, together with a compositional decomposition of system dispersion with and without the AfD. The two channels do not cover the same party set in every period, which matters here as it does for the trends of Section 4.8.3: in LP 18 the manifesto channel carries six parties against four in the speech channel, and the two additional programmes are those of the AfD and the FDP, so part of any cross-channel gap in that period is compositional rather than substantive. H2c owns the comparison of levels, H3 the change of those levels over time, and H2b the mapping of one channel onto the other within a party. A slope below one and a narrower spread are not the same finding, since a flatter mapping compresses the individual party while residual variation can leave the spread across parties where it was (Section 2.5).

4.8.3 Hypothesis H3

H3 describes how the polarisation of the party system evolves over time, with an increase expected in particular in the Europe and Migration domains and around the AfD's parliamentary entry in 2017 (Section 2.5). As an analysis of directional positions, it runs on scheme B and is read as a description of trends rather than as a test. System-level polarisation is the Dalton-weighted standard deviation of the party position scores defined in Section 4.8.2. Per legislative period, domain, and channel, the weights are renormalised to sum to one and polarisation is the square root of the weighted variance around the weighted mean position (Dalton, 2008). The quantity is undefined where fewer than two parties carry a score. Because the same formula returns the weighted mean position as the centre of that dispersion, we read the two together, the dispersion stating how far apart the parties stand and the mean stating where the system as a whole sits. The manifesto channel is weighted by party-list vote shares at the respective election and the speech channel by seat shares at the start of the legislative period, before any defections, so that each channel weights a position by the weight the party actually carried there (Bundeswahlleiterin, 2025; Deutscher Bundestag, n.d.). The two channels do not cover the same party set in every period, which matters for every pre/post comparison. The AfD enters the manifesto channel in LP 18, with a 2013 manifesto and a vote share of 4.7%, but the speech channel only in LP 19, so LP 18 counts six parties in the manifesto channel and four in the speech channel.

The evolution over time is then described by two complementary objects. First, we fit an OLS trend of polarisation on legislative period per domain and channel and report slope, standard error, and *p* as descriptive quantities, with eight data points each. No significance language attaches to them. Second, a pre/post-2017 contrast compares mean polarisation across LPs 13 to 18 with LPs 19 to 20, an openly unbalanced split of six against two periods. Finally, the trends are refit without the AfD, with the weights of the remaining parties renormalised, as the pre-committed sensitivity arm (6) of Section 4.9. Published work finds that the success of a radical right competitor also moves the established parties (Abou-Chadi and Krause, 2020), so the refit cannot separate the two sources. What it isolates is the compositional part, the contribution the AfD's own position and weight make to a trend, while whatever its entry set in motion among the other parties remains in the refit and is not attributed to it. A causal design around that threshold is beyond the scope of this thesis.

4.8.4 Analysis Overview

Table 4.1 summarises the declarations of this section, giving for each analysis its layer, objects, benchmark, specification, and group. The grouping binds the remainder of the thesis. Chapter 5 reports every analysis under the group declared here, and no analysis is moved between groups after the fact. The per-cell values and intervals behind H1a are tabulated in [Appendix F.2](#), the six bootstrap bands of the specification cross in [Appendix F.3](#), and the benchmark tables in [Appendices F.4](#) and [F.5](#).

Table 4.1: Analysis overview: layer, objects, benchmarks, specification, and group per analysis.

Analysis	Layer	Object	Benchmark / reference	Specification	Group
H1a	A	per-cell JSD with 95% bootstrap CI	double reference (speech side and manifesto side); nearest-manifesto test (derived baseline 21.1%, expected rank 2.95)	full filter $\times$ 305 excl $\times$ native decoding temperature	main
H1b	B	per-cell divergence across the nine directional sub-buckets + within-domain composition	608/607 re-split as pre-committed sensitivity (arm 5)	full filter $\times$ 305 incl $\times$ native decoding temperature, residual excluded, renormalised	additional
H2a	A	per-bucket decomposition + signed per-bucket differences	even-spread reference ($1/9 = 11.1\%$ per bucket); sign consistency of the per-bucket differences across parties	as H1a	main
H2b	B	polarisation score per cell; OLS speech $\sim$ manifesto (slope and intercept)	45-degree reference (slope = 1)	as H1b	additional
H2c	B	Dalton-weighted dispersion per channel (level, incl. all vs excl-AfD composition)	cross-channel comparison	as H1b; weights: votes / seats	additional
H3	B	Dalton-weighted polarisation over time; descriptive OLS trends; pre/post-2017 contrast	all vs excl-AfD slope (arm 6)	as H1b; weights: votes / seats	additional

Note: Specifications refer to Section 4.7.2. The manifesto of each legislative period is that of the election from which the period emerged (Section 3.3).

4.9 Robustness Design

The robustness battery translates the pre-commitment architecture of Section 4.2 into six numbered arms: for each arm, this section states what varies, what the variation tests, and the criterion fixed before the analysis, which is the operational answer to the problem of undisclosed analytic flexibility ([Gelman and Loken, 2014](#)). Chapter 5 reports the outcomes of every arm as they came out, with full tables in [Appendix F](#), and the numbering below is binding: the results table in Section 5.5 and [Appendix F](#) reference the arms by it. Not every arm carries

a binary criterion, and we state this rather than invent thresholds after the fact: arms (1) and (2) carry pre-committed rank-correlation criteria, arm (3) carries the pre-stated redundancy thresholds of Section 4.6.2, and arms (4) to (6) are sensitivity designs that report the shift a design variant induces. Table 4.2 summarises the battery.

Table 4.2: The pre-committed robustness battery. Acceptance criteria were fixed before the analysis; the results table in Section 5.5 and Appendix F reference the arms by their numbers.

Arm	Varies	Tests	Acceptance criterion	Results reported in
(1)	temperature $\tau \in \{0.5, 1, 2\}$ applied to the sentence posteriors (Section 4.7.2)	dependence on the calibration sharpness inherited from the classifier	Spearman $\rho \geq 0.90$ between the cell rankings of each τ pair, for H1a and H1b separately	Section 5.5
(2)	argmax assignment with thresholds 0.4 and 0.5, sub-threshold mass to a residual bucket	soft versus hard aggregation, the defining choice of Section 4.7.2	Spearman $\rho \geq 0.85$, same basis	Section 5.5
(3)	filter ladder crossed with the 305 treatment (3×2)	marginal effect and redundancy of the two corrections of Section 4.6	redundancy thresholds fixed in advance: the exclusion counts as redundant if it removes less than 15% of the DPS divergence the filter removes and the filter-only specification converges to the old primary to within 15% of the correction magnitude	Section 4.6.2 and Appendix F.3; Section 5.5 cross-references
(4)	tightened prefilter of at least eight words per sentence on top of the full filter	sensitivity of H2a to sentence length	sensitivity: rank agreement with the primary specification reported, no binary criterion	Section 5.5
(5)	migration axis re-split to the multiculturalism pair alone (restrictive 608, liberal 607)	contamination of the restrictive pole by codes 601 and 602 (Section 4.7.1)	sensitivity: directional composition and the cell-level spread of the JSD differences reported (cell bases as in Section 4.8.1)	Section 5.2; Section 5.5 cross-references
(6)	polarisation trends refit without the AfD, weights renormalised	AfD entry effect in the trends of H3	sensitivity: slope shift reported	Section 5.4; Section 5.5 cross-references

Two clarifications prevent misreadings of the table. First, arm (2) runs on code 305 included for construction reasons, so its JSD levels sit on the inclusive basis, with a soft-aggregation mean of 0.0510 rather than the 0.0297 of the primary specification; the arm therefore compares rank orders across the 38 cells, never levels, and Section 5.5 reports it accordingly. Second, arm (4) varies sentence length only: its originally envisaged speaker-role and short-speech components are already enforced upstream during corpus extraction and would be no-ops, which is why the arm reports the rank agreement of the tightened corpus with the primary specification without a binary criterion.

The battery closes with what it takes to reproduce it. The evaluation stage is deterministic: the fixed bootstrap seed of Section 4.7.4 reproduces every interval, and identical runs reproduce

every number in this thesis exactly. The analysis tier, from the classified sentences to every table and figure, runs in R without GPUs or external services; only the classification stage of Sections 4.4 and 4.6.1 requires GPU inference and API access. [Appendix G](#) documents the software versions, the pipeline overview, and the runbook, together with the integrity checks referenced in Section 4.7.4.

Results

5.1 Analysis Base and Reading Conventions

The hypotheses are read as stated in Section 2.5. H1a and H2a carry the main argument, and H1b, H2b, H2c and H3 deepen it. Findings on the directional layer rest on the weaker validation floor of Section 4.5 and are therefore read with more caution, and the trend analyses of Section 5.4 describe developments without attaching formal tests to them. This chapter reports the results of the manifesto–speech comparison across 38 party-period cells: eight each for CDU/CSU, SPD and Grüne, seven for FDP, five for Linke and two for AfD. The exclusions are discussed in Section 4.8. Each cell aggregates between 41,562 and 279,947 sentences (median 106,641). The results reported here follow the primary specification of Section 4.7.2: the full filter at native decoding temperature throughout, with code 305 excluded on the salience layer (Aggregation A) and retained on the directional layer (Aggregation B), where it maps to no directional bucket and drops out under within-domain renormalisation. All deviations from this specification are marked as robustness arms.

The Jensen–Shannon divergence, our principal measure of manifesto–speech congruence (Section 4.7.3), carries no absolute reading, which is why every congruence claim in this chapter is measured against the between-party reference scale introduced in Section 4.8.1. That scale frames the size of a divergence between a lower and an upper reference point: at the lower end, how far apart two parties' speech agendas typically stand from one another, and at the upper end, how far apart their manifestos stand. The numerical reference points are introduced in Section 5.2.1. Directional findings on Aggregation B are read differently, as shifts in composition rather than against this salience scale.

5.2 Agenda Congruence (H1)

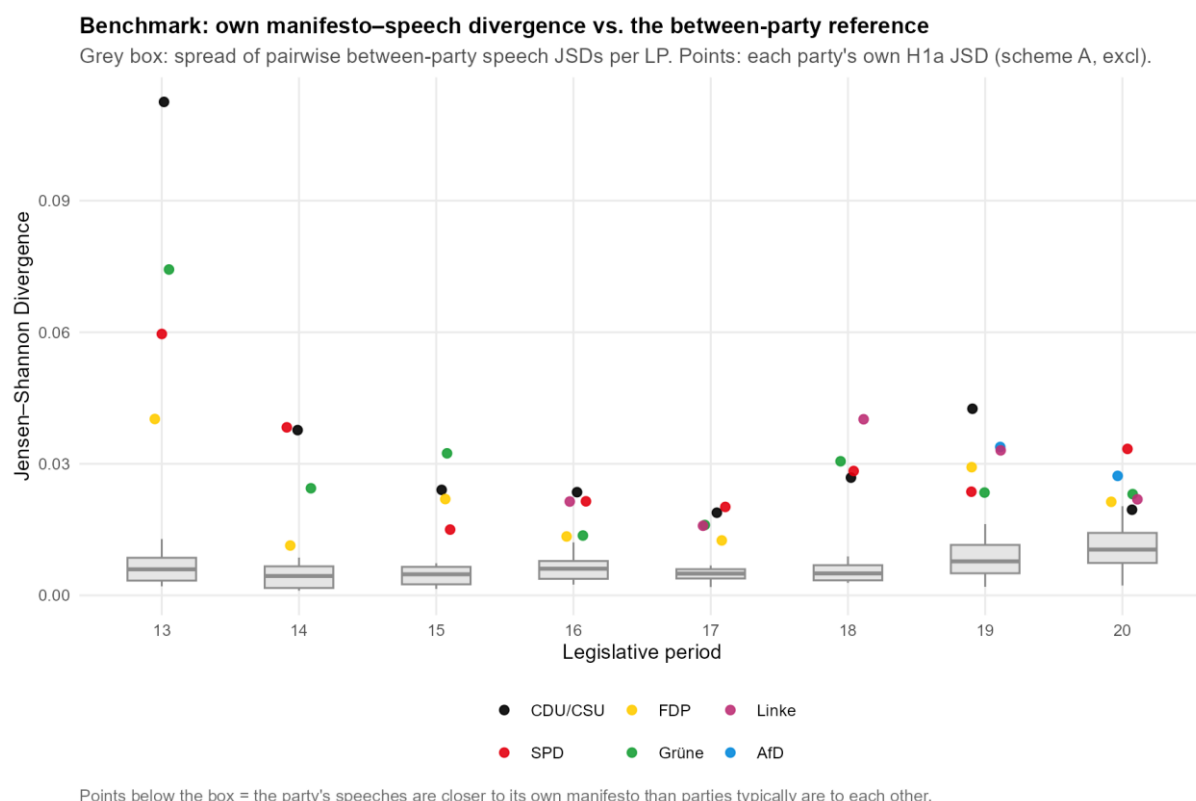
5.2.1 H1a

The main unit of analysis in H1a is the Jensen–Shannon divergence. Over the 38 cells, the JSDs are distributed with a mean of 0.0297 and a median of 0.0238. The smallest value is found for the FDP in LP 14 at 0.0114 and the largest for the CDU/CSU in LP 13 at 0.1125. Because a JSD carries no absolute reading, two reference points give it a scale: a lower reference of 0.0074, the mean distance between speech agendas, and an upper reference of 0.0520, the mean distance between manifestos, about seven times higher. Neither is a bound in the strict sense, since three cells, all in LP 13, exceed the upper reference. Against these, the mean within-party divergence is four times as large as the distance between agendas but only 0.57 times the distance between manifestos.

Figure 5.1 makes this comparison visible period by period. For each legislative period it draws the spread of the pairwise distances between parties' speech agendas as a grey box, whose edges mark the quartiles of that between-party reference, and overlays each party's own manifesto–speech divergence as a coloured point. Were a party's speeches as close to its own manifesto as parties' agendas are to one another, its point would fall inside the grey box. Every one of the 38 points sits above the third quartile of its own period, all 38 lie above the pooled third quartile of the between-party agenda distances (0.0093), and 13 (34.2%) rise above even the largest distance observed between any two agendas (0.0303). The boxes sit low and remain stable from LP 14 to LP 18, and rise in LPs 19 and 20.

Figure 5.1. Own manifesto–speech divergence against the between-party reference, by legislative period. Grey boxes give the quartiles of the pairwise between-party speech JSDs within each period; coloured points give each party's own manifesto–speech JSD. A point inside the box would mean a party's speeches are as

close to its own manifesto as parties' agendas are to one another. Aggregation A, primary specification, n = 38 cells.¹

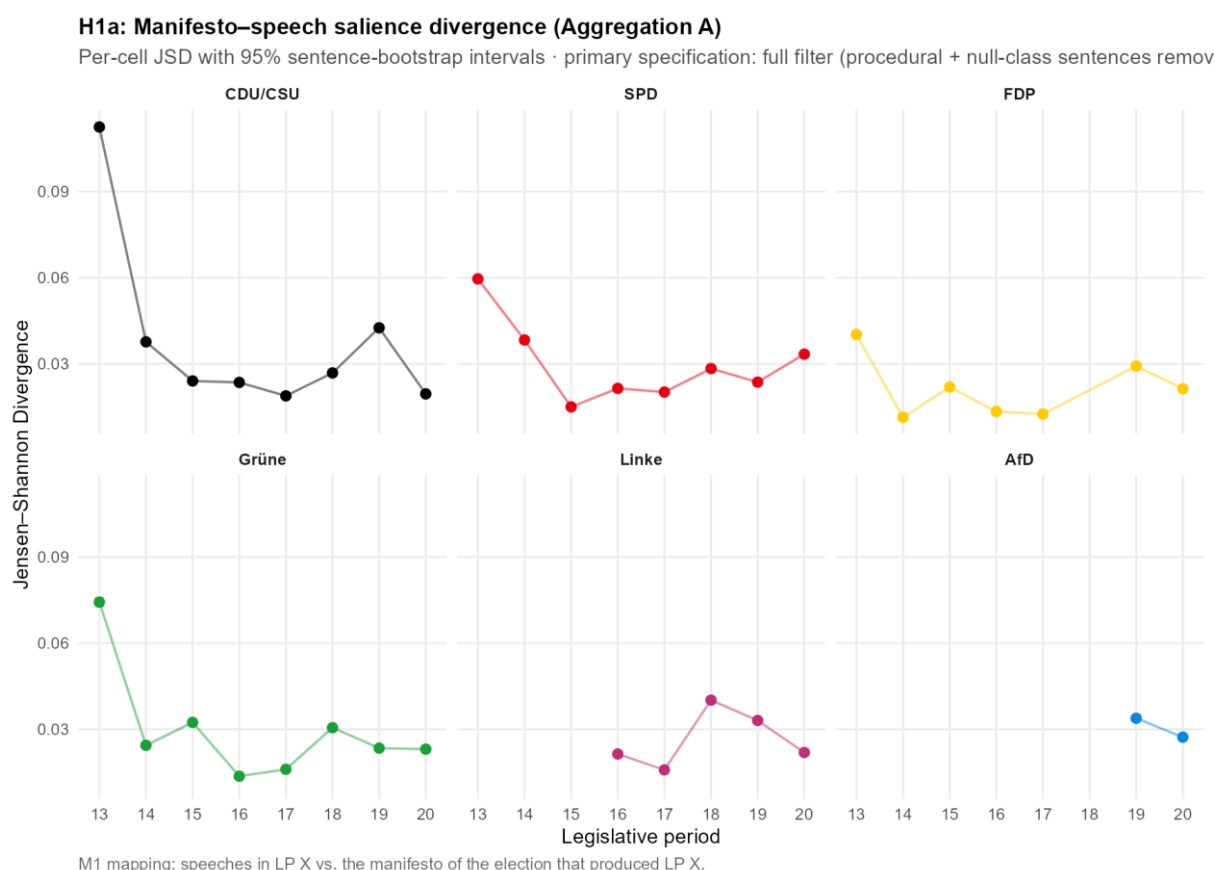


Each cell value is reported with a 95% sentence bootstrap interval, and across the 38 cells these intervals are narrow, with a mean width of 0.0013 against point estimates ranging from 0.0114 to 0.1125. As Section 4.7.4 sets out, they are read as lower bounds on the total uncertainty, which is why we locate each cell on the reference scale rather than comparing the overlap of intervals between cells. On this scale the within-party divergence is intermediate rather than small, as its mean sits almost exactly halfway between the two reference points. On average, plenary speech neither reproduces the manifesto nor drifts as far from it as parties' programmes stand from one another. On the reference benchmark, the divergence stays below the programmatic differentiation between parties, but not by a wide margin. H1a as stated has two parts, and the two benchmarks separate them: the reference scale supports the first, since a party's speeches stay closer to its own manifesto than parties' programmes stand to one another, while the nearest-manifesto test of the next paragraph qualifies the second, since a party's own programme is recognisable far above chance but is not reliably the nearest.

Across parties the JSD values are relatively similar, with means of CDU/CSU 0.0382, SPD 0.0300, Grüne 0.0297, Linke 0.0265, FDP 0.0214, and AfD 0.0305 (n = 2). Their movement over time is not flat, however. Figure 5.2 shows this in six panels, one per party, each plotting that party's divergence for every legislative period. In each of the four parties observed since

1994 the points fall into a broadly similar shape: highest in LP 13, dropping to a trough somewhere between LP 14 and LP 17 depending on the party, and rising moderately again towards the end of the window. Averaged across parties, the legislative period means fall from 0.0717 in LP 13 to 0.0167 in LP 17 and return to around 0.03 (0.0315 in LP 18, 0.0310 in LP 19, 0.0244 in LP 20). Two features of the diagrams are worth noting. First, for each observation the bootstrap interval is depicted, but its bars are so short that they are barely visible against the points, so the shape of each party's series is not an artefact of sentence-level sampling, which is the only source of uncertainty these intervals capture. Second, the re-rise is broad but mild, its one exception being the CDU/CSU in LP 19, at 0.0426, the tallest point outside LP 13 anywhere in the figure.

Figure 5.2. Manifesto–speech salience divergence per party and legislative period, with 95% sentence-bootstrap intervals. One panel per party; each legislative period is paired with the manifesto of the election from which it emerged (Section 3.3). Aggregation A, primary specification.2



The LP-13 column is what stands out. In four of the diagrams the 1994 cell is the highest the party reaches, at 4.4 times the CDU/CSU's own median, 3.1 times for the Grüne, 2.3 for the SPD, and 1.9 for the FDP. The three highest points in Figure 5.1 are the LP-13 cells of the CDU/CSU, the Grüne and the SPD; the FDP's LP-13 cell sits high but not exceptionally so. Because the peak recurs in every party present in 1994 rather than in one, we read it as a property of the period, not as four coincident outliers. That property sits on the manifesto side and at the

edge of the observation window. The 1994 programmes are the earliest we observe, and they stand farther from each party's later, more settled manifestos of 1998 to 2021, which the genre-stable plenary speech most resembles, than any other election in the window. The step to the 1994 manifesto is unremarkable by the long-run standard of the three parties with a long coded record; for the Grüne, whose pre-1994 record is not captured by the party identifier used here, it is the largest of the observed transitions but small in absolute terms. The elevation is therefore a boundary effect of the window, not an anomalous 1994 document and not a coding or pipeline artefact, as [Appendix F.10](#) shows. Excluding the window-boundary LP-13 cells lowers the mean JSD from 0.0297 to 0.0247, barely moves the median (0.0238 to 0.0235), and leaves both reference-scale comparisons in place, so no congruence claim rests on that period.

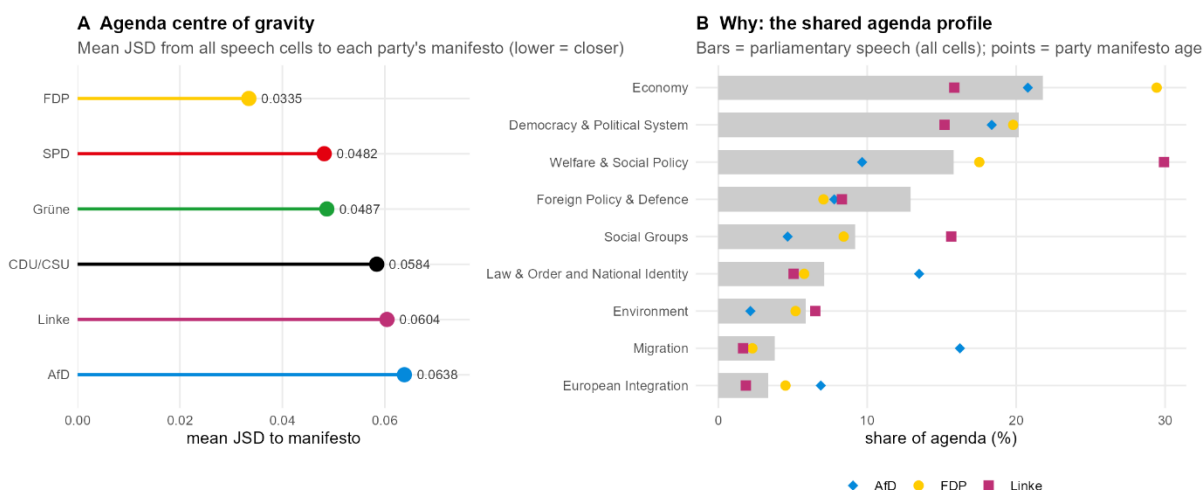
The other benchmark discussed in Section 4.8.1 is the nearest-manifesto test, which asks whether a party's own manifesto is the nearest to that party's speech agenda in a given LP. A party's own manifesto is the nearest in 42.1% of cells against a 21.1% chance baseline, and its mean rank is 1.87 against a chance expectation of 2.95; identification thus succeeds far above chance yet fails in a majority of cells, and a party's own manifesto never ranks worse than fourth. What is interesting here is the strong party heterogeneity in how often a party's own manifesto is the closest to its agenda: FDP 7/7, AfD 2/2, Grüne 4/8, CDU/CSU 2/8, SPD 1/8, Linke 0/5. Which manifestos the non-matching agendas gravitate towards instead follows a roughly consistent pattern. Across the cells in which a party's speeches are not closest to its own programme, the FDP manifesto is by a clear margin the most frequent nearest neighbour, and it is the nearest manifesto in 18 of the 38 cells overall. This is not an artefact of the FDP appearing often as a candidate: if we take the mean divergence from each of the 38 speech cells to every party's manifesto in the same period, the FDP programme sits closest to the pool of parliamentary speech at 0.0257, ahead of the SPD (0.0379), the Grüne (0.0396), the CDU/CSU (0.0463), the Linke (0.0524) and the AfD (0.0638), and it stays first when each party's distance to its own speech is excluded. Figure 5.3 sets this out in two panels. Panel A ranks the party programmes by that mean distance to the whole body of parliamentary speech, and the FDP sits clearly closest, with the Linke and the AfD farthest. Panel B shows why: the bars give the chamber's average agenda across the nine salience buckets, comparatively heavy on governance and economy and light on welfare and migration, and the overlaid points give the manifesto agendas of the FDP and of the two most distant programmes, so that the FDP is seen to track the parliamentary profile while the welfare-centred Linke and the migration-centred AfD depart from it most. It should be stressed that this is a statement about salience, that is, about which topics are emphasised, and not about position: the agenda of debate resembles the FDP's agenda,

it does not show speech adopting FDP positions. How much of this resemblance reflects the institutional genre of a legislative chamber rather than a substantive lean is taken up in the conclusion (Chapter 6).

Figure 5.3. Where the parliamentary agenda sits among the party programmes. Panel A ranks each party's manifesto by its mean divergence to the body of parliamentary speech in the same period (lower = the programme the chamber's agenda most resembles). Panel B gives the chamber's mean agenda across the nine salience buckets (bars) against the manifesto agendas of the FDP and of the two most distant programmes (points). Salience, not position. Aggregation A, primary specification, 38 speech cells.³

The parliamentary agenda leans toward the FDP's programmatic topic profile

Salience layer (Aggregation A); primary specification: full filter, code-305 mass subtracted, native temperature



Centre of gravity = mean JSD from each of the 38 speech cells to a party's manifesto in the same period; manifesto candidates restricted to parties in parliament that period. Salience, not position.

5.2.2 H1b

Overall, the mean JSD across the B buckets is 0.0506 (median 0.0520), with party-specific means of Grüne 0.0744, Linke 0.0676, AfD 0.0596, FDP 0.0440, CDU/CSU 0.0361 and SPD 0.0342. To analyse the directional differences between the two channels we compare the domain-pooled within-domain shares of manifestos and speeches. On migration, the restrictive share rises from 42.9% in manifestos to 68.4% in speeches.¹⁴ On the EU stance, the contra-EU share rises from 17.6% to 36.0%. Economy divides into three categories: speech is more market-liberal (23.2% to 29.2%) and correspondingly less interventionist (state intervention 32.9% to 29.3%, general economy 43.9% to 41.5%), so the market-liberal gain comes at the expense of interventionist rather than general economic language. Welfare is essentially unchanged between the channels (retrenchment 5.8% versus 7.4%).

The largest and clearest shift is on migration, where speeches carry a far higher restrictive share than the corresponding manifestos. Two reservations bound how this is read. First, a non-negligible part of the gap reflects general national-identity rather than multiculturalism

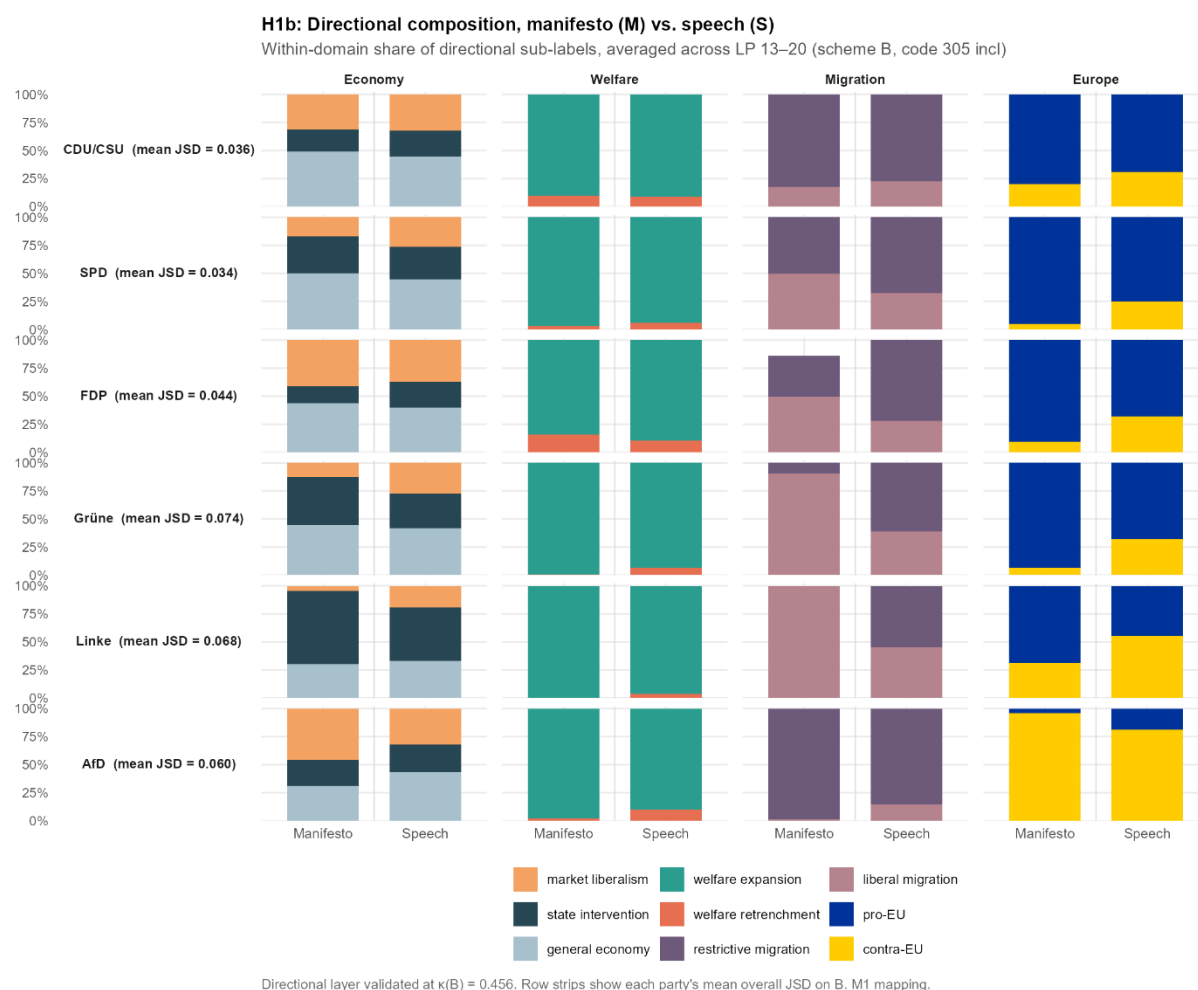
¹⁴ The FDP's 1994 manifesto carries no migration-coded material, so its migration score is undefined; all migration shares are computed over the 37 cells that carry migration content on the manifesto side.

language: re-splitting the migration bucket so that only the multiculturalism codes (607 and 608) define the restrictive and liberal poles (robustness arm 5) lowers the restrictive speech share from 68.4% to 48.6% and the manifesto share from 42.9% to 28.0%, even though the mean migration divergence barely moves (a matched mean change of -0.0062 across 36 cells). The composition shift is therefore material even where the divergence level is unchanged. Second, the direction itself is imperfectly identified: against the silver reference the classifier assigns "restrictive" too readily (precision 0.466, recall 0.753), so part of the speech-side overhang may be an assignment artefact. The same reservation attaches to the EU stance, where the classifier over-assigns contra-EU relative to the silver reference (precision 0.525), while the pro-EU pole is among the most reliably identified labels in the scheme. The welfare direction is too weakly identified ($F1 = 0.148$) to bear any interpretation beyond the near-identity just noted.

Figure 5.4 shows that this restrictive and EU-sceptic lean is not a uniform drift but a compression. The parties with the most liberal migration programmes move furthest towards the restrictive pole (the Grüne from 9.6% to 61.1%, the Linke from 0.3% to 54.4%, the FDP from 35.9% to 72.1%), while the two parties whose programmes are already restrictive move in the other direction, the CDU/CSU marginally (82.2% to 77.6%) and the AfD more clearly (99.0% to 85.4%); contra-EU follows the same pattern. The manifesto range of 0.3% to 99.0% on the restrictive share collapses to 54.4% to 85.4% in speech, a convergence of the parties onto a shared, restrictive-leaning speech profile rather than a shift of every party in the same direction. This cell-spanning compression is the same phenomenon that Section 5.3 measures on the position scale as cross-channel convergence.

Figure 5.4. Directional composition of manifestos and speeches within the four directional domains, averaged across LP 13 to LP 20. Bars give the within-domain share of each directional sub-label; row strips give each party's mean overall divergence on Aggregation B. The FDP's migration bar on the manifesto side

does not fill the panel because the party's 1994 manifesto carries no migration-coded material. Aggregation B, code 305 retained; directional layer validated at $\kappa(B) = 0.456.4$



H1b is read here as a composition claim rather than as an identification claim, since no nearest-manifesto counterpart is computed on the directional layer, which the weaker validation floor of Section 4.5 would not carry. These findings are read as consistent with, rather than as demonstrating, a systematic pattern: plenary speech leans more restrictive on migration and more EU-sceptic than the same parties' manifestos, is modestly more market-liberal, and is almost indistinguishable on welfare, with part of the migration and EU lean attributable to coding behaviour we can name.

5.3 What Drives and Moderates Divergence (H2)

The second hypothesis family is concerned with what drives divergence, with emphasis on topic differences and how these in turn differ by parties.

5.3.1 H2a

Table 5.1: Per-bucket decomposition of the manifesto–speech divergence. Mean contribution of each of the nine salience buckets to the cell JSD, with standard deviation across cells and share of the mean JSD. Contributions sum exactly to the mean JSD. Aggregation A, primary specification, n = 38 cells.

Bucket	mean contribution	sd	share
Democracy & Political System	0.0062	0.0088	20.9%
Welfare & Social Policy	0.0048	0.0041	16.1%
Foreign Policy & Defence	0.0047	0.0037	15.8%
Law & Order and National Identity	0.0032	0.0065	10.6%
Environment	0.0030	0.0046	10.0%
Migration	0.0029	0.0040	9.6%
Economy	0.0019	0.0026	6.5%
Social Groups	0.0016	0.0024	5.3%
European Integration	0.0015	0.0018	5.2%

The breakdown of the mean divergence across the nine topic buckets, as reported in Table 5.1, shows the mean per-bucket contributions, summing exactly to the mean JSD. The Democracy & Political System bucket holds the largest part of the divergences between speeches and manifestos at 0.0062 (20.9%) of the mean JSD. As discussed in Section 4.6, this bucket is already cut substantially by the two measures we implemented to alleviate the cross-domain topic-classification artefact. Both corrections therefore pull against the expectation under test: they remove mass from precisely the bucket H2a predicts will lead, and it leads all the same. This contribution should nevertheless be treated as only partly substantive, since the bucket probably still absorbs some forced misclassification. Second and third largest are Welfare at 0.0048 (16.1%) and Foreign Policy & Defence at 0.0047 (15.8%). The smallest contribution comes from European Integration at 0.0015 (5.2%). Figure 5.5 shows the individual party-period cells behind these averages. Their contributions are right-skewed, and while the most extreme cells are the LP-13 cells discussed in Section 5.2, the typical cell sits close to its bucket average, so the ranking does not rest on those few outliers.

Figure 5.5. Contribution of each salience bucket to the manifesto–speech divergence. Bars give the mean per-bucket contribution across the 38 cells; points give the individual party-period cells. Contributions sum to the cell JSD. Aggregation A, primary specification.⁵

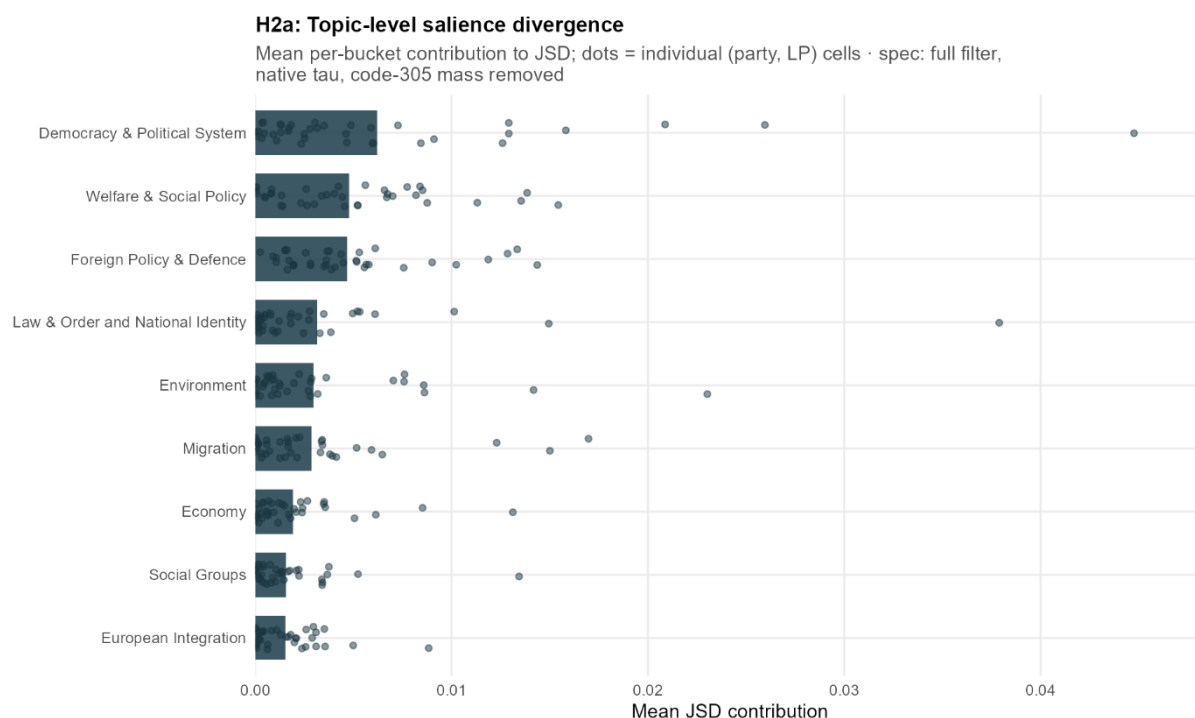


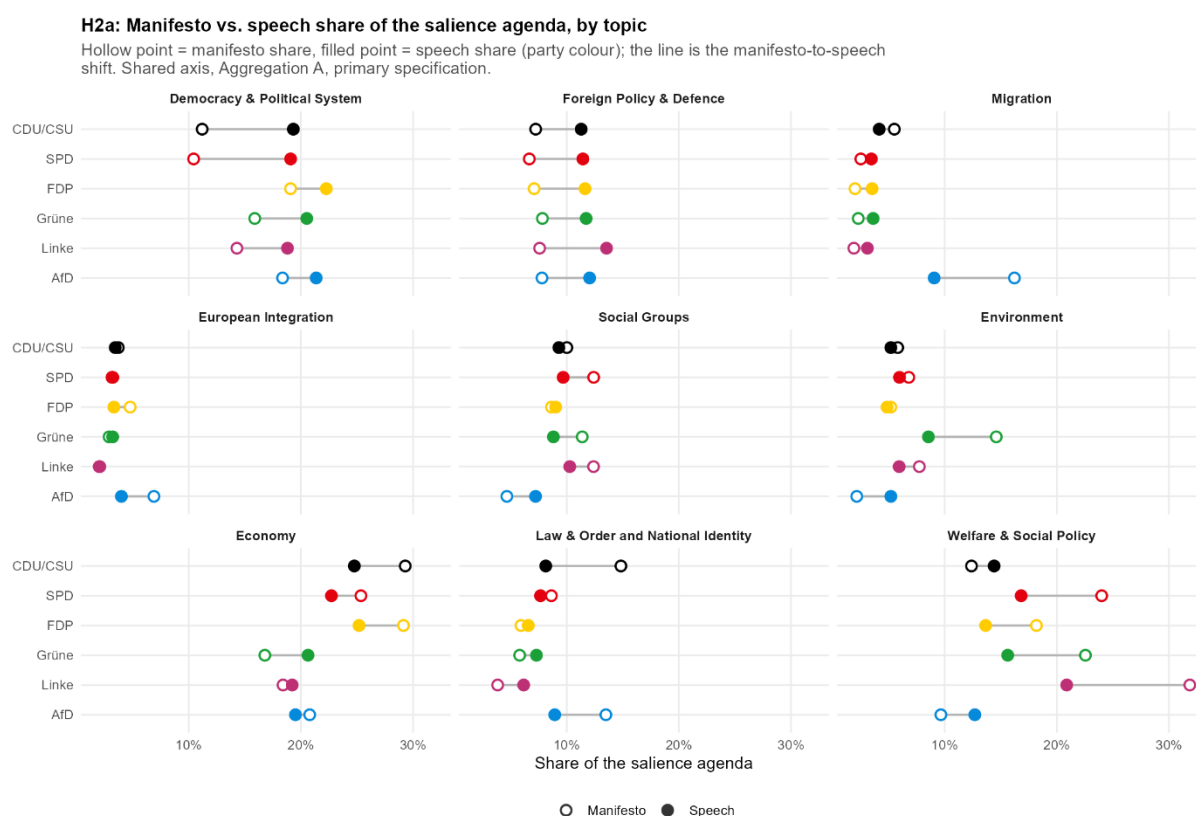
Figure 5.5. Per-bucket contribution to the manifesto–speech divergence (H2a). Bars give each salience bucket's mean contribution to the cell JSD, dots the individual party-period cells ($n = 38$). Aggregation A, primary specification: full filter, code-305 mass subtracted, native temperature. The artefact correction is set out in Section 4.6, and the exact values are in Table 5.1.

Beyond how much each topic contributes, the divergence also has a direction, and that direction is heterogeneous across topics. On some it is strikingly uniform across parties, while on others it is very party-dependent. Two topics take up a larger share of every party's plenary speech than of its manifesto: Democracy & Political System — the language of government — and Foreign Policy & Defence. The first rises from about 15% of the manifesto agenda to about 20% of the speech agenda, the second from about 7% to about 12%, and both rise for all six parties without exception. These are the only two topics on which every party moves the same way, so the shift is not the work of one or two parties but a property of the chamber. Figure 5.6 shows this directly: in the Democracy & Political System and Foreign Policy & Defence panels, every party's speech share sits above its manifesto share, whereas in the other panels the parties move in both directions.

The reverse movement, where a topic takes up less of the speech than of the manifesto, is real but far less uniform, and here the levels are what matter. Welfare is the clearest case. It fills a large part of the left's programmes, 31.9% of the Linke's manifesto agenda, 24.0% of the SPD's and 22.6% of the Grüne's, against only 12.4% of the CDU/CSU's and 9.7% of the AfD's. In

speech these shares move towards one another. The Linke still gives welfare the most room of any party at 20.9%, but the wide spread across the manifestos narrows to a band between roughly 13% and 21%, as the parties that emphasise welfare most in their programmes pull back while the CDU/CSU and the AfD give it a little more room than before. What at first glance looks like the left downplaying welfare is therefore better read as the chamber converging on a common level. The remaining topics differ only in party-specific ways, such as the CDU/CSU giving law-and-order and national-identity language less room in speech than in its programme. We therefore read the divergence as a genre signature of parliamentary debate. Speaking from government and about the state and foreign affairs pulls every party's speech agenda in the same direction, whereas on the other topics it varies from party to party, on some issues along ideological lines. What this signature implies for reading legislative speech against manifestos is taken up more broadly in Chapter 6.

Figure 5.6. Manifesto and speech share of the salience agenda, by topic and party. Hollow points give the manifesto share, filled points the speech share; the connecting line is the manifesto-to-speech shift. The axis is shared across panels, so levels are comparable between topics. Aggregation A, primary specification, party means over legislative periods.⁶



Levels, not differences: a party can move down on a topic and still lead it (the AfD's migration share falls from 16% to 9% but stays far the highest; the Linke's welfare share falls from 32% to 21% and stays the highest). Both channels show every party over-weighting system-and-procedure and foreign-policy content.

5.3.2 H2b

The goal of this analysis is to see how positioning in manifestos carries over into the positions parties take in plenary speech, broken down by domain. The main measure we introduce to

analyse this is a position score in $[-1, 1]$ that captures a party's directional position within a given domain. The spread of these positions across parties, which we take up in H2c, is what constitutes system polarisation. We code one pole of each domain as positive, namely the market-liberal, retrenchment, restrictive and EU-sceptic side, and the opposing pole as negative. Concretely we calculate for each domain: domain score = (positive pole – negative pole) / (sum of both poles). Note that in the economy domain we include only the two directional sub-categories and exclude the neutral one, since Economy is the only domain in scheme B that carries a non-directional category at all (Section 4.7.1). These scores are calculated for speeches from the classifier's directional sub-label shares and for manifestos from the MARPOR category shares. Because it is a ratio of the two directional shares within the domain, the score is independent of how salient that domain is, so it measures direction only and not the salience analysed under H1.

To compare the two channels, we implement a simple OLS model regressing the speech scores on the manifesto scores:

$$speech_{pol} = \beta_0 + \beta_1 * manifesto_{pol} + \varepsilon$$

The unit of observation is the cell (party-period), pooled over all parties and all LPs ($n = 38$, all valid party-period cells; for migration, $n = 37$).¹⁵ The slope β_1 measures how strongly the between-party differences in manifesto position carry over into differences in speech position. The intercept β_0 can be read as the estimated speech position of a party with a directionally perfectly balanced manifesto, which would mean a manifesto score of zero. R^2 reports how closely the cells fall along the fitted line, that is, the share of the variance in speech position that manifesto position accounts for. It is separate from the slope, since a line can be flat and still fit tightly.

Two effects have to be kept apart, since they show up in different coefficients. A slope below one is moderation, meaning the spread between parties shrinks from manifesto to speech, so they speak from positions closer to one another. That is not a move towards the centre, only towards each other. Where the compressed positions sit is read off the intercept together with the slope. The link between the two is exact, since the fitted line runs through the two-channel means, so the mean speech position equals $\beta_0 + \beta_1 \cdot \text{mean manifesto position}$. The level shift is

¹⁵ This is because the FDP's 1994 manifesto contains no migration-coded material, so its manifesto score is undefined and the cell drops from the migration regression.

then the gap between the two means and depends on slope, intercept and starting level together, which is why we read compression from the slope and the level from the intercept.

In every domain the slope falls well below one, so moderation holds throughout. Economy sits at 0.389, Europe at 0.481, Welfare at 0.267 and Migration at 0.239, and manifesto position still predicts speech position well, with R^2 between 0.53 and 0.78, so the parties are drawn closer together while the broad ordering largely survives; it is not preserved pair by pair, however. The 38 cells are six parties observed over eight periods rather than independent draws, so these fits are read as descriptions of the pooled cloud, not as estimates with a sampling interpretation (Section 4.8.2). Migration is the tightest fit despite the flattest slope, so its parties keep their relative standing almost intact even as their spread is squeezed hardest.

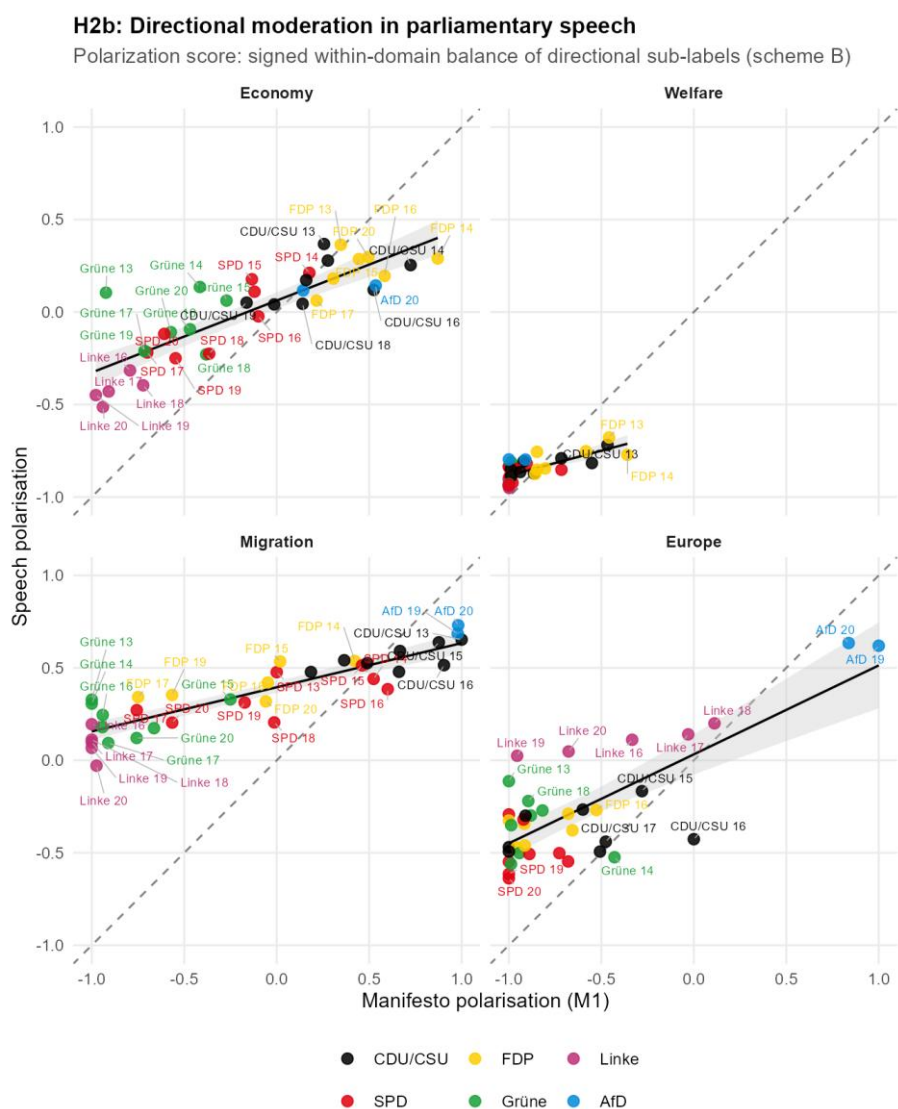
But the four domains differ in where the compressed positions sit. Welfare compresses in place, its party mean barely moving from -0.88 in manifestos to -0.85 in speech while the standard deviation of the positions falls from 0.18 to 0.07 , so the parties converge onto a level that stays firmly on the expansion side. Migration does the opposite and relocates as a block towards the restrictive pole, from a manifesto mean of -0.14 to a speech mean of $+0.36$, its intercept of $+0.40$ lifting even a balanced manifesto across the line. This is a comparison pooled across cells; its development over time is taken up in Section 5.4. Economy compresses around a near-neutral level, from -0.14 to $+0.01$, and Europe compresses towards the centre while its mean stays on the pro-EU side, from -0.65 to -0.28 .

Figure 5.7 plots speech position against manifesto position for the four domains, with the 45-degree line as reference. In every domain the fitted line runs flatter than the diagonal, which is the moderation, and it is migration whose cloud is lifted farthest above the line, clear across zero onto the restrictive side, which is the level shift.¹⁶

Figure 5.7. Speech position against manifesto position, by domain. Each point is a party-period cell; the axes give the signed within-domain directional balance in each channel on the fixed range $[-1, 1]$. The dashed line is the 45-degree reference (perfect consistency), the solid line the per-domain OLS fit. A flatter cloud is

¹⁶ For a figure with adjusted scaling for readability of dot party pairs refer to Figure F.9.1.

compression, a cloud off the diagonal is a level shift. Aggregation B; directional layer validated at $\kappa(B) = 0.456.7$



Axes fixed to the score range [-1, 1] with equal aspect, so the dashed line is a true 45-degree diagonal (perfect consistency). Solid: per-domain OLS. Slope < 1 is moderation, seen as a flatter cloud; a cloud sitting off the diagonal is a level shift (see migration, above the diagonal). Welfare is a tight far-left cluster, so its labels are thinned. Directional layer validated at $\kappa(B) = 0.456$.

This result is read with caution, because the finding at issue, that migration shifts towards the restrictive pole, rests on one of the least reliably assigned labels in the scheme: restrictive-migration precision against the silver reference is 0.466, and contra-EU is only a little higher at 0.525. Only the welfare-retrenchment label is weaker still, and no reading in this chapter rests on it. Part of the migration shift may therefore be over-assignment of the restrictive label rather than a real relocation. The manifesto side is coded from MARPOR and is untouched, and the concern is the direction of the shift rather than its compression.

5.3.3 H2c

This hypothesis asks whether the directional positions disperse less across parties in plenary speech than in their manifestos. This is the cross-channel convergence that the compression of

directional shares in Section 5.2.2 already pointed to, now measured on the position scale rather than in shares. To do this, we take the party-period position scores of Section 5.3.2 and compute their Dalton-weighted standard deviation across parties, for each legislative period, domain and channel, then compare the two channels directly. See Section 4.8.3 for the methodological discussion of the measures we employ here. While H2b analyses how much is carried over from manifestos to speech as a slope below one, suggesting that the spread between parties narrows, the approach taken in H2c measures that decline directly as the dispersion of those same positions across parties in each channel, weighted by the vote or seat share the party carries there. We read these dispersions as levels and compare them between channels here, while their change over time is discussed in Section 5.4.

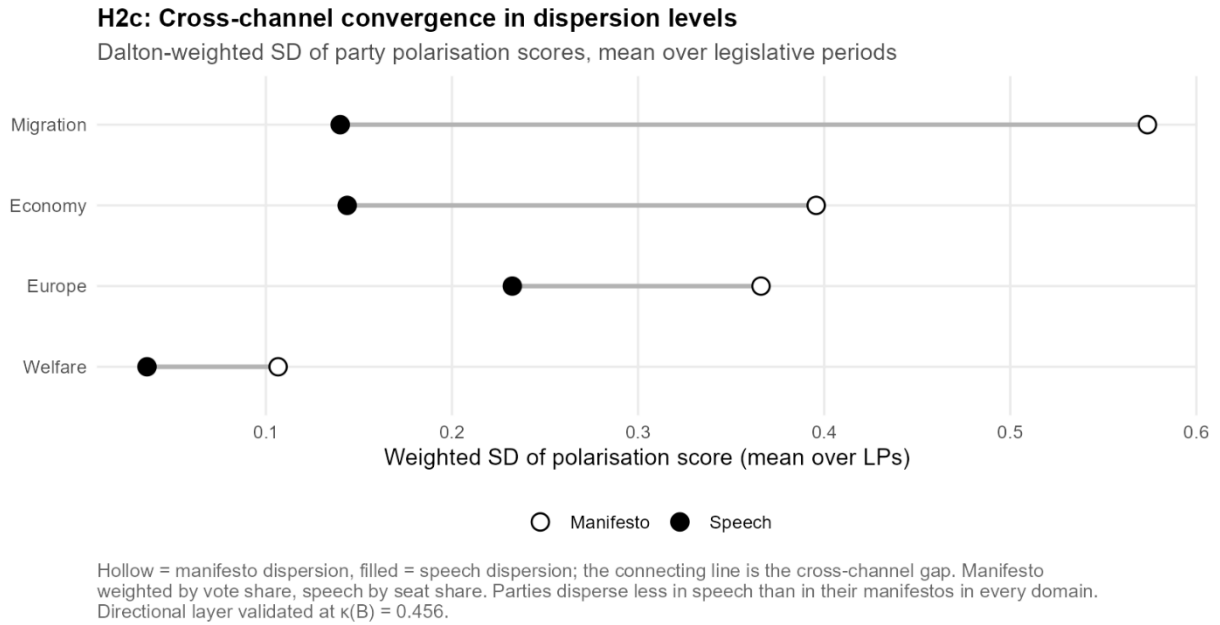
The analysis suggests that there is a decline in the weighted SDs of the position scores for each domain going from manifesto to speeches, although the size of the gap varies considerably across domains. Figure 5.8 plots the two channels side by side for each domain, the filled speech marker sitting to the left of the hollow manifesto marker throughout. Averaged over all periods, dispersion falls from 0.386 on the manifesto side to 0.138 on the speech side in the Economy domain, from 0.551 to 0.136 on Migration, from 0.104 to 0.035 on Welfare, and from 0.345 to 0.222 on Europe. Read as a mechanism, the pattern is the same in every domain. Parties show clearly separated positions in their programmes and then speak from a much narrower band, so the programmatic distances between them contract once the debate moves into the chamber. Migration shows this most sharply because its parties start furthest apart, from the widest manifesto spread of the four down to about a quarter of it. This convergence onto a shared band is what H2 anticipated as a response to the common parliamentary agenda; why the chamber compresses positions in this way is taken up in Chapter 6.

The convergence holds not only on average but period by period, with speech dispersing less than manifestos in 31 of the 32 domain-period pairs.¹⁷ The one exception is Europe in LP 13, where the 1994 manifestos are barely dispersed on Europe at all, an emergence that Section 5.4 traces. Europe's high spread at the other end of the window has a clear compositional source. In LP 19 and LP 20 the manifesto dispersion is at its highest, 0.677 and 0.567, and most of it comes from the AfD. Recomputed without that party it falls to 0.014 and 0.078, and the speech dispersion roughly halves, from 0.396 and 0.392 to 0.188 and 0.187. It is the presence of one

¹⁷ The dispersion series is computed on the 38-cell analysis sample, so the PDS cells of LPs 13 to 15 are outside it in both channels. The party-continuity decision of Section 4.8 applies to the assignment of manifestos to parties across the 2007 merger rather than to which parties held seats.

party at the EU-sceptic end of the scale, not a widening among the established parties, that accounts for the Europe spread in these periods.

Figure 5.8. Inter-party dispersion of directional positions, manifesto against speech. Dalton-weighted standard deviation of party position scores, averaged over legislative periods; hollow = manifesto (weighted by vote share), filled = speech (weighted by seat share). Aggregation B; directional layer validated at $\kappa(B) = 0.456.8$



5.4 Polarisation over Time (H3)

H3 asks whether system-level polarisation — measured by the Dalton-weighted standard deviation of position scores, the same measure used in Section 5.3.3 — increases over the observation period, in particular in the Europe and Migration domains and around the AfD's parliamentary entry in 2017. We analyse the changes in this measure to describe domain-specific trends over time but refrain from claiming causality since we read these as trends rather than as tests. A full discussion of the measure utilised here is found in Section 4.8.3.

To analyse the time trends we fit a simple OLS regression of polarisation on the legislative period, separately for each domain and channel, with the period entering as a numeric variable from 13 to 20.

$$polarisation = \beta_0 + \beta_1 * LP + \varepsilon$$

The unit of observation is the period ($n = 8$). The slope β_1 is the mean change in weighted dispersion per period, while the intercept β_0 has no meaningful reading, being the extrapolated value at period zero. R^2 reports how closely the eight points follow a straight line, and standard errors are reported but not read as significance tests, in line with Section 5.1. To localise the AfD's contribution, each fit is run a second time without the party, its weight redistributed over

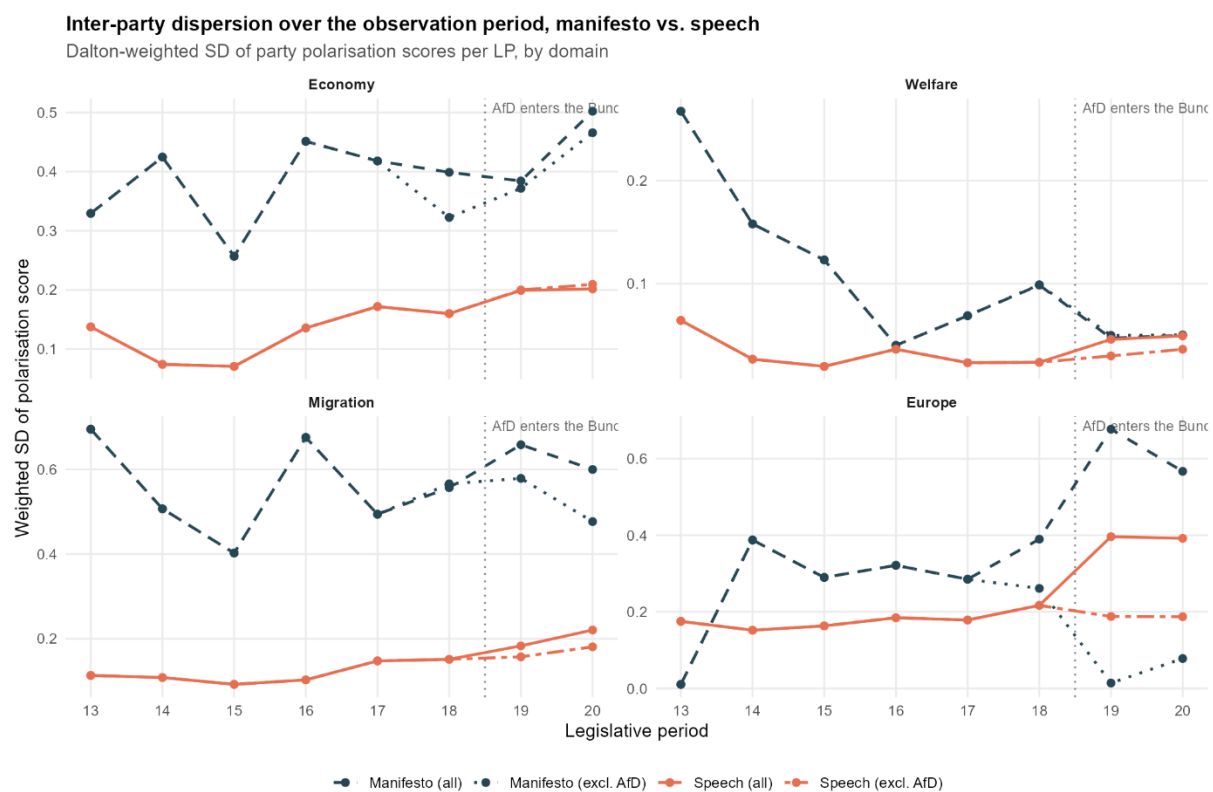
the others (arm 6, Section 4.9), so that only the periods in which it carries weight change. The difference between the two slopes then isolates the share of a trend that hangs on the party's entry: where the estimates coincide, the trend is independent of it, and where they diverge, it is carried by the changed composition of the system.

Europe is the one domain where the fits register a clear rise in both channels, at a manifesto slope of 0.0769 per legislative period (SE 0.0139, $p = 0.001$) and a speech slope of 0.0407 (SE 0.0080, $p = 0.002$). Figure 5.9 shows that the rise is concentrated in the last two periods rather than steady, and that the window opens with Europe almost undispersed in the 1994 manifestos, so the conflict axis emerges within the window rather than being present at its start. The two periods that carry the jump follow the euro crisis and the AfD's foundation as an anti-euro party in 2013, with its entry into the Bundestag in 2017. Refitting without the AfD removes most of both trends, the manifesto slope flipping from 0.0769 to -0.0079 and the speech slope falling from 0.0407 to 0.0113, the largest exclusion effects in the battery (Figure 5.10). The sign-flip locates the rise in the AfD's own position rather than in any movement of the established parties, confirming on the trend what Section 5.3.3 showed on the level.

This is less surprising than it looks. That a weighted dispersion measure jumps when one party enters at the pole opposite all the others is arithmetically expected, and Europe is the least salient of the four domains, so its scores rest on little text. What the decomposition adds is the quantification: almost the entire rise is compositional. The established parties do not move apart on this domain — in the manifesto channel they in fact converge sharply, their dispersion excluding the AfD falling from 0.261 in LP 18 to 0.014 in LP 19, which is what turns the refitted slope negative. That is the caution a system-level polarisation series calls for: a rising curve need not mean parties are drifting apart, and here it coincides with the established parties moving closer together. The speech side may in addition be inflated by the over-assignment of contra-EU, taken up below, while the manifesto series is MARPOR-coded and unaffected.

Figure 5.9. Inter-party dispersion over the observation period, by domain and channel. Dalton-weighted standard deviation of party position scores per legislative period, with and without the AfD. The AfD enters

the speech channel at LP 19 and the manifesto channel at LP 18. Note the free y-axis per panel. Aggregation B; directional layer validated at $\kappa(B) = 0.456.9$



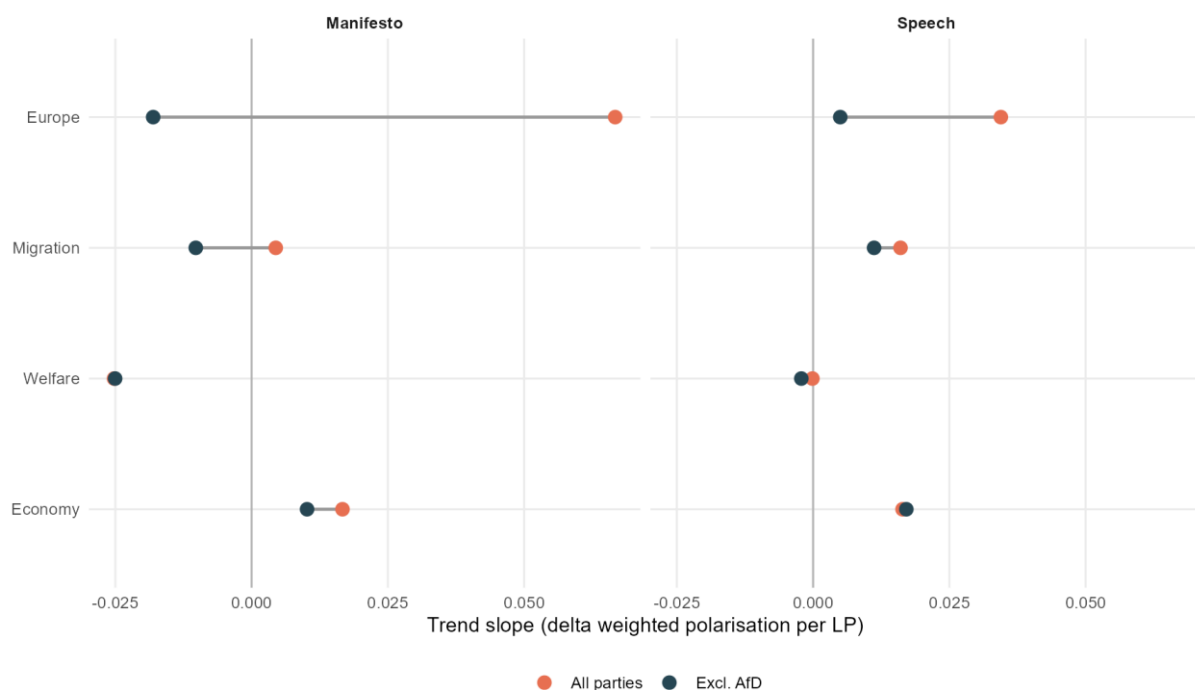
Manifesto weighted by vote share, speech by seat share. AfD enters the speech channel at LP 19 (2017); the manifesto channel includes its 2013 manifesto from LP 18. Directional layer validated at $\kappa(B) = 0.456$.

Figure 5.10. Polarisation trend slopes, all parties against excl-AfD. OLS slope of Dalton-weighted polarisation on legislative period, per domain and channel; coinciding points indicate a trend that does not

depend on the AfD. Standard errors and p values are given in Table 5.2 and are read descriptively. Aggregation B.10

H3 robustness: polarisation trend slope, all parties vs excl-AfD

OLS slope of Dalton-weighted polarisation on LP, per domain and channel. AfD: speech channel from LP 19; manifesto char



Coinciding points = the time trend is not an AfD-entry artefact. The LEVEL/compositional decomposition is in H2c. Directional layer validated at $\kappa(B) = 0$

The other domains do not follow the Europe pattern, which is why we report all four in Table 5.2. In the economy domain the speech slope is $+0.0195$ ($p = 0.016$) and stays practically unchanged without the AfD ($+0.0201$), so speech-side polarisation in economic policy rises with or without the party. In welfare the movement sits on the manifesto side, where the slope is -0.0240 ($p = 0.022$) and equally robust to the exclusion (-0.0238), while the speech series stays flat ($+0.0002$ against a standard error of 0.0026 , indistinguishable from zero at this resolution). This programmatic de-polarisation also explains the near-meeting of the two channels at the close of the window, at 0.047 against 0.046 in LP 19 and 0.050 against 0.049 in LP 20, visible in the welfare panel of Figure 5.9: it is the manifesto dispersion falling to the already low level of speech, not speech rising to it. In migration the speech slope of $+0.0181$ ($p = 0.001$) drops to $+0.0132$ ($p = 0.001$) without the AfD, so part of the rise hangs on the party, but a part remains that does not. The migration manifesto slope is close to zero on either side of the exclusion, at $+0.0154$ with all parties and $+0.0007$ without the AfD, so we read it as flat. The overall positive trend on the speech side for migration *is*, on closer inspection, largely attributable to the periods from LP 17 (2009 to 2013) onwards, beginning before the 2015

migration crisis and steepening over the periods that contain and follow it. In Figure 5.10 all these pairs lie almost on top of each other, which is the visual counterpart of the same point: these trends are not an artefact of the AfD's entry. While the Europe trend is likely compositional, the movements in the other domains are weaker but harder to attribute to composition, since they persist within the established party set. The two clearest movements among the established parties run in opposite directions, rising polarisation in economic policy speech and declining polarisation in welfare programmes.

The residual rise the preceding paragraph leaves on migration invites one further question, because the dispersion measure is invariant to the level. If every party shifted by the same amount the weighted standard deviation would not move at all, so a modest rise is compatible with the chamber drifting as a block as much as with the parties moving apart. On migration the difference is not academic, since the two readings imply opposite things about where the chamber sits. The weighted mean position, which the same formula returns alongside the dispersion, separates them. Speech dispersion more than doubles across the window, from 0.104 in LP 13 to 0.221 in LP 20, while the weighted mean moves in the opposite direction, from +0.554 to +0.344, that is, towards the liberal pole rather than the restrictive one. Disaggregated by party, the pattern is a fan rather than a drift. Between LP 13 and LP 20 every established party moves towards the liberal pole — the Grüne from +0.33 to +0.12, the SPD from +0.48 to +0.20, the FDP from +0.59 to +0.32 — and the CDU/CSU moves least, from +0.65 to +0.54. The Linke, which enters the analysis sample in LP 16, moves from +0.20 to −0.03 and is the only party whose balance ends on the liberal side of zero. The AfD enters at +0.69, above the entire previous range, and moves to +0.73. Removing the AfD leaves the dispersion rising to 0.181 rather than 0.221, so about two thirds of the increase is carried by the parties already present. This is what the residual consists of, and it is the counterpart to Europe, where the same decomposition removed the rise almost entirely.

The same reading applies beyond migration. Across the window the weighted mean moves by −0.22 on economy and by −0.12 on welfare in speech while the corresponding dispersions move by +0.07 and −0.01, and on the manifesto side the migration mean moves by −0.54 while its dispersion falls slightly. In each case the level moves further than the spread, and a series reporting only the spread would have missed it. Figure F.9.2 reports these means for all four domains and both channels, and Figure F.9.4 sets the migration trajectories beside the system series. Whether the parties moved and whether they moved apart are separate questions, and only the second is what a dispersion series reports.

Table 5.2: Polarisation trend slopes by domain and channel. OLS slope of the Dalton-weighted dispersion of party position scores on legislative period (n = 8 periods per fit), with the standard error in parentheses, alongside the same fit with the AfD excluded and its weight redistributed (arm 6, Section 4.9). Δ slope = slope (all parties) – slope (excl. AfD). p values are reported as descriptive quantities and are not read as significance tests (Section 4.8.3). Aggregation B; directional layer validated at $\kappa(B) = 0.456$.

Domain	Channel	Slope (SE)	p	Slope excl. AfD	p	Δ slope
Economy	Manifesto	+0.0208 (0.0102)	0.086	+0.0143	0.242	+0.0065
Economy	Speech	+0.0195 (0.0059)	0.016	+0.0201	0.014	−0.0007
Welfare	Manifesto	−0.0240 (0.0078)	0.022	−0.0238	0.023	−0.0002
Welfare	Speech	+0.0002 (0.0026)	0.932	−0.0018	0.446	+0.0020
Migration	Manifesto	+0.0154 (0.0183)	0.431	+0.0007	0.969	+0.0147
Migration	Speech	+0.0181 (0.0032)	0.001	+0.0132	0.001	+0.0049
Europe	Manifesto	+0.0769 (0.0139)	0.001	−0.0079	0.725	+0.0848
Europe	Speech	+0.0407 (0.0080)	0.002	+0.0113	0.011	+0.0295

Table 5.2: Descriptive OLS trend slopes of the Dalton-weighted polarisation on the legislative period, per domain and channel (LP 13 to 20, n = 8 points per fit), for all parties and refit without the AfD (arm 6, Section 4.9). Δ slope is the difference between the two estimates and localises the AfD's contribution to a trend. The manifesto channel is weighted by vote shares and the speech channel by seat shares, with the AfD carrying weight in the manifesto channel from LP 18 and in the speech channel from LP 19 (Section 4.8.3). Standard errors in parentheses. The p values are reported for transparency and are not read as significance tests (Section 5.1). Directional layer: weaker validation floor (Section 4.5).

The trend slopes compress each path into a single line, but, as Figure 5.9 shows, the Europe path is closer to a jump, which is also why the fits are not perfectly linear. As a model-free view we compare the mean dispersion before and after the AfD's entry, LP 13 to 18 against LP 19 to 20. For Europe the manifesto mean rises from 0.2527 to 0.6219 (+0.3692) and the speech mean from 0.1650 to 0.3942 (+0.2291). The remaining domains move far less, +0.0878 for migration and +0.0836 for economy on the speech side and −0.0738 for welfare on the manifesto side. With six pre-entry and only two post-entry periods, this is a description, not a test, and the pre period already carries an AfD contribution in the manifesto channel from LP 18 (Section 4.8.3), so the manifesto contrast is understated if anything.

The speech side of the Europe series carries the caveat we flagged above. Against the silver reference the classifier over-assigns contra-EU (precision 0.525, F1 = 0.634), applying the label about one and a half times as often as the Sonnet reference does, while the gold standard contains too few contra-EU sentences to quantify it at all (n = 3). This is the mechanism that can inflate a speech-side polarisation signal, since hostile rhetoric, in particular from the AfD,

may be read as EU rejection. The asymmetry matters for how far the two sides can be trusted: the pro-EU pole reaches an F1 of 0.800 against the contra-EU pole's 0.634, so the pro-EU side of the series is the more reliable one. The manifesto series does not run through the classifier, and since the sign flip sits on the manifesto side, the core reading of this section does not depend on this label.

5.5 Robustness

This section reports the robustness battery of Section 4.9. All six arms were specified in advance; arms (1) to (3) carry pre-committed acceptance criteria, while arms (4) to (6) are sensitivity designs that report the shift a variant induces. The outcomes are reported as they came out. Arms 1 and 2 are applied to H1a on Aggregation A and to H1b on Aggregation B, so both aggregations are tested at the layer at which they are constructed. They are not repeated for H2 and H3: those analyses use a different functional of the same distributions, a within-domain ratio rather than a divergence, so the rank stability established here does not transfer to them automatically, and their results carry the corresponding caution on top of the validation floor of Section 4.5.

Table 5.3: Pre-committed robustness arms 1 and 2. Spearman rank correlation between the 38 cell rankings of each pair of variants, computed separately for H1a on Aggregation A and H1b on Aggregation B. ✓ = the pre-committed acceptance criterion of Section 4.9 is met, ✗ = it is missed; the criterion is $\rho \geq 0.90$ for the temperature arm and $\rho \geq 0.85$ for the threshold arm. Arm 2 runs on the code-305-inclusive basis for construction reasons and compares rank orders, never levels (Section 4.9).

Arm	Comparison	H1a (A)	H1b (B)
Temperature (1)	sharp vs native	0.970 ✓	0.956 ✓
	sharp vs flat	0.579 ✗	0.706 ✗
	native vs flat	0.729 ✗	0.843 ✗
Threshold (2)	soft vs hard 0.4	0.943 ✓	0.861 ✓
	soft vs hard 0.5	0.929 ✓	0.772 ✗
	hard 0.4 vs hard 0.5	0.996 ✓	0.964 ✓

Table 5.3: The two robustness arms that carry a pre-committed rank-correlation criterion (arms 1 and 2 of Section 4.9). Each entry is the Spearman rank correlation between the rankings of the 38 party-period cells under two specifications, reported for the salience layer (H1a, Aggregation A) and the directional layer (H1b, Aggregation B) separately. The temperature arm rescales the sentence posteriors before aggregation, with sharp $\tau = 0.5$, native $\tau = 1$ and flat $\tau = 2$. The threshold arm replaces the soft aggregation of the primary specification with argmax assignment at cut-offs of 0.4 and 0.5. The criteria were fixed before the analysis, $\rho \geq 0.90$ for the temperature arm and $\rho \geq 0.85$ for the threshold arm. A tick marks a pass and a cross a miss. The threshold arm is computed on code 305 included for construction reasons, so it compares rank orders across the cells and not divergence levels (Section 4.9). Full per-cell results are in Appendix F.6 for the temperature arm and F.7 for the threshold arm.

Two things are stable across both layers. The temperature effect between the sharp and native variants holds at 0.970 and 0.956, so neither ranking depends on the calibration sharpness inherited from the classifier within that range. Sharp against native is the comparison that carries that claim, since both are calibrations under which the pipeline could plausibly have run. The flat variant is not one of them. It fails on both layers, and it inflates the divergence itself, from 0.0297 to 0.0368 on H1a and from 0.0506 to 0.0852 on H1b. The pre-committed criterion is missed, and we report it as missed. Read after the fact, the failure is uninformative about the primary specification rather than a threat to it: raising the temperature to 2 pushes every posterior towards the uniform distribution, which is not a calibration the pipeline could plausibly have run under. What the arm establishes in retrospect is that the measure responds at all rather than returning the same ordering whatever it is fed. The two layers part company on the threshold arm. On the salience layer the choice between soft and hard aggregation is close to inconsequential, from 0.929 to 0.996, while on the directional layer it is not, with the soft-to-hard comparison at the higher cut-off falling to 0.772. This is the same fragility that led us to read every scheme-B result with more caution. We read it as reflecting the finer-grained directional distinctions the B scheme draws, which rest on the less reliable code-level signal noted in Section 4.5.

The per-cell detail behind these correlations is given in the appendix. [Appendix F.6](#) plots the temperature sweep cell by cell, showing that the flat variant visibly reshuffles the ordering that the sharp and native variants preserve. [Appendix F.7](#) plots each hard-threshold variant against the soft baseline; in it the points track the identity line up to a level shift, which is what the rank correlations summarise. The remaining arms are reported where their design concern arises. Arm 3, which crosses the filter ladder with the code-305 treatment, is reported with the corrections themselves in Section 4.6.2 and Appendices F.1 and F.3, where the two corrections are shown to be complementary rather than redundant.

Arm 4, a tightened prefilter of at least eight words per sentence, was specified as a sensitivity check on H2a. It is subsumed by the full filter, which already removes the sentences it targets, and is therefore not reported separately. Arm 5, the re-split of the migration axis to the multiculturalism codes alone, is reported with the directional composition in Section 5.2.2, where it lowers the restrictive shares without moving the migration divergence. Arm 6, the refit of the polarisation trends without the AfD, is reported with the trends themselves in Section 5.4, where it localises the European rise to the party's own position and weight.

The corrections also do not buy their lower headline numbers at the price of wider uncertainty. Across the six bands that cross the filter ladder with the code-305 treatment at native temperature, the sentence bootstrap intervals do not widen as the corrections are applied. The mean interval width is 0.0013 for all three code-305-excluded bands against 0.0016 to 0.0017 for the included bands. For the code-305-included bands the mean divergence falls from 0.0715 without a filter to 0.0641 under the procedural filter and 0.0510 under the full filter; for the excluded bands the corresponding values are 0.0333, 0.0322 and 0.0297. The excluded bands are, if anything, the tighter ones. The full table of the six bands is given in [Appendix F.3](#).

In sum, the results are robust to moderate temperature variation on both layers and to the argmax alternative on the salience layer, but they are not robust to extreme flattening, and the directional layer inherits part of that fragility, consistent with its weaker validation.

5.6 Summary of Findings

H1a holds in the sense the reference scale defines: a party's speeches stay closer to its own manifesto than party programmes stand to one another, at 0.57 times that distance, though its own programme is the nearest of all programmes in only 42.1% of cells, against a chance baseline of 21.1%. H1b is read as a composition claim and its findings are consistent with a systematic pattern rather than demonstrating one: the resemblance between the two channels holds on economy, where speech is modestly more market-liberal, and welfare is almost unchanged between them, while it does not hold on migration or Europe, where speech leans distinctly more restrictive and more EU-sceptic than the programmes do. H2a holds on two of its three parts and in part on the third: the divergence concentrates rather than spreads, the Democracy & Political System bucket leads at 20.9% of the mean divergence even after both corrections, and the direction is uniform across all six parties on two topics but party-specific on the remaining seven. H2b holds: the slope falls well below one in every domain, and migration adds a level shift to that compression. H2c holds: dispersion is lower in speech than in manifestos in all four domains and in 31 of the 32 domain-period pairs. H3 holds only in part and only descriptively, since polarisation rises on Europe in both channels but the rise is largely compositional: excluding the AfD reverses the manifesto trend outright and cuts the speech trend to roughly a quarter of its size, while the smaller rises on economy and migration in speech survive the exclusion. Every finding on the directional layer carries the weaker validation floor of Section 4.5.

Conclusion

How does the agenda a party announces in its manifesto compare with the agenda it practises in parliament? To answer this question, this thesis has compared the manifestos and the plenary speech of the parties represented in the Bundestag in one shared category scheme, across 38 party-period cells spanning eight legislative periods from 1994 onwards. The distance between the two agendas is measured as the Jensen–Shannon divergence between their topic distributions, a distance in emphasis rather than position, and it is read against a scale derived from the parties themselves: how far their speech agendas typically stand from one another at the lower end, and how far their manifestos do at the upper end. The observation window also contains the parliamentary entry of a challenger party, which lets us follow how a new competitor registers in both channels. The answer is two-sided. Parties practise their announced agendas to a recognisable degree, and the institution they speak in bounds how much of the manifesto carries into the chamber.

On that scale, the divergence sits between the two references, at roughly four times the typical distance between parties' speech agendas yet only 0.57 times the typical distance between their manifestos. A party's own manifesto is identifiable in its speech agenda far above chance, but it is the nearest manifesto in only 42.1% of cells against a chance baseline of 21.1%. The identification varies strongly across parties, with the FDP and the AfD matching their own programme in every period, the Linke in none, and the remaining parties in between. The congruence between the two channels is therefore real but institutionally capped, as the review led us to expect. Plenary speech neither reproduces the manifesto nor escapes it.

The divergence that remains is not noise but structure. Decomposed by topic, one pattern holds across the whole spectrum. Speeches over-weight institutional content, the Democracy & Political System bucket rising from about 15% of the manifesto agenda to about 20% of the speech agenda and Foreign Policy & Defence from about 7% to about 12%, and both rise in every one of the six parties. The movement in the other direction is party-specific, and where it is substantial it takes the same shape. On welfare, the parties that give it most room in their programmes pull back while the CDU/CSU and the AfD give it slightly more, so the wide programmatic spread narrows into a much tighter band in speech. Migration repeats the pattern around a different party, the AfD and the CDU/CSU pulling back from the two heaviest programmatic emphases while the remaining parties raise theirs a little, and the AfD remains the most migration-heavy party in both channels. What looks like parties playing down their signature issues is better read as the chamber converging on a common level. The directional

layer adds three observations, each read with the caution the validation calls for. Speech positions follow manifesto positions at a slope well below one, so the distance between parties shrinks as the channel changes while their broad ordering survives, which is compression and not a move to the centre. Dispersion tells the same story, smaller in speech than in manifestos in all four domains and falling to about a quarter of the manifesto spread on migration, the same compression that the directional shares of Section 5.2.2 show on a different scale. Migration alone adds a level shift to that compression, its party mean moving from -0.14 in manifestos to $+0.36$ in speech and crossing from the liberal to the restrictive side, though the classifier assigns the restrictive label too readily against the silver reference at a precision of 0.466, so part of that shift may be an artefact we can name.

This shape is not accidental: it is exactly the pattern the literature predicts. The two-sided result follows from two forces that work on plenary speech from opposite directions. On one side, the agenda of the chamber is set by procedure and formed by all parties together as a party-system agenda that no party can ignore (Green-Pedersen and Mortensen, 2010; Sieberer, 2006). On the other, parties have a strong interest in speaking on the issues they are most closely associated with, and their leaderships allocate floor time in the service of that message (Proksch and Slapin, 2012). Of the channels available to a party, the plenary is the one in which a pre-given agenda constrains its own agenda most (Ivanusch, 2025). The compression of positions follows from the same shared agenda and from the events the chamber must respond to, which draw parties towards common ground without any of them abandoning its position or disturbing the established order, narrowing the range of differences rather than removing it. A second reading of the same compression is available, since coalition parties have reason to demonstrate unity as much as to differentiate (Sagarzazu and Klüver, 2017), and the cells we pool contain parties in office and parties out of it. What speaks against reading the whole pattern as an effect of office is that the institutional shift holds for every one of the six parties, including the Linke and the AfD, neither of which was part of a federal government in any period of the window. Plenary speech is therefore not a neutral rendering of a party agenda but that agenda after the institution has filtered it. Congruence accordingly is read between two reference points, the distance between parties' speech agendas below and the distance between their manifestos above, and without such a frame of reference a congruence value cannot be interpreted at all, which is why the double reference travels beyond this study as a scale for comparing programmes with parliamentary speech. The attention-based mandate survives the channel switch as an ordering of emphases while their levels shift markedly. Each force has its literature,

the institutional logic accounting for the moderation between parties and the ownership logic for the party profile that persists in spite of it.

Which single programme the filtered agenda of debate most resembles has a clear answer, and it is the FDP's, the nearest manifesto in 18 of the 38 speech cells, on average the closest programme to the pooled parliamentary speech (0.0257), and first even when own-party pairs are set aside (Section 5.2.1, Figure 5.3). What this means must be disentangled carefully. To be clear, this is a by-product of a benchmark built to test identification, and it concerns which topics debate emphasises in the chamber, not the positions taken, since topic weights carry no direction. Interpreted descriptively, it tells us how the agenda profile of the plenary debate itself is shaped, weighted towards economy and governance and comparatively light on welfare and migration. Concretely, this does not show that the agenda of debate is shaped by the FDP but that the FDP's programmes happen to resemble it most closely, while the more migration-focused and welfare-focused programmes do not. This is the institutional filter discussed above seen from the other side, showing which programme sits closest to its result. On the directional layer the resemblance holds only for economy, where the market-liberal share rises from 23.2% to 29.2% from manifestos to speeches on average. It does not hold for migration or Europe, where speech is more restrictive and more EU-sceptic than the parties' manifestos. The chamber converges on an FDP-shaped agenda, at most with a modest economic-liberal tilt, not on FDP positions. If any programmatic style speaks parliament's language, it is the FDP's, and whether that style travels more easily into debate is a question this design can raise but not settle.

So far the discussion has taken the party system as stable, but the window contains a change to the system itself: the entry of a challenger. We describe how the entry of the AfD in 2017 registers in the directional positions of the other parties across the four domains. The clearest movement appears on the directional layer, where polarisation in the European domain rises in both channels. The axis itself emerges inside the window, since Europe is almost undispersed in the 1994 manifestos, so the rise starts from close to nothing. That rise is almost entirely attributable to the entrant, since refitting the trend without the AfD flips the manifesto slope from +0.0769 to -0.0079 and cuts the speech slope to roughly a quarter of its size, while the established parties do not move apart on this domain — in the manifesto channel they converge sharply on one another. The rise is therefore composition in the sense of Section 2.4 rather than a response of any kind. It is also an instance of a transnational cleavage transforming party systems through rising parties rather than through the conversion of the established ones (Hooghe and Marks, 2018). But the entry registers strongly on the European domain alone.

Economic polarisation in speech rises over the window as well, but it rises with and without the party. On migration a smaller but steady increase in speech polarisation likewise survives the exclusion, and it sets in two legislative periods before the entry. In welfare the movement runs the other way and sits on the manifesto side, where polarisation declines across the window, again regardless of the exclusion.

The European movement is therefore the larger one but also the one that decomposes into an arrival, while the trends in the other domains are smaller and reflect movement among the parties already present, which makes them the more substantive finding. Migration is the case in which the exclusion removes part of the rise and leaves part of it standing, and what remains is not identifiable from these data. The parties may have been responding to a changed environment, since the chamber must address problems as they arise, or they may have been responding to one another, which is the contagion mechanism reported for the electoral arena (van Spanje, 2010; Abou-Chadi, 2016). The design we implemented cannot separate the two, since it describes trends rather than identifying what produces them. What it can rule out is one direction of the second reading. Contagion in the positional sense would mean the established parties moving towards the position the challenger holds, and on migration none of them does, since every established party moves towards the liberal pole in speech across the window, the CDU/CSU least of all. The rise in dispersion therefore comes from the established parties moving apart from one another rather than converging on the entrant, and the direction of that movement is away from the pole the entrant would later occupy, which is the pattern Lewandowsky et al. (2022) report for this chamber rather than the convergence contagion would produce. Europe is the contrast to this rather than the parallel, since there the exclusion removes most of the rise rather than a part of it: the manifesto trend reverses outright as the established parties converge on one another, and only a small residual drift remains in speech. On migration the movement is under way before the AfD takes its seats, so the pattern holds without the entry having to be its cause. The caution this calls for runs in more than one direction. A rising system polarisation score does not differentiate between a new pole and movement among the parties already there, and a flat one does not differentiate a system at rest from a system moving as a block. Recomputing without the entrant settles the first question but not the second, and it leaves open in which direction the parties already present have moved, which is what separates contagion from distancing. This is why we treat the recomputation as a constitutive part of the analysis rather than as a mere robustness check, and why the weighted mean belongs beside the dispersion. Migration makes the second point on the programmatic

side, where the directional balance of the party system moves by just over half a scale unit across the window while its dispersion barely changes.

This thesis contributes to the literature in three ways. First, on a substantive level, to our knowledge this is the first systematic measurement of agenda congruence between election programmes and plenary speech for the German party system, carried out in one shared category scheme, extended by a directional layer, at the level of the party-period cell, and over a window long enough to contain the entry of a challenger to the established order. Second, on a methodological level, we show that the cross-domain application of a manifesto-trained classifier fails in two systematic ways, namely through forced assignment where the label space contains no null class and through the inflation of a sink category that absorbs part of those forced codes and is over-assigned beyond them, so that neither correction can substitute for the other. To our knowledge neither mode has been named and quantified in political science applications of such classifiers. Both are structural rather than random and do not average out with corpus size, so they call for correction rather than caveat, and finding them at all required a corpus-representative reference rather than a hand-coded sample alone. Third, two instruments introduced along the way carry beyond this application, the double reference that makes a text distance readable and the recomputation of a polarisation series without the entering party.

Our results themselves are bounded, and most of the limits were fixed with the design rather than discovered after it. The salience layer captures what parties talk about and not what they advocate, so agenda congruence is not positional congruence. The weaker validation of the directional layer was established before the analysis, and the classifier assigns the restrictive and the contra-EU direction more generously than the silver reference warrants, the latter at a precision of 0.525, and the welfare direction bears no interpretation, so part of the restrictive and EU-sceptic lean in speech may belong to the classifier rather than to the parties. The silver reference that carries this diagnosis is itself a model-based standard, anchored in the gold sample rather than a ground truth (Section 4.5.2). Our main results do not rest on this layer. Even after both corrections the largest single contribution to the divergence remains only partly substantive, since the bucket that carries it is the one in which both failure modes concentrate, and the speech side gives every sentence the same total weight however many topics it carries, which may thin out secondary topics relative to the quasi-sentence coding of the manifestos (Section 4.3.1). A Jensen–Shannon divergence has no meaning of its own, so every congruence claim is anchored to the reference scale rather than to an adjective, and the bootstrap intervals

are lower bounds on the total uncertainty. None of the analyses identifies causal effects, and the trend fits describe the whole window with only two periods after the entry and the AfD present in two cells. The cells are also not distinguished by whether a party held office in the period, so the divergence we report pools governing and opposition parties, and any part of it that follows from office rather than from the chamber cannot be separated within this design. Beyond that, the findings hold for one country, one chamber and one observation window, with the elevation of its earliest period diagnosed as a boundary effect, and for a manifesto side that enters as fixed MARPOR aggregates. Apart from the pooling of governing and opposition parties, which is named only here, none of these limits appears for the first time in this chapter, each being stated in the chapter that produced it, and no claim is pressed beyond them.

A number of questions lie beyond the scope of this thesis, and for some of them the design we implemented is already equipped. The literature expects governing and opposition parties to differ in the latitude they have to place an agenda of their own, since a government must respond to problems as they arise while an opposition remains freer to stay on the issues that favour it (Green-Pedersen and Mortensen, 2010), and because four of the six parties change between government and opposition at least once across our window, that claim can be tested on the cells we already have. A second extension would raise the directional layer onto a firmer validation footing, since targeted training data on the few axes the validation identifies least reliably would let those hypotheses be tested rather than explored. A third concerns migration, where our results leave open whether parties were responding to a changing environment or to one another, and separating the two would require either external event and salience series entering as covariates or an identification strategy such as the regression discontinuity design Abou-Chadi and Krause (2020) apply to the electoral arena. The latter needs variation across many entries and elections, which one country cannot provide, and this points to the extension with the widest reach, running the same comparison across several parliaments. The core of the method travels, since manifestos are coded in one scheme across countries and the classifier itself is multilingual (Section 4.4), while the correction layer — the procedural filter, the null-class filter and the segmenter — is built for German parliamentary language and would have to be rebuilt per chamber, and what such a comparison would deliver goes well beyond additional cases. It would show whether the genre signature we describe is a property of this chamber or of parliamentary debate as such, it would answer whether one programmatic style sits closest to the language of parliament everywhere or only here, and it would test the mechanism itself, since the floor that a shared agenda sets should vary with how far a chamber controls its own order of business, and the Bundestag, where agenda setting is consensual and government

control weak by design, marks only one point on that scale. Whatever those comparisons find, the answer for this chamber stands. Each party carries its own agenda and its own message into parliament. Both are tempered by a level the institution sets for everyone alike.

Erklärung an Eidesstatt

Übersicht verwendeter Hilfsmittel

Die folgende Übersicht führt alle generativen und nicht-generativen KI-Werkzeuge auf, die bei der Erstellung dieser Arbeit verwendet wurden, mit Angabe von Umfang, Zweck und den betroffenen Abschnitten. Die eingesetzten Klassifikations- und Annotationsmodelle sind zugleich Gegenstand der Arbeit und werden in Kapitel 4 sowie Appendix G im Detail dokumentiert.

Werkzeug und Version	Umfang	Zweck	Abschnitte
ManifestoBERTa (manifesto-project/manifestoberta- xlm-roberta-56policy- topics-context-2024-1- 1)	Klassifikation des gesamten Redekorpus von über sechs Millionen Sätzen	Forschungsinstrument: thematische Zuordnung der Plenarreden zu den 56 MARPOR-Kategorien	4.4, Appendix G
Claude Sonnet (claude- sonnet-4-6, Anthropic, Batch-API)	Blindkodierung von zwei disjunkten Zufallsstichproben zu je 10.000 Sätzen	Forschungsinstrument: Silber-Standard-Validierung und Erzeugung der Trainingslabels für den Null- Klassen-Filter	4.5.2, 4.6.1, Appendix D, Appendix G
deepset/gbert-large	Fine-Tuning auf den 20.000 silberkodierten Sätzen, Anwendung auf das Gesamtkorpus	Forschungsinstrument: binäre Erkennung nicht kodierbarer Sätze	4.6.1, Appendix G
SoMaJo (de_CMC, Proisl and Uhrig, 2016)	Segmentierung des Gesamtkorpus	Regelbasierte, deterministische Satzsegmentierung, kein generatives Modell	4.3.1, Appendix A
Claude Opus 4.8; Claude Opus 5; Claude Fable, Anthropic	Als unterstützendes Hilfsmittel bei Recherche, Kodieren,	Unterstützung bei Literaturrecherche und Strukturierung, Erstellung	Kapitel 1 bis 6, Analyseskripte

	Grammatik und Gegenlesen, Logiküberprüfung, Überprüfung mathematischer Feinheiten.	von Gliederungs- und Schreibplänen, kritische Durchsicht von Methodenkapitel und Argumentation, Unterstützung bei der Erstellung und Prüfung der R- und Python-Skripte. Sämtliche Vorschläge wurden eigenständig geprüft, überarbeitet und in den eigenen Text überführt; kein Textabschnitt wurde unbearbeitet übernommen und die Arbeit basiert auf eigenen Ideen und Konzepten.	
R und Python	Gesamte Auswertungspipeline	Datenaufbereitung, Aggregation, Messung, Robustheitsanalysen; Paketliste in Appendix G	Kapitel 4 und 5, Appendix G

Eidesstattliche Erklärung

Ich versichere hiermit ehrenwörtlich, dass ich die vorliegende Arbeit selbstständig und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt habe. Die aus fremden Quellen direkt oder indirekt übernommenen Gedanken sind als solche kenntlich gemacht. Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Im Falle der Nutzung von generativen Modellen zur Erstellung von Texten, Abbildungen, Berechnungen, Skripten und anderen Leistungen verantworte ich die Auswahl, Übernahme und sämtliche Ergebnisse des von mir verwendeten generierten Outputs vollumfänglich selbst.

Im Verzeichnis „Übersicht verwendeter Hilfsmittel“ habe ich alle verwendeten generativen Modelle mit ihrem Produktnamen benannt und angegeben, wie, in welchem Umfang und zu welchem Zweck diese benutzt wurden.

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Appendix

Appendix A: Speech-Side Sentence Segmentation — Implementation Details (cf. Section 4.3.1)

This appendix documents the segmentation procedure applied to the speech corpus, implemented in `02_resegment_somajo.py` using SoMaJo version 2.4.3 under its `de_CMC` tokenisation guidelines. Three implementation choices bear on reproduction.

(a) *Surface reconstruction.* SoMaJo returns each sentence as a list of tokens, each carrying a `space_after` flag. We reconstructed the sentence surface by concatenating tokens and inserting a single space only where `space_after` is set, followed by a trailing `strip()`. This step is required because ManifestoBERTa operates on natural-language strings rather than token lists, and a naive whitespace join would yield corrupted surfaces (e.g. "z . B .") that the model never encountered in training. Because the GermaParl2/CWB representation stores punctuation as detached tokens with explicit spacing information, the `space_after` annotation is precisely what is needed to reattach punctuation to its host token and restore the original typography deterministically — the same detached-punctuation problem that defeats naive regular-expression splitting (Section 4.3.1) is resolved here.

(b) *Camel-case splitting disabled.* SoMaJo's default behaviour, oriented towards computer-mediated communication, may split tokens on internal capitalisation (hashtag-style segmentation). We set `split_camel_case=False`, because parliamentary prose contains legitimate intra-capital tokens — party and programme abbreviations such as *GroKo* and *BAföG*, and comparable proper nouns — that must not be fragmented, as fragmentation would corrupt token counts and degrade downstream classification. We note that `de_CMC` is SoMaJo's only German guideline set; adaptation to running parliamentary prose is therefore achieved through option settings such as this one, not through model selection.

(c) *Non-alphabetic fragment removal.* After boundaries are set and surfaces reconstructed, every candidate sentence passes an alphabetic-content filter prior to classification. A candidate is retained if and only if it contains at least one alphabetic character (Python's `str.isalpha()`, which covers the German umlauts and ß); candidates failing this test are discarded. The filter targets non-linguistic residue produced by the protocol layout — lines consisting solely of punctuation, isolated page or agenda-item numbers, tabular and formatting artefacts, and symbol-only lines — none of which carries codeable content, yet each of which would otherwise receive a forced label from ManifestoBERTa and add spurious mass to the speech-side distribution. Because the criterion is defined over alphabetic presence rather than sentence length, short but genuinely content-bearing sentences (e.g. *Ja.*, *Nein.*) are retained by construction. The step is deliberately confined to non-linguistic residue: genre-specific but alphabetic material — salutations, applause annotations such as (*Beifall*), and comparable floor language — passes unchanged and is handled separately by the procedural-sentence filter (Section 4.3.2). Taken together, surface reconstruction and residue removal reduced the sentence count from 5,977,336 under naive splitting to 5,751,659, a net decline of approximately 3.8%. The decline combines residue removal with the rejoining of false boundaries at abbreviations and ordinals; the pipeline does not log the two contributions separately.

Appendix B: Procedural-Sentence Filter — Hand-Coded Typology

The table lists the 500 most frequent normalised sentence forms in the speech corpus, each assigned a typology by hand on the basis of surface form alone, before any inspection of classification or divergence effects (pre-commitment). A form is flagged as procedural when its typology is not KEPT; forms coded KEPT (kept, shown in grey) are retained as substantive language. Of the 500 forms, 228 are flagged as procedural and 272 are kept. Normalisation

masks named appellations to a placeholder (e.g. "Herr Präsident!" → "herr <name>"); the identical routine builds this list and matches the corpus.

Table B.1: Hand-coded typology of the 500 most frequent normalised sentence forms. Typology labels: salutation (salutation), thanks/ closing (thanks / closing), procedural (procedural staging), interjection (interjection), structuring (structuring / enumeration), KEPT (kept, i.e. non-procedural).

Rank	Count	Normalized form	Example (original)	Typology
1	37,151	liebe kolleginnen und kollegen	Liebe Kolleginnen und Kollegen!	salutation
2	34,144	herr <name>	Herr Präsident!	salutation
3	33,384	frau <name>	Frau Präsidentin!	salutation
4	28,819	vielen dank	Vielen Dank.	thanks/ closing
5	21,752	meine damen und herren	Meine Damen und Herren!	salutation
6	13,026	sehr geehrte frau <name>	Sehr geehrte Frau Präsidentin!	salutation
7	12,950	herzlichen dank	Herzlichen Dank.	thanks/ closing
8	11,470	sehr geehrter herr <name>	Sehr geehrter Herr Rexrodt!	salutation
9	10,248	meine sehr verehrten damen und herren	Meine sehr verehrten Damen und Herren!	salutation
10	9,124	zweitens	Zweitens.	structuring
11	7,984	danke schön	Danke schön.	thanks/ closing
12	6,951	meine sehr geehrten damen und herren	Meine sehr geehrten Damen und Herren!	salutation
13	5,307	drittens	Drittens.	structuring
14	5,166	sehr geehrte damen und herren	Sehr geehrte Damen und Herren!	salutation
15	4,423	erstens	Erstens.	structuring
16	4,051	danke	Danke.	thanks/ closing
17	3,508	ich danke ihnen	Ich danke Ihnen.	thanks/ closing
18	2,994	ja	Ja.	interjection
19	2,960	verehrte kolleginnen und kollegen	Verehrte Kolleginnen und Kollegen!	salutation
20	2,705	liebe kollegen	Liebe Kollegen!	salutation
21	2,417	sehr geehrte kolleginnen und kollegen	Sehr geehrte Kolleginnen und Kollegen!	salutation
22	2,297	liebe kolleginnen	Liebe Kolleginnen!	salutation
23	2,152	vielen dank für ihre aufmerksamkeit	Vielen Dank für Ihre Aufmerksamkeit.	thanks/ closing
24	2,127	meine lieben kolleginnen und kollegen	Meine lieben Kolleginnen und Kollegen!	salutation
25	2,064	viertens	Viertens.	structuring
26	1,992	werte kolleginnen und kollegen	Werte Kolleginnen und Kollegen!	salutation
27	1,875	vielen dank, frau <name>	Vielen Dank, Frau Präsidentin.	thanks/ closing
28	1,870	nein	Nein.	interjection
29	1,822	vielen dank, herr <name>	Vielen Dank, Herr Präsident!	thanks/ closing
30	1,562	ich komme zum schluss	Ich komme zum Schluss.	thanks/ closing

Rank	Count	Normalized form	Example (original)	Typology
31	1,487	warum	Warum?	<i>KEPT</i>
32	1,451	schönen dank	Schönen Dank.	thanks/ closing
33	1,301	kolleginnen und kollegen	Kolleginnen und Kollegen!	salutation
34	1,157	vielen dank für die aufmerksamkeit	Vielen Dank für die Aufmerksamkeit.	thanks/ closing
35	1,116	meine sehr verehrten kolleginnen und kollegen	Meine sehr verehrten Kolleginnen und Kollegen!	salutation
36	1,074	herzlichen dank für ihre aufmerksamkeit	Herzlichen Dank für Ihre Aufmerksamkeit.	thanks/ closing
37	921	fünftens	Fünftens.	structuring
38	918	herzlichen dank für die aufmerksamkeit	Herzlichen Dank für die Aufmerksamkeit.	thanks/ closing
39	894	das ist die wahrheit	Das ist die Wahrheit.	<i>KEPT</i>
40	847	vielen herzlichen dank	Vielen herzlichen Dank.	thanks/ closing
41	845	gerne	Gerne.	<i>procedural</i>
42	799	ich bedanke mich	Ich bedanke mich..	thanks/ closing
43	739	das ist richtig	Das ist richtig.	<i>KEPT</i>
44	694	ich danke ihnen für ihre aufmerksamkeit	Ich danke Ihnen für Ihre Aufmerksamkeit.	thanks/ closing
45	684	ich danke für ihre aufmerksamkeit	Ich danke für Ihre Aufmerksamkeit.	thanks/ closing
46	638	verehrte frau <name>	Verehrte Frau Präsidentin Vollmer!	salutation
47	636	ja, bitte	Ja, bitte.	procedural
48	632	bitte	Bitte.	procedural
49	606	bitte schön	— Bitte schön.	procedural
50	606	meine herren	Meine Herren!	salutation
51	585	ja, gerne	Ja, gerne.	procedural
52	585	sehr verehrte frau <name>	Sehr verehrte Frau Präsidentin!	salutation
53	545	danke sehr	Danke sehr.	thanks/ closing
54	517	liebe gäste	Liebe Gäste!	salutation
55	510	ich bedanke mich für ihre aufmerksamkeit	Ich bedanke mich für Ihre Aufmerksamkeit.	thanks/ closing
56	473	das wollen wir nicht	Das wollen wir nicht.	<i>KEPT</i>
57	470	darum geht es	Darum geht es.	<i>KEPT</i>
58	451	meine sehr geehrten kolleginnen und kollegen	Meine sehr geehrten Kolleginnen und Kollegen!	salutation
59	451	liebe frau <name>	Liebe Frau Beer!	salutation
60	435	ich danke ihnen für die aufmerksamkeit	Ich danke Ihnen für die Aufmerksamkeit.	thanks/ closing
61	434	zweiter punkt	Zweiter Punkt.	structuring
62	433	das ist die realität	Das ist die Realität!	<i>KEPT</i>
63	425	das geht nicht	Das geht nicht.	<i>KEPT</i>
64	413	sechstens	Sechstens.	structuring
65	413	selbstverständlich	Selbstverständlich!	interjection
66	408	das ist gut so	Das ist gut so.	<i>KEPT</i>
67	408	ich danke für die aufmerksamkeit	Ich danke für die Aufmerksamkeit.	thanks/ closing

Rank	Count	Normalized form	Example (original)	Typology
68	408	sehr verehrte damen und herren	Sehr verehrte Damen und Herren!	salutation
69	405	werte kollegen	Werte Kollegen!	salutation
70	401	das gegenteil ist der fall	Das Gegenteil ist der Fall.	KEPT
71	381	sehr geehrte präsidentin	Sehr geehrte Präsidentin!	salutation
72	375	das ist das problem	Das ist das Problem.	KEPT
73	369	sehr geehrte kollegen	Sehr geehrte Kollegen!	salutation
74	361	nichts	Nichts!	interjection
75	361	meine kolleginnen und kollegen	Meine Kolleginnen und Kollegen!	salutation
76	358	ich bedanke mich für die aufmerksamkeit	Ich bedanke mich für die Aufmerksamkeit.	thanks/ closing
77	358	danke, herr <name>	Danke, Herr Präsident Hirsch.	thanks/ closing
78	356	das ist falsch	Das ist falsch.	KEPT
79	348	fehlanzeige	Fehlanzeige!	KEPT
80	336	liebe kollegen und kolleginnen	Liebe Kollegen und Kolleginnen!	salutation
81	331	meine verehrten kolleginnen und kollegen	Meine verehrten Kolleginnen und Kollegen!	salutation
82	327	so weit, so gut	So weit, so gut.	KEPT
83	321	meine damen	Meine Damen!	salutation
84	316	worum geht es	Worum geht es?	KEPT
85	313	sehr verehrte kolleginnen und kollegen	Sehr verehrte Kolleginnen und Kollegen!	salutation
86	307	ganz herzlichen dank	Ganz herzlichen Dank.	thanks/ closing
87	305	das ist der richtige weg	Das ist der richtige Weg.	KEPT
88	302	geschätzte kolleginnen und kollegen	Geschätzte Kolleginnen und Kollegen!	salutation
89	296	bitte sehr	Bitte sehr.	procedural
90	296	danke, frau <name>	Danke, Frau Präsidentin.	thanks/ closing
91	290	verehrter herr <name>	Verehrter Herr Präsident!	salutation
92	290	das ist gut	Das ist gut.	KEPT
93	285	danke für die aufmerksamkeit	Danke für die Aufmerksamkeit.	thanks/ closing
94	280	gut	Gut.	interjection
95	278	das ist der entscheidende punkt	Das ist der entscheidende Punkt.	KEPT
96	277	sehr gerne	Sehr gerne.	procedural
97	276	meine damen und herren kollegen	Meine Damen und Herren Kollegen!	salutation
98	273	sehr verehrter herr <name>	Sehr verehrter Herr Präsident!	salutation
99	269	genau	-- Genau.	interjection
100	268	besten dank	Besten Dank.	thanks/ closing
101	268	verehrte damen und herren	Verehrte Damen und Herren!	salutation
102	264	das ist der punkt	Das ist der Punkt.	KEPT
103	257	das stimmt	Das stimmt.	KEPT
104	254	herzlichen dank dafür	Herzlichen Dank dafür!	thanks/ closing

Rank	Count	Normalized form	Example (original)	Typology
105	253	das ist der unterschied	Das ist der Unterschied.	<i>KEPT</i>
106	248	das darf nicht sein	Das darf nicht sein.	<i>KEPT</i>
107	243	das ist wichtig	Das ist wichtig.	<i>KEPT</i>
108	241	das ist so	— Das ist so.	<i>KEPT</i>
109	240	vielen dank, meine damen und herren	Vielen Dank, meine Damen und Herren.	thanks/ closing
110	239	was heißt das	Was heißt das?	<i>KEPT</i>
111	239	glück auf	Glück auf!	thanks/ closing
112	237	meine sehr verehrten damen	Meine sehr verehrten Damen!	salutation
113	235	ich danke	Ich danke.	thanks/ closing
114	233	dritter punkt	Dritter Punkt.	structuring
115	227	das ist auch gut so	Das ist auch gut so.	<i>KEPT</i>
116	213	das ist nicht der fall	Das ist nicht der Fall.	<i>KEPT</i>
117	212	natürlich	— Natürlich.	interjection
118	212	das ist nicht in ordnung	Das ist nicht in Ordnung.	<i>KEPT</i>
119	210	der zweite punkt	Der zweite Punkt.	structuring
120	210	warum ist das so	Warum ist das so?	<i>KEPT</i>
121	209	das kann nicht sein	Das kann nicht sein.	<i>KEPT</i>
122	202	erster punkt	Erster Punkt.	structuring
123	200	danke für ihre aufmerksamkeit	Danke für Ihre Aufmerksamkeit.	thanks/ closing
124	197	ich komme zum ende	Ich komme zum Ende.	thanks/ closing
125	197	ein weiterer punkt	Ein weiterer Punkt.	structuring
126	197	das ist der falsche weg	Das ist der falsche Weg.	<i>KEPT</i>
127	196	richtig	Richtig!	interjection
128	196	gern	Gern.	procedural
129	195	ja, natürlich	Ja, natürlich.	interjection
130	195	vielen dank dafür	Vielen Dank dafür!	thanks/ closing
131	194	das ist ein skandal	Das ist ein Skandal.	<i>KEPT</i>
132	193	zum zweiten	Zum zweiten.	structuring
133	192	letzter punkt	Letzter Punkt.	structuring
134	190	liebe zuhörerinnen und zuhörer	Liebe Zuhörerinnen und Zuhörer!	salutation
135	187	siebtens	Siebtens.	structuring
136	187	zum schluss	Zum Schluss.	thanks/ closing
137	186	das muss sich ändern	Das muss sich ändern.	<i>KEPT</i>
138	185	ja, selbstverständlich	Ja, selbstverständlich.	<i>interjection</i>
139	184	lieber herr <name>	Lieber Herr Kollege Pinger!	salutation
140	183	das ist unser ziel	Das ist unser Ziel.	<i>KEPT</i>
141	180	so einfach ist das	So einfach ist das.	<i>KEPT</i>

Rank	Count	Normalized form	Example (original)	Typology
142	178	okay	— Okay.	interjection
143	174	doch	— Doch.	interjection
144	172	ich komme zum schluß	Ich komme zum Schluß.	thanks/ closing
145	171	aber gerne	Aber gerne.	procedural
146	171	liebe bürgerinnen und bürger	Liebe Bürgerinnen und Bürger!	salutation
147	170	das ist das entscheidende	Das ist das Entscheidende.	KEPT
148	169	herzlichen dank, herr <name>	Herzlichen Dank, Herr Präsident.	thanks/ closing
149	163	das lehnen wir ab	Das lehnen wir ab.	KEPT
150	160	so geht es nicht	So geht es nicht.	KEPT
151	159	vielen dank. - herr <name>	Vielen Dank. - Herr Präsident!	thanks/ closing
152	157	herzlichen glückwunsch	Herzlichen Glückwunsch.	thanks/ closing
153	156	das ist die situation	Das ist die Situation.	KEPT
154	156	danke schön, herr <name>	Danke schön, Herr Präsident.	thanks/ closing
155	155	das ist unverantwortlich	Das ist unverantwortlich.	KEPT
156	154	das ist unsere aufgabe	Das ist unsere Aufgabe.	KEPT
157	154	was bedeutet das	Was bedeutet das?	KEPT
158	153	das stimmt nicht	Das stimmt nicht.	KEPT
159	152	liebe zuschauerinnen und zuschauer	Liebe Zuschauerinnen und Zuschauer!	salutation
160	152	liebe zuschauer	Liebe Zuschauer!	salutation
161	151	das machen wir nicht mit	Das machen wir nicht mit.	KEPT
162	151	was machen sie	Was machen Sie?	KEPT
163	150	das müssen wir ändern	Das müssen wir ändern.	KEPT
164	149	das ist das eine	— Das ist das eine.	KEPT
165	149	recht hat er	Recht hat er.	KEPT
166	147	herzlichen dank, frau <name>	Herzlichen Dank, Frau Kollegin.	thanks/ closing
167	146	das ist der erste punkt	— Das ist der erste Punkt.	KEPT
168	145	nein, danke	Nein, danke.	thanks/ closing
169	145	das passt nicht zusammen	Das passt nicht zusammen.	KEPT
170	145	liebe kolleginnen und kolle gen	Liebe Kolleginnen und Kolle gen!	salutation
171	144	das geht so nicht	Das geht so nicht.	KEPT
172	144	verehrte kollegen	Verehrte Kollegen!	salutation
173	142	das ist wahr	Das ist wahr.	KEPT
174	141	sehr geehrte damen und herren abgeordnete	Sehr geehrte Damen und Herren Abgeordnete!	salutation
175	140	das ist gar keine frage	Das ist gar keine Frage.	KEPT
176	140	aber selbstverständlich	Aber selbstverständlich.	interjection
177	140	ein letzter punkt	Ein letzter Punkt.	structuring
178	140	werte frau <name>	Werte Frau Präsidentin!	salutation

Rank	Count	Normalized form	Example (original)	Typology
179	139	meine sehr geehrten herren und damen	Meine sehr geehrten Herren und Damen!	salutation
180	138	so ist es	So ist es.	interjection
181	136	das ist überhaupt keine frage	Das ist überhaupt keine Frage.	KEPT
182	136	guten morgen, herr <name>	Guten Morgen, Herr Präsident!	salutation
183	135	das ist völlig klar	Das ist völlig klar.	KEPT
184	134	entschuldigung	Entschuldigung!	procedural
185	134	was ist passiert	Was ist passiert?	KEPT
186	133	das ist ganz wichtig	Das ist ganz wichtig.	KEPT
187	130	das wissen sie	Das wissen Sie.	KEPT
188	130	recht herzlichen dank	Recht herzlichen Dank.	thanks/ closing
189	130	punkt	Punkt!	interjection
190	130	sehr geehrte gäste	Sehr geehrte Gäste!	salutation
191	128	vielen dank. - frau <name>	Vielen Dank. - Frau Präsidentin!	thanks/ closing
192	127	das wissen wir	Das wissen wir.	KEPT
193	127	sehr geehrter präsident	Sehr geehrter Präsident!	salutation
194	127	das ist ihre politik	Das ist Ihre Politik.	KEPT
195	127	woran liegt das	Woran liegt das?	KEPT
196	125	das ist doch absurd	Das ist doch absurd.	KEPT
197	125	sehr verehrte kolleginnen	Sehr verehrte Kolleginnen!	salutation
198	123	ich komme zum schluss, herr <name>	Ich komme zum Schluss, Herr Präsident.	thanks/ closing
199	122	dem ist nichts hinzuzufügen	Dem ist nichts hinzuzufügen.	KEPT
200	121	und warum	Und warum?	KEPT
201	121	das ist klar	Das ist klar.	KEPT
202	121	das ist doch die wahrheit	Das ist doch die Wahrheit!	KEPT
203	120	liebe kolleginnen, liebe kollegen	Liebe Kolleginnen, liebe Kollegen!	salutation
204	120	das sind die fakten	Das sind die Fakten.	KEPT
205	118	meine sehr verehrten damen und herren kollegen	Meine sehr verehrten Damen und Herren Kollegen!	salutation
206	117	schade	— Schade!	KEPT
207	117	warum sage ich das	Warum sage ich das?	KEPT
208	117	das ist nicht hinnehmbar	Das ist nicht hinnehmbar.	KEPT
209	116	jawohl	Jawohl.	KEPT
210	116	meine damen und herren, ich komme zum schluss	Meine Damen und Herren, ich komme zum Schluss.	thanks/ closing
211	116	liebe gäste auf den tribünen	Liebe Gäste auf den Tribünen!	salutation
212	115	das dürfen wir nicht zulassen	Das dürfen wir nicht zulassen!	KEPT
213	115	letzter satz	Letzter Satz.	structuring
214	114	ja, bitte schön	Ja, bitte schön.	procedural

Rank	Count	Normalized form	Example (original)	Typology
215	114	darauf können wir stolz sein	Darauf können wir stolz sein.	KEPT
216	114	danke schön, frau <name>	Danke schön, Frau Präsidentin!	thanks/ closing
217	114	ich komme zum schluss, frau <name>	Ich komme zum Schluss, Frau Präsidentin.	thanks/ closing
218	113	das ist nicht akzeptabel	Das ist nicht akzeptabel.	KEPT
219	113	das ist zu wenig	Das ist zu wenig.	KEPT
220	113	dafür herzlichen dank	Dafür herzlichen Dank!	thanks/ closing
221	112	im gegenteil	Im Gegenteil.	KEPT
222	112	das ist mit uns nicht zu machen	Das ist mit uns nicht zu machen.	KEPT
223	111	warum nicht	Warum nicht?	KEPT
224	111	darin arbeiten wir	Darin arbeiten wir.	KEPT
225	111	das wollen wir ändern	Das wollen wir ändern.	KEPT
226	110	liebe kolleginnen und liebe kollegen	Liebe Kolleginnen und liebe Kollegen!	salutation
227	109	darauf können sie sich verlassen	Darauf können Sie sich verlassen.	KEPT
228	109	das ist fakt	Das ist Fakt.	KEPT
229	109	genau darum geht es	Genau darum geht es.	KEPT
230	109	das funktioniert nicht	Das funktioniert nicht!	KEPT
231	108	ja, gern	Ja, gern.	procedural
232	108	das ist in ordnung	Das ist in Ordnung.	KEPT
233	108	damit muss schluss sein	Damit muss Schluss sein!	KEPT
234	108	geschätzte frau <name>	Geschätzte Frau Präsidentin!	salutation
235	107	achtens	Achtens.	structuring
236	106	das tun wir	Das tun wir.	KEPT
237	105	werter herr <name>	Werter Herr Präsident!	salutation
238	105	das ist ein wichtiger punkt	Das ist ein wichtiger Punkt.	KEPT
239	105	geehrte kolleginnen und kollegen	Geehrte Kolleginnen und Kollegen!	salutation
240	104	das wissen sie ganz genau	Das wissen Sie ganz genau.	KEPT
241	103	ich bitte sie	Ich bitte Sie.	KEPT
242	103	liebe landsleute	Liebe Landsleute!	salutation
243	102	punkt zwei	Punkt zwei.	structuring
244	102	das ist absurd	Das ist absurd!	KEPT
245	101	herzlichen dank fürs zuhören	Herzlichen Dank fürs Zuhören.	thanks/ closing
246	100	zweites beispiel	Zweites Beispiel.	structuring
247	100	das ist notwendig	Das ist notwendig.	KEPT
248	100	sehr geehrte abgeordnete	Sehr geehrte Abgeordnete!	salutation
249	100	zitat ende	Zitat Ende.	KEPT
250	99	bitte schön, herr <name>	Bitte schön, Herr Kirschner.	salutation
251	99	darauf kommt es an	Darauf kommt es an.	KEPT

Rank	Count	Normalized form	Example (original)	Typology
252	99	darauf sind wir stolz	Darauf sind wir stolz.	KEPT
253	97	das ist völlig richtig	Das ist völlig richtig.	KEPT
254	97	herr <name>, vielen dank	Herr Präsident, vielen Dank.	thanks/ closing
255	97	das ist auch richtig so	-- Das ist auch richtig so.	KEPT
256	96	das ist keine frage	Das ist keine Frage.	KEPT
257	96	das ist ein ganz wichtiger punkt	Das ist ein ganz wichtiger Punkt.	KEPT
258	96	letzte bemerkung	Letzte Bemerkung.	KEPT
259	96	wunderbar	-- Wunderbar.	KEPT
260	96	das lassen wir ihnen nicht durchgehen	Das lassen wir Ihnen nicht durchgehen.	KEPT
261	95	ich freue mich darauf	Ich freue mich darauf.	KEPT
262	92	so geht das nicht	So geht das nicht.	KEPT
263	92	frau <name>, ich komme zum schluss	Frau Präsidentin, ich komme zum Schluss.	thanks/ closing
264	92	liebe bürger	Liebe Bürger!	salutation
265	91	vielen dank fürs zuhören	Vielen Dank fürs Zuhören.	thanks/ closing
266	90	was ist zu tun	Was ist zu tun?	KEPT
267	89	das wissen sie auch	Das wissen Sie auch.	KEPT
268	88	das ist ganz klar	Das ist ganz klar.	KEPT
269	87	das wird nicht funktionieren	Das wird nicht funktionieren.	KEPT
270	87	das zweite	Das zweite.	KEPT
271	87	werte gäste	Werte Gäste!	salutation
272	86	das haben wir getan	Das haben wir getan.	KEPT
273	86	das ist nicht richtig	Das ist nicht richtig.	KEPT
274	86	das kann es nicht sein	Das kann es nicht sein!	KEPT
275	86	auch das gehört zur wahrheit	Auch das gehört zur Wahrheit.	KEPT
276	85	nein, herr <name>	— Nein, Herr Präsident.	procedural
277	84	das gegenteil ist richtig	Das Gegenteil ist richtig.	KEPT
278	84	das wissen wir alle	Das wissen wir alle.	KEPT
279	84	stimmen sie unserem antrag zu	Stimmen Sie unserem Antrag zu!	KEPT
280	84	dafür setzen wir uns ein	Dafür setzen wir uns ein.	KEPT
281	84	nächster punkt	Nächster Punkt.	structuring
282	84	das ist aber nicht der fall	Das ist aber nicht der Fall.	KEPT
283	84	darin sind wir uns einig	Darin sind wir uns einig.	KEPT
284	84	das ist ihr problem	Das ist Ihr Problem.	KEPT
285	84	aber was machen sie	Aber was machen Sie?	KEPT
286	83	alles gute	Alles Gute.	KEPT
287	82	bitte, herr <name>	Bitte, Herr Dreßen.	salutation
288	82	das finde ich gut	Das finde ich gut.	KEPT

Rank	Count	Normalized form	Example (original)	Typology
289	82	das ist eine tatsache	Das ist eine Tatsache.	KEPT
290	82	sehr geehrte herren und damen	Sehr geehrte Herren und Damen!	salutation
291	82	geschätzter herr <name>	Geschätzter Herr Präsident!	salutation
292	81	wir sind dazu bereit	Wir sind dazu bereit.	KEPT
293	81	ich weiß es nicht	Ich weiß es nicht.	KEPT
294	81	wir sind auf einem guten weg	Wir sind auf einem guten Weg.	KEPT
295	81	herr <name>, ich komme zum schluss	Herr Präsident, ich komme zum Schluss.	thanks/ closing
296	80	ich bin sofort fertig	Ich bin sofort fertig.	thanks/ closing
297	80	das werden wir tun	Das werden wir tun.	KEPT
298	80	beides gehört zusammen	Beides gehört zusammen.	KEPT
299	80	und das ist gut so	Und das ist gut so.	KEPT
300	80	das haben sie nicht getan	Das haben Sie nicht getan.	KEPT
301	80	wissen sie was	Wissen Sie was?	KEPT
302	79	so ist das	— So ist das.	interjection
303	79	packen wir es an	Packen wir es an!	KEPT
304	79	gut so	Gut so!	KEPT
305	79	verehrtes präsidium	Verehrtes Präsidium!	salutation
306	78	das tun wir auch	Das tun wir auch.	KEPT
307	78	hört	Hört!	KEPT
308	78	vielen dank für die frage	Vielen Dank für die Frage!	thanks/ closing
309	77	und was machen sie	Und was machen Sie?	KEPT
310	77	das kann doch nicht wahr sein	Das kann doch nicht wahr sein.	KEPT
311	77	sehr gern	Sehr gern.	procedural
312	77	das muss man sich einmal vorstellen	Das muss man sich einmal vorstellen.	KEPT
313	76	das ist das ergebnis ihrer politik	Das ist das Ergebnis Ihrer Politik!	KEPT
314	76	ganz im gegenteil	Ganz im Gegenteil.	KEPT
315	76	das begrüßen wir	Das begrüßen wir.	KEPT
316	76	das ist gut und richtig	Das ist gut und richtig.	KEPT
317	76	ja, sehr gerne	Ja, sehr gerne.	procedural
318	76	das wäre der richtige weg	Das wäre der richtige Weg.	KEPT
319	75	es tut mir leid	— Es tut mir leid.	interjection
320	75	warum eigentlich	Warum eigentlich?	KEPT
321	75	was haben sie gemacht	Was haben Sie gemacht?	KEPT
322	75	meine verehrten damen und herren	Meine verehrten Damen und Herren!	salutation
323	75	das ist schade	Das ist schade.	KEPT
324	75	lassen sie mich zum schluss kommen	Lassen Sie mich zum Schluss kommen.	thanks/ closing
325	74	das ist ein problem	Das ist ein Problem.	KEPT

Rank	Count	Normalized form	Example (original)	Typology
326	74	wir wollen das nicht	Wir wollen das nicht.	KEPT
327	74	das ist ein erfolg	Das ist ein Erfolg.	KEPT
328	74	das ist ein großer erfolg	Das ist ein großer Erfolg!	KEPT
329	74	das ist ungerecht	Das ist ungerecht.	KEPT
330	73	wir sind auf dem richtigen weg	Wir sind auf dem richtigen Weg.	KEPT
331	73	da machen wir nicht mit	Da machen wir nicht mit.	KEPT
332	73	daran müssen wir arbeiten	Daran müssen wir arbeiten.	KEPT
333	73	das kann doch wohl nicht wahr sein	Das kann doch wohl nicht wahr sein!	KEPT
334	73	ich danke ihnen fürs zuhören	Ich danke Ihnen fürs Zuhören.	thanks/ closing
335	73	frau <name>, vielen dank	Frau Präsidentin, vielen Dank!	thanks/ closing
336	73	damit muss endlich schluss sein	Damit muss endlich Schluss sein.	KEPT
337	72	dazu stehen wir	Dazu stehen wir.	KEPT
338	72	das ist doch der punkt	Das ist doch der Punkt.	KEPT
339	72	ein letztes	Ein Letztes.	structuring
340	72	das unterstützen wir	Das unterstützen wir.	KEPT
341	72	ich freue mich auf die beratungen	Ich freue mich auf die Beratungen.	KEPT
342	71	das ist doch das problem	Das ist doch das Problem.	KEPT
343	70	herzlichen dank, meine damen und herren	Herzlichen Dank, meine Damen und Herren.	thanks/ closing
344	70	die zeit drängt	Die Zeit drängt.	KEPT
345	69	das kann nicht funktionieren	Das kann nicht funktionieren.	KEPT
346	69	hören sie zu	— Hören Sie zu!	KEPT
347	69	ein zweiter punkt	Ein zweiter Punkt.	structuring
348	68	das darf nicht passieren	Das darf nicht passieren.	KEPT
349	68	ich fasse zusammen	Ich fasse zusammen.	KEPT
350	68	das kann doch nicht sein	Das kann doch nicht sein.	KEPT
351	68	das ist dringend notwendig	Das ist dringend notwendig.	KEPT
352	68	meine damen und her ren	Meine Damen und Her ren!	salutation
353	67	ja, ja	— Ja, ja.	interjection
354	67	dafür stehen wir	Dafür stehen wir.	KEPT
355	67	da haben sie recht	Da haben Sie recht.	KEPT
356	67	genau das wollen wir nicht	Genau das wollen wir nicht.	KEPT
357	67	das kann doch nicht ihr ernst sein	Das kann doch nicht Ihr Ernst sein.	KEPT
358	67	ja, genau	-- Ja, genau.	interjection
359	67	ja, klar	Ja, klar.	interjection
360	67	liebe gäste auf der tribüne	Liebe Gäste auf der Tribüne!	salutation
361	66	aber gern	Aber gern.	procedural
362	66	zum ersten	Zum ersten.	structuring

Rank	Count	Normalized form	Example (original)	Typology
363	66	das werden wir nicht mitmachen	Das werden wir nicht mitmachen.	KEPT
364	66	das ist das eigentliche problem	Das ist das eigentliche Problem.	KEPT
365	66	gar nichts	Gar nichts!	KEPT
366	66	das muss geändert werden	Das muss geändert werden.	KEPT
367	65	liebe zuhörer	Liebe Zuhörer!	salutation
368	64	sehr geehrte kollegen und kolleginnen	Sehr geehrte Kollegen und Kolleginnen!	salutation
369	64	dem ist nicht so	Dem ist nicht so.	KEPT
370	64	herzlichen dank und glück auf	Herzlichen Dank und Glück auf!	thanks/ closing
371	63	das ist die wirklichkeit	Das ist die Wirklichkeit.	KEPT
372	63	das ist das ziel	Das ist das Ziel.	KEPT
373	63	ich glaube das nicht	Ich glaube das nicht.	KEPT
374	63	vielen dank, herr <name>, dass sie die zwischenfrage zulassen	Vielen Dank, Herr Kollege, dass Sie die Zwischenfrage zulassen.	thanks/ closing
375	62	zweite bemerkung	Zweite Bemerkung.	KEPT
376	62	aha	— Aha.	interjection
377	62	und drittens	Und drittens.	structuring
378	62	verehrte präsidentin	Verehrte Präsidentin!	salutation
379	62	das ist richtig so	Das ist richtig so.	KEPT
380	61	aber natürlich	— Aber natürlich.	KEPT
381	61	das ist die eine seite	Das ist die eine Seite.	KEPT
382	61	vierter punkt	Vierter Punkt.	structuring
383	61	das ist der entscheidende unterschied	Das ist der entscheidende Unterschied.	KEPT
384	61	das reicht nicht	Das reicht nicht!	KEPT
385	61	genau das tun wir	-- Genau das tun wir.	KEPT
386	60	verehrte damen	Verehrte Damen!	salutation
387	60	einverstanden	Einverstanden.	KEPT
388	60	dabei bleibt es	Dabei bleibt es.	KEPT
389	60	das werden wir nicht zulassen	Das werden wir nicht zulassen.	KEPT
390	60	das ist eine schande	Das ist eine Schande.	KEPT
391	60	punkt eins	-- Punkt eins.	structuring
392	60	eine letzte bemerkung	Eine letzte Bemerkung.	KEPT
393	60	das kann so nicht weitergehen	Das kann so nicht weitergehen.	KEPT
394	60	danke fürs zuhören	Danke fürs Zuhören.	thanks/ closing
395	60	verehrte gäste	Verehrte Gäste!	salutation
396	59	darum geht es doch	Darum geht es doch.	KEPT
397	59	ich will ihnen auch sagen, warum	Ich will Ihnen auch sagen, warum.	KEPT
398	59	ich bin gleich fertig	Ich bin gleich fertig.	thanks/ closing
399	59	das ist das erste	Das ist das erste.	KEPT

Rank	Count	Normalized form	Example (original)	Typology
400	59	davon bin ich überzeugt	Davon bin ich überzeugt.	KEPT
401	59	das verstehe ich nicht	Das verstehe ich nicht.	KEPT
402	59	der dritte punkt	Der dritte Punkt.	structuring
403	59	das ist inakzeptabel	Das ist inakzeptabel.	KEPT
404	59	das ist eine gute nachricht	Das ist eine gute Nachricht.	KEPT
405	59	das machen wir	Das machen wir.	KEPT
406	59	das tun sie aber nicht	Das tun Sie aber nicht.	KEPT
407	59	das muss aufhören	Das muss aufhören.	KEPT
408	59	liebe besucherinnen und besucher	Liebe Besucherinnen und Besucher!	salutation
409	58	ja, herr <name>	— Ja, Herr Geis.	procedural
410	58	das ist nun einmal so	Das ist nun einmal so.	KEPT
411	58	das kann nicht richtig sein	Das kann nicht richtig sein.	KEPT
412	58	das wollen wir	Das wollen wir.	KEPT
413	58	deshalb lehnen wir ihren antrag ab	Deshalb lehnen wir Ihren Antrag ab.	KEPT
414	58	das ist unglaublich	Das ist unglaublich.	KEPT
415	58	meine sehr geehrten damen und herren abgeordnete	Meine sehr geehrten Damen und Herren Abgeordnete!	salutation
416	57	nein, nein	— Nein, nein.	interjection
417	57	da sind wir uns einig	Da sind wir uns einig.	KEPT
418	57	das ist unerträglich	Das ist unerträglich.	KEPT
419	57	das ist unbestritten	Das ist unbestritten.	KEPT
420	56	sehr gut	Sehr gut!	interjection
421	56	ich bitte um ihre zustimmung	Ich bitte um Ihre Zustimmung.	procedural
422	56	das haben wir gemacht	Das haben wir gemacht.	KEPT
423	56	das werden wir auch tun	Das werden wir auch tun.	KEPT
424	56	liebe damen und herren	Liebe Damen und Herren!	salutation
425	55	ich glaube, das ist der richtige weg	Ich glaube, das ist der richtige Weg.	KEPT
426	55	ein letzter satz	Ein letzter Satz.	structuring
427	55	das tun sie nicht	Das tun Sie nicht.	KEPT
428	55	nichts ist passiert	Nichts ist passiert.	KEPT
429	55	ja, das ist richtig	Ja, das ist richtig.	KEPT
430	55	das ist entscheidend	Das ist entscheidend.	KEPT
431	55	verehrte kolleginnen	Verehrte Kolleginnen!	salutation
432	55	vielen dank, liebe kolleginnen und kollegen	Vielen Dank, liebe Kolleginnen und Kollegen.	thanks/ closing
433	54	neuntens	Neuntens.	structuring
434	54	das ist auch richtig	Das ist auch richtig.	KEPT
435	54	das halten wir für falsch	Das halten wir für falsch.	KEPT
436	54	was heißt das konkret	Was heißt das konkret?	KEPT

Rank	Count	Normalized form	Example (original)	Typology
437	54	natürlich nicht	Natürlich nicht.	KEPT
438	54	das ist bedauerlich	Das ist bedauerlich.	KEPT
439	54	darin führt kein weg vorbei	Darin führt kein Weg vorbei.	KEPT
440	54	das ist der eine punkt	Das ist der eine Punkt.	KEPT
441	54	so sieht es aus	So sieht es aus.	KEPT
442	54	was tun sie	Was tun Sie?	KEPT
443	54	das ist sehr wichtig	Das ist sehr wichtig.	KEPT
444	54	liebe kolleginnen und kol legen	Liebe Kolleginnen und Kol legen!	salutation
445	54	genau so ist es	Genau so ist es.	KEPT
446	54	das muss unser ziel sein	Das muss unser Ziel sein.	KEPT
447	54	liebe zuhörerinnen und zuhörer auf den tribünen	Liebe Zuhörerinnen und Zuhörer auf den Tribünen!	salutation
448	54	ja, vielen dank	Ja, vielen Dank.	thanks/ closing
449	53	aber bitte	Aber bitte.	procedural
450	53	so	So!	KEPT
451	53	was können wir tun	Was können wir tun?	KEPT
452	53	was ist daraus geworden	Was ist daraus geworden?	KEPT
453	53	das ist ihr gutes recht	Das ist ihr gutes Recht.	KEPT
454	53	das ist verantwortungslos	Das ist verantwortungslos.	KEPT
455	53	schämen sie sich	Schämen Sie sich!	KEPT
456	52	ich halte das für richtig	Ich halte das für richtig.	KEPT
457	52	ich verstehe das nicht	Ich verstehe das nicht.	KEPT
458	52	und heute	Und heute?	KEPT
459	52	schönen dank für die aufmerksamkeit	Schönen Dank für die Aufmerksamkeit.	thanks/ closing
460	52	das ist die entscheidende frage	Das ist die entscheidende Frage.	KEPT
461	52	auch das ist richtig	Auch das ist richtig.	KEPT
462	52	das kann so nicht bleiben	Das kann so nicht bleiben.	KEPT
463	52	ich glaube, nicht	Ich glaube, nicht.	KEPT
464	52	ja, ich komme zum schluss	Ja, ich komme zum Schluss.	thanks/ closing
465	52	das muss ein ende haben	Das muss ein Ende haben.	KEPT
466	52	liebe kolleginnen und kollegen der demokratischen fraktionen	Liebe Kolleginnen und Kollegen der demokratischen Fraktionen!	salutation
467	51	das ist alles richtig	Das ist alles richtig.	KEPT
468	51	zum dritten	Zum dritten.	structuring
469	51	mitnichten	Mitnichten.	KEPT
470	51	darum geht es nicht	Darum geht es nicht.	KEPT
471	51	das wissen auch sie	Das wissen auch Sie.	KEPT
472	51	das ist selbstverständlich	Das ist selbstverständlich.	KEPT

Rank	Count	Normalized form	Example (original)	Typology
473	51	das ist eine gute sache	Das ist eine gute Sache.	KEPT
474	51	erstes beispiel	Erstes Beispiel.	structuring
475	51	das haben sie nicht gemacht	Das haben Sie nicht gemacht.	KEPT
476	51	ich sage ihnen auch, warum	Ich sage Ihnen auch, warum.	KEPT
477	51	handeln sie endlich	Handeln Sie endlich!	KEPT
478	51	wohl kaum	-- Wohl kaum.	KEPT
479	51	ja, das stimmt	-- Ja, das stimmt.	KEPT
480	51	hier müssen wir ansetzen	Hier müssen wir ansetzen.	KEPT
481	51	liebe kolleginnen und kollegen, ich komme zum schluss	Liebe Kolleginnen und Kollegen, ich komme zum Schluss.	thanks/ closing
482	50	das ist wohl wahr	Das ist wohl wahr.	KEPT
483	50	das freut mich	Das freut mich..	KEPT
484	50	das ist nicht so	Das ist nicht so.	KEPT
485	50	darüber müssen wir reden	Darüber müssen wir reden.	KEPT
486	50	weit gefehlt	Weit gefehlt!	KEPT
487	50	das ist zu begrüßen	Das ist zu begrüßen.	KEPT
488	50	gerne, herr <name>	Gerne, Herr Präsident.	procedural
489	50	das ist schön	Das ist schön.	KEPT
490	50	das stimmt einfach nicht	Das stimmt einfach nicht.	KEPT
491	50	vielen dank, herr <name>. - liebe kolleginnen und kollegen	Vielen Dank, Herr Präsident. - Liebe Kolleginnen und Kollegen!	thanks/ closing
492	50	genau das gegenteil ist der fall	Genau das Gegenteil ist der Fall.	KEPT
493	50	das können wir nicht hinnehmen	Das können wir nicht hinnehmen.	KEPT
494	50	zum ersten punkt	Zum ersten Punkt.	structuring
495	50	das ist die wahrheit, meine damen und herren	Das ist die Wahrheit, meine Damen und Herren.	KEPT
496	50	so viel dazu	So viel dazu.	KEPT
497	49	ich weiß, wovon ich rede	— Ich weiß, wovon ich rede.	KEPT
498	49	das ist die aufgabe	Das ist die Aufgabe.	KEPT
499	49	ich meine, nein	Ich meine, nein.	KEPT
500	49	das ist sehr bedauerlich	Das ist sehr bedauerlich.	KEPT

Appendix C: Realised Distribution of the Gold Sample across Strata

The gold sample was drawn stratified on the classifier's predicted bucket at the finest level the analysis distinguishes: the B-bucket where the predicted code carries one, the A-bucket otherwise. Each stratum was drawn with a floor of ten sentences, and the DPS bucket was

deliberately oversampled. Because the draw is path-dependent, it cannot be reproduced; the table below documents the realised allocation. "Coded" reports the sentences assigned to each stratum and coded by hand, "carried" those matched to the final corpus version, which is the basis of every figure reported in Section 4.5.1.

Table C.1: Gold sample allocation by stratum, as drawn and as carried onto the final corpus version.

Stratum	Level	Coded	Carried
Marktliberalismus	B (Economy)	13	11
Staatsintervention	B (Economy)	10	10
Wirtschaft Allgemein	B (Economy)	12	11
Sozialstaat Ausbau	B (Welfare)	10	9
Sozialstaat Begrenzung	B (Welfare)	10	10
Migration restriktiv	B (Migration)	16	15
Migration liberal	B (Migration)	12	12
Pro-EU	B (Europe)	13	12
Contra-EU	B (Europe)	10	9
Economy (residual)	residual	11	11
Welfare & Social Policy (residual)	residual	13	13
Democracy & Political System	A	56	53
Foreign Policy & Defence	A	18	18
Environment	A	18	18
Law & Order and National Identity	A	18	18
Social Groups	A	19	19
Total		259	249

Ten sentences could not be matched to the final corpus version. Three of them fall in the DPS stratum, the remainder are spread thinly across six further strata. Two strata, Sozialstaat Ausbau and Contra-EU, drop from ten to nine sentences and therefore no longer meet the design floor. At the level of the nine A-buckets of the human coding, the smallest cell retains twelve sentences.

Appendix D: Silver-Standard Prompt

The following contains the exact prompt used for both silver-standard sample A and B

Codieranweisung: MARPOR-Kodierung einzelner Bundestags-Sätze

(Silber-Standard, v2 — mit Fehltrennungs-Flag; ergänzt v1 um die symmetrische Behandlung von Segmentierungsbrüchen analog zum menschlichen Gold-Standard.)

Rolle und Aufgabe

Du bist ein erfahrener, regeltreuer Inhaltskodierer nach dem Schema des Comparative Manifesto Project (CMP/MARPOR, 4. Auflage des Kodierhandbuchs). Deine Aufgabe: Du ordnest jeden vorgelegten, bereits segmentierten Satz aus einer Bundestagsrede **genau einem** der 56 MARPOR-Codes (101–706) **oder dem Code 000** zu — und markierst zusätzlich, ob die vorgelegte Einheit überhaupt eine gültige Kodiereinheit ist (siehe „Fehltrennung“).

Du kodierst die **im Satz vertretene Aussage** — die Botschaft, die der Sprecher übermitteln will — nicht die mutmaßliche Partei des Sprechers und niemals nach deiner eigenen politischen Wertung. Du wendest die Regeln konsistent an; im Zweifel hat die Regel Vorrang vor der Intuition.

Die Kodiereinheit

Die Sätze sind bereits in Kodiereinheiten (Quasi-Sätze) zerlegt. Du ****schneidest nicht**** und fasst nicht zusammen. Du kodierst die ****eine dominante Aussage**** des vorgelegten Satzes. Verstehe die Botschaft, aber lies nicht zu viel hinein — Texte sind keine zu entschlüsselnden Codes, sondern werden so kodiert, wie sie geschrieben sind.

Grundfrage und Vorgehen

Stelle bei jedem Satz die Frage: ****Welche Botschaft vertritt dieser Satz? Welche Position übermitteln er?*** Lies den Satz sorgfältig und vergib den ****spezifischsten passenden**** Code. Die Kategoriendefinitionen sind nicht abschließend — sie geben die Grundidee; ordne jede Aussage zu, die zur allgemeinen Idee der Kategorie gehört.

Das Kategoriensystem (56 Codes + 000)

Domäne 1 — Außenbeziehungen

- ****101 Spezielle Außenbeziehungen: Positiv**** — wohlwollende Erwähnung bestimmter Länder, zu denen eine besondere Beziehung besteht; Kooperation mit oder Hilfe für sie.
- ****102 Spezielle Außenbeziehungen: Negativ**** — negative Erwähnung solcher Länder.
- ****103 Anti-Imperialismus**** — gegen imperiales/dominierendes Verhalten eines Staates über andere; für Selbstbestimmung/Entkolonialisierung.
- ****104 Militär: Positiv**** — Bedeutung von äußerer Sicherheit/Verteidigung; Erhalt/Erhöhung der Militärausgaben, Modernisierung, Bündnisverpflichtungen, Wehrpflicht.
- ****105 Militär: Negativ**** — gegen Militär/Kriegseinsatz; Abrüstung, Senkung der Ausgaben, reduzierte Wehrpflicht, „Übel des Krieges“.
- ****106 Frieden**** — Bekenntnis zu Frieden und friedlicher Konfliktlösung ****ohne**** Militärbezug; Verhandlungen, Beendigung von Kriegen.

- ****107 Internationalismus: Positiv**** — internationale Kooperation, Entwicklungshilfe, Weltordnung, internationale Gerichte, Unterstützung für UN/internationale Organisationen.
- ****108 Europäische Union: Positiv**** — wohlwollende Erwähnung der EU; Beitritt/Verbleib, Ausweitung der EU-Kompetenzen, Stärkung des Europäischen Parlaments.
- ****109 Internationalismus: Negativ**** — gegen internationale Kooperation; nationale Souveränität/Unabhängigkeit, Isolation, Unilateralismus.
- ****110 Europäische Union: Negativ**** — gegen die EU; Opposition zu EU-Politiken oder zum Nettobeitrag.

Domäne 2 — Freiheit und Demokratie

- ****201 Freiheit und Menschenrechte**** — persönliche Freiheit, Bürgerrechte, Meinungs-/Presse-/Versammlungsfreiheit, Freiheit von staatlichem/bürokratischem Zwang, Individualismus.
- ****202 Demokratie**** — Unterstützung der Demokratie als alleinige Ordnung; Bürgerbeteiligung, direkte/repräsentative Demokratie, Rechtsstaat, Gewaltenteilung, Unabhängigkeit der Gerichte.
- ****203 Verfassung: Positiv**** — Erhalt des Status quo der Verfassung; Unterstützung bestimmter Verfassungsaspekte; Verfassung als Argument.
- ****204 Verfassung: Negativ**** — Opposition zur Verfassung; Rufe nach Verfassungsänderungen/-reformen.

Domäne 3 — Politisches System

- ****301 Föderalismus/Dezentralisierung**** — Unterstützung für Föderalismus/Dezentralisierung, Subsidiarität, Autonomie subnationaler Ebenen, lokale Bräuche/Expertise.
- ****302 Zentralisierung**** — gegen Entscheidungen auf unteren Ebenen; Einheitsstaat, mehr zentrale Verwaltung.
- ****303 Verwaltungseffizienz**** ***(Auffangkategorie)*** — Effizienz/Sparsamkeit in Regierung und Verwaltung, Behördenreform, Bürokratieabbau. ****Spezifische Politik hat Vorrang.****

- ****304 Politische Korruption**** — Notwendigkeit, Korruption und Amtsmissbrauch zu beseitigen.
- ****305 Staatsautorität**** ***(Auffangkategorie)*** — Regierungsfähigkeit der eigenen Partei / fehlende Kompetenz anderer; Wunsch nach starker/stabiler Regierung. ****Spezifische Politik hat Vorrang.****

Domäne 4 — Wirtschaft

- ****401 Freier Markt**** — freier Markt/Kapitalismus, Laissez-faire, Überlegenheit privaten Unternehmertums, Privateigentum, individuelle Initiative.
- ****402 Anreize**** — angebotsorientiert, Hilfe für Unternehmen, Subventionen/Steuererleichterungen zur Förderung von Unternehmertum.
- ****403 Marktregulierung**** — fairer/offener Markt, Verbraucherschutz, gegen Monopole, Verteidigung kleiner gegen große Unternehmen, soziale Marktwirtschaft.
- ****404 Wirtschaftsplanung**** — staatliche Wirtschaftsplanung (konsultativ/indikativ).
- ****405 Korporatismus/Mischwirtschaft**** — Zusammenarbeit von Regierung, Arbeitgebern und Gewerkschaften in der Wirtschaftsplanung.
- ****406 Protektionismus: Positiv**** — Schutz interner Märkte: Zölle, Quoten, Exportsubventionen.
- ****407 Protektionismus: Negativ**** — Freihandel, offene Märkte, Abbau von Marktschutz.
- ****408 Wirtschaftsziele**** ***(Auffangkategorie)*** — breite/allgemeine wirtschaftliche Ziele ohne spezifischen Bezug. ****Spezifische Politik hat Vorrang.****
- ****409 Keynesianismus**** — nachfrageorientiert, Hilfe für Konsumenten, öffentliche Nachfrage/Sozialausgaben, Konjunkturprogramme.
- ****410 Wirtschaftswachstum**** — Wachstumsparadigma, Förderung der Produktion.
- ****411 Technologie und Infrastruktur**** — Modernisierung von Industrie, Verkehr/Kommunikation, Forschung in der Wirtschaft, öffentliche Infrastrukturausgaben (Straßen, Breitband). ***(Technische Ausbildung hier, nicht 506.)***
- ****412 Gelenkte Wirtschaft**** — direkte staatliche Kontrolle (Preise, Mindestlöhne).
- ****413 Verstaatlichung**** — staatliches Eigentum an Industrie/Boden.

- ****414 Wirtschaftsorthodoxie**** — Haushaltsdisziplin, Defizitabbau, Sparsamkeit, starke Währung, traditionelle Wirtschaftsinstitutionen.
- ****415 Marxismus**** — positive Bezüge auf marxistisch-leninistische Ideologie/Terminologie.
- ****416 Anti-Wachstum/Nachhaltigkeit**** — Ablehnung von „jedes Wachstum ist gut“, nachhaltige Entwicklung, Wachstum mit Umwelt-/Gesellschaftsschäden.

Domäne 5 — Wohlfahrt und Lebensqualität

- ****501 Umweltschutz**** — Schutz von Umwelt/Klima, „grüne“ Politik, Erhalt von Natur/Wäldern/Nationalparks, Tierschutz.
- ****502 Kultur**** — staatliche Förderung von Kultur/Freizeit/Kunst/Sport (Museen, Bibliotheken, Vereine).
- ****503 Gleichheit**** — soziale Gerechtigkeit, faire Behandlung aller, Schutz Benachteiligter, Abbau von Klassenschranken, Ende von Diskriminierung.
- ****504 Sozialstaat-Ausbau**** — Einführung/Erhalt/Ausweitung sozialer Dienste/Sicherung: Gesundheit, Kinderbetreuung, Pflege, Renten, sozialer Wohnungsbau. *(Bildung ausgenommen.)*
- ****505 Sozialstaat-Begrenzung**** — Begrenzung von Sozialausgaben; private vor staatlicher Fürsorge (Subsidiarität).
- ****506 Bildungs-Ausbau**** — Ausbau/Verbesserung von Bildung auf allen Stufen. *(Technische Ausbildung → 411.)*
- ****507 Bildungs-Begrenzung**** — Begrenzung von Bildungsausgaben, Studiengebühren, mehr Privatschulen.

Domäne 6 — Gesellschaftliches Gefüge

- ****601 Nationale Lebensweise: Positiv**** — Nation/Geschichte, Patriotismus, Nationalismus, Nationalstolz, etablierte nationale Ideen; Aussetzung von Freiheiten zum Schutz des Staates.
- ****602 Nationale Lebensweise: Negativ**** — gegen Patriotismus/Nationalismus/Nationalstaat/Nationalstolz.

- **603 Traditionelle Moral: Positiv** — traditionelle/religiöse Werte, Stabilität der traditionellen Familie, Rolle der Religion.
- **604 Traditionelle Moral: Negativ** — gegen traditionelle Moral; Scheidung/Abtreibung, moderne Familienformen, Trennung von Kirche und Staat.
- **605 Recht und Ordnung: Positiv** — strikte Strafverfolgung, härteres Vorgehen gegen Kriminalität, Unterstützung der Polizei, innere Sicherheit.
- **606 Bürgersinn: Positiv** — nationale Solidarität, gesellschaftlicher Zusammenhalt, Gemeinnutzen, öffentliches Interesse.
- **607 Multikulturalismus: Positiv** — kulturelle Vielfalt/Pluralität, Erhalt religiöser/sprachlicher Eigenständigkeit.
- **608 Multikulturalismus: Negativ** — kulturelle Integration/Homogenität, Anpassung.

Domäne 7 — Soziale Gruppen *(Auffangdomäne, außer 703)*

- **701 Arbeitnehmer: Positiv** — Arbeiterschaft/Arbeitslose, Gewerkschaften, gute Behandlung von Beschäftigten (Jobs, Bedingungen, Löhne, Renten) — als Gruppe, ohne spezifische Politik.
- **702 Arbeitnehmer: Negativ** — gegen Arbeitnehmergruppen/Gewerkschaften, „Machtmissbrauch“ der Gewerkschaften.
- **703 Landwirtschaft und Bauern: Positiv** — spezifische Politik für Landwirtschaft/Bauern (nur wenn Landwirtschaft das zentrale Anliegen ist). *(Sonderregel: 703 schlägt andere Kategorien.)*
- **704 Mittelschicht und Berufsgruppen** — Mittelschicht, Professionen (Ärzte, Anwälte), Angestellte, Dienstleistungssektor. **Soziale**, keine ökonomische Kategorie.
- **705 Benachteiligte Minderheiten** — Behinderte, Homosexuelle, Immigranten u. ä. — nur, wenn nicht anderweitig kodierbar (503, 504, 604 ...).
- **706 Nicht-ökonomische demografische Gruppen** — Frauen, Studierende, Altersgruppen — nur, wenn nicht unter anderen Kategorien erfasst (503, 504 ...).

****000 Keine sinnvolle Kategorie**** — Aussagen, die von keiner Kategorie erfasst werden oder völlig bedeutungslos sind (siehe Abschnitt „Wann 000“). ****Nicht**** für Segmentierungsbrüche — dafür das Fehltrennungs-Flag (siehe unten).

Entscheidungsregeln

Wende diese Regeln in der angegebenen Priorität an.

1. Spezifisch schlägt Auffangkategorie

****303, 305, 408 und 701–706 (außer 703) sind Auffangkategorien.**** Verwende sie ****nur**, wenn keine spezifischere Politikategorie passt.** Sie sind zu vermeiden, sobald ein konkreter Politikcode greift — gelten aber sehr wohl, wenn nichts Spezifisches passt.

- „Wir heben die Renten an“ → ****504**** (konkrete Sozialpolitik), nicht 706.

- „Menschen mit Behinderungen erhalten besseren Eingliederungshilfe-Anspruch“ → ****504**/503** (konkrete Leistung), nicht 705.

- „Wir würdigen die Arbeit der Gewerkschaften“ ohne konkrete Politik → ****701**** (kein spezifischer Code → Auffangkategorie gilt).

- „Wir brauchen eine effizientere Verwaltung“ ohne konkrete Maßnahme → ****303****; mit konkreter Maßnahme (z. B. Stellenabbau zum Sparen) → der spezifische Wirtschaftscod.

****Besonders für 305:**** „klingt nach Staat/Autorität“ ist ****kein**** hinreichender Grund. 305 nur, wenn der Satz wirklich von Regierungs-/Oppositionskompetenz oder dem Wunsch nach starker/stabiler Regierung handelt ****und**** keine konkrete Politik trägt. Sobald eine Sachpolitik im Satz steht (auch ein anderer Demokratie-Code wie 303 oder 202), kodiere diese. Prozedurales/Verfahren → 000 (nicht 305). Verlasse dich nicht auf Autoritäts-Vokabular allein.

2. Ziel schlägt Mittel

Bei „Wir wollen A erreichen, indem wir B tun“ kodierst du ****A**** (das Ziel); B ist nur Werkzeug.

- „Wir ändern das Grundgesetz, ****um**** den Klimaschutz zu verankern“ → ****501**** (Ziel = Umwelt), nicht 204. Die Verfassungsänderung ist nur Mittel.

- ****Ausnahme:**** Wird das Mittel als ****einzige/zwingende**** Option betont, wird es zur eigenständigen Botschaft. („Um Frieden zu erreichen, müssen wir die NATO verlassen und das Militär abrüsten“ → Frieden 106 ****und**** Militär-negativ 105.)

3. Kritik am Ist-Zustand = Unterstützung des Ideals

Eine kritische Bemerkung über den ****aktuellen Zustand**** von X ist meist ****Unterstützung**** für das Ideal X, kein Statement gegen X.

- „Unsere Demokratie funktioniert nicht mehr gut genug!“ → ****202**** (Sorge um die Demokratie, positiv), nicht gegen Demokratie.

- „Unser Sozialstaat lässt immer mehr Menschen im Stich“ → ****504**** (pro Sozialstaat), nicht 505. Nur explizites „weniger/kürzen“ → 505.

4. Einleitungs- und Verbindungssätze erben den Kontext — nicht 000

Überleitende, einführende oder verbindende Sätze sind keine eigenständigen Aussagen, sondern Teil eines fortlaufenden Arguments. Sie bekommen den ****Code des umgebenden Arguments**** (bzw. des Absatzes), ****nicht**** 000 — ****vorausgesetzt, die Einheit ist grammatisch vollständig.**** (Ist die Einheit dagegen zerrissen, → Fehltrennung, nicht erben.)

- Kontext „Wir investieren massiv in den sozialen Wohnungsbau.“ → Satz „Ich möchte das an zwei Beispielen verdeutlichen.“ → ****504**** (erbt das Wohnungsbau-Argument), nicht 000.

- Nutze dafür den vorhergehenden und folgenden Satz (siehe Regel 8).

5. Agrar-Sonderregel

Steht die Wahl zwischen ****703**** und einer anderen (oft wirtschaftlichen) Kategorie, wähle ****703**** — aber nur, wenn der Satz ****tatsächlich**** um Landwirtschaft/Bauern geht. Das bloße Wort „Bauer“ macht noch keine 703.

6. Einseitige Kategorien → 000

Einige Codes existieren nur positiv (Umwelt 501/416, Landwirtschaft 703). Eine klar ****gegenläufige**** Aussage hat keinen passenden Code → ****000****, sofern kein anderer Code greift.

- Eine reine „Agrarpolitik schadet“-Aussage ohne anderen Bezug → ****000****.
- Eine Anti-Umwelt-Aussage, die ökonomisch gerahmt ist, bekommt den Wirtschaftscode; ist sie es nicht → 000.

7. Wie geschrieben kodieren — keine eigene Wertung

Kodiere die ****behauptete**** Position, auch wenn sie dir falsch oder unrealistisch erscheint.

- „Wir erhöhen die Militärausgaben, ****um Frieden zu sichern****“ → ****106**** (Frieden, das genannte Ziel).
- Hintergrundwissen nur zur Auflösung mehrdeutiger Sätze; persönliche Meinung ****nie****.

8. Kontext-Hierarchie bei Mehrdeutigkeit

Ist der Satz allein unklar, nutze Kontext in dieser Reihenfolge — die nächstliegende Ebene zuerst: 1) der Rest des Satzes, 2) der ****vorhergehende und folgende Satz****. (Weitere Ebenen wie Absatz, Kapitel oder Gesamttext stehen bei isolierten Redesätzen nicht zur Verfügung.)
Nutze Kontext nur, um den Satz zu ****verstehen****, nicht, um einen an sich klaren Satz umzuetikettieren.

Wann 000 — eng auslegen

Grundsatz: Verwende, wann immer möglich, einen sinnvollen Code (101–706). ****000 nur****, wenn:

1. die Aussage ****völlig bedeutungslos**** ist und auch ****im Kontext nicht**** kodierbar (prüfe zuerst Regel 4!), ****oder****
2. eine Politikposition vorliegt, die das Schema nicht kennt (inklusive der falschen Richtung einer einseitigen Kategorie, Regel 6).

****Nicht**** 000 für Einleitungs-/Verbindungssätze innerhalb einer Sachpassage — die erben (Regel 4). ****Nicht**** 000 für Segmentierungsbrüche — die bekommen das Fehltrennungs-Flag (siehe unten).

000 ist eine ****vollständige, wohlgeformte**** Äußerung ohne Policy-Gehalt: Anrede, Dank, Aphorismus, Sprichwort, sarkastische oder meta-kommentierende Bemerkung („Nach dem Ende ist vor dem Anfang.“, „Sehr geehrte Frau Präsidentin!“, „Ich glaube, uns ist gar nicht klar, was wir hier beschließen.“). Entscheidend: die Einheit ist intakt, sie trägt nur keine Politik.

Besonderheit Bundestagsreden

Reden enthalten mehr Verfahrens-, Zwischenruf- und vor allem ****Segmentierungs****-Material als Parteiprogramme. Entscheide in ****drei**** Schritten:

1. ****Ist die vorgelegte Einheit grammatisch zerrissen**** (Bruch durch fehlerhafte Segmentierung)? → ****Fehltrennung-Flag**** (siehe nächster Abschnitt). Dies zuerst prüfen.
2. ****Sonst — ist der Satz in ein laufendes Sachargument eingebettet**** (vollständig, aber ohne eigene Botschaft)? → ****Kontext erben**** (Regel 4).
3. ****Sonst — freistehendes, wohlgeformtes Verfahrensrauschen ohne Policy**** (echte Zwischenrufe/Heckling, Anreden, Geschäftsordnungs-/Verfahrensfloskeln, inhaltsleere Ausrufe)? → ****000****.

Leitfrage in einem Satz: ****zerrissen** → Fehltrennung; eingebettet → erben; freistehend & inhaltsleer → 000.******

Fehltrennung — Segmentierungsbruch (eigenes Flag, nicht 000)

Manche vorgelegten „Sätze“ sind ****keine gültige Kodiereinheit****, weil die automatische Satztrennung sie falsch geschnitten hat. Das ist ein ****Datenqualitäts****-Problem, kein Inhaltsproblem — und wird mit einem ****eigenen Flag**** markiert, damit solche Einheiten später symmetrisch zum menschlichen Gold-Standard ausgeschlossen werden können (dort war „Fehltrennung“ eine eigene Kategorie, getrennt von 000).

****Typische Auslöser**** (Abkürzungen/Zahlen über die Satzgrenze zerrissen, mitten in der Konstruktion abgeschnitten, verwaiste Teilsätze, fehlattribuierte Dialog-Fragmente):

- Eine ****Ordnungszahl oder ein Datum**** ist zerrissen: `context\_before` endet auf „... zum 1.“ und der Satz beginnt mit „Januar 2025 in Kraft ...“.
- Eine ****Abkürzung oder ein Aktenzeichen**** ist zerrissen: `context\_before` endet auf „... Verordnung Nr.“ und der Satz beginnt mit „2019/1234 ...“; oder der Satz endet auf „... bzw.“ / „... F.D.P.“ / „... z. B.“ und läuft im `context\_after` weiter.
- Ein ****verwaister Teilsatz**** ohne eigenständige Aussage, dessen Hauptsatz im Kontext liegt und dessen Botschaft durch den Bruch nicht mehr eindeutig rekonstruierbar ist.
- Ein ****Dialog-/Zwischenruf-Fragment**** mit führendem „-“ oder „--“, bei dem die Einheitengrenze erkennbar falsch liegt und kein eigenständiger Satz vorliegt.

****Abgrenzungen:****

- ****Fehltrennung vs. 000:**** 000 = vollständige, wohlgeformte Äußerung ohne Policy. Fehltrennung = gebrochene/zerrissene Einheit. Faustregel: **grammatisch vollständig, nur inhaltlich leer** → 000; **Einheit selbst zerrissen** → Fehltrennung. (Ein führendes „-“ allein

macht noch keine Fehltrennung — ist der Satz danach vollständig und nur inhaltsleer, bleibt es 000.)

- **Fehltrennung vs. Regel 4 (erben):** Regel 4 erbt nur **grammatisch vollständige** Überleitungssätze. **Zerrissene** Einheiten werden **nicht** geerbt, sondern als Fehltrennung markiert.

- **Botschaft überlebt den Bruch:** Ist trotz unsauberer Trennung **eine** klare Policy-Botschaft erkennbar (z. B. fehlt nur das Subjekt, der Sinn ist eindeutig), kodiere den **Inhalt** und setze ``fehltrennung = false``. Fehltrennung nur, wenn der Bruch eine eindeutige Kodierung **verhindert**.

Ausgabe bei Fehltrennung: ``marpor = "000" und `fehltrennung = true``. (Der Ausschluss erfolgt downstream über das Flag, nicht über den Code; ``flag_unsure`` bleibt hier ``false``, da keine inhaltliche Code-Konkurrenz vorliegt.)

Hinweis: Auf einem sauber re-segmentierten Korpus (z. B. nach SoMaJo mit deutscher Abkürzungsliste) sind die meisten dieser Fälle — vor allem zerrissene Abkürzungen, Aktenzeichen und Ordnungszahlen — bereits behoben. ``fehltrennung = true`` sollte dann **selten** feuern. Nutze es **nicht** als Mülleimer für schwer kodierbare, aber **gültige** Sätze — die bekommen einen Code (ggf. mit ``flag_unsure = true``).

Konfidenz und Unsicherheit

Gib zu jedem Satz eine Konfidenz an:

- **high** — eindeutige Einzelbotschaft; ein spezifischer Code passt klar. (Auch ein klarer Fehltrennungs- oder 000-Fall kann „high“ sein.)

- **medium** — kodierbar, aber Urteilsache (Abwägung Spezifisch-vs-Auffang, Kontext nötig, Ziel-vs-Mittel). **Geerbte Überleitungssätze (Regel 4) und Grenzfälle „000 vs. erben vs. Fehltrennung“** sind typischerweise höchstens medium.

- **low** — echt mehrdeutig: mehrere plausible Codes oder schwaches Signal. Vergib „low“ **nicht** reflexhaft nur, weil ein Satz kein Lehrbuchbeispiel ist.

Setze `flag_unsure = true` bei genuiner Code-Konkurrenz **zwischen gültigen Codes** (z. B. 602 vs. 607, 701 vs. 405). Bei Fehltrennung bleibt `flag_unsure = false`. `note`: höchstens ein kurzer Satz, nur bei Grenzfällen.

Ausgabeformat

Gib **ausschließlich** ein JSON-Array zurück — ein Objekt pro Eingabesatz, in derselben Reihenfolge. **Kein** Fließtext, **keine** Markdown-Codezäune, **keine** Kommentare außerhalb des JSON.

Schema pro Objekt:

```

```
{ "gs_id": "<id>", "marpor": "<NNN>", "fehltrennung": <true|false>,
 "flag_unsure": <true|false>, "confidence": "high"|"medium"|"low", "note": "<kurz oder
leer>" }
```

```

- `marpor` ist ein dreistelliger Code als String: einer von 101–706 oder `"000"`. Bei `fehltrennung = true` immer `"000"`.

- `fehltrennung` markiert einen Segmentierungsbruch (ungültige Einheit), **getrennt** von inhaltlichem 000.

- `flag_unsure` markiert inhaltliche Code-Konkurrenz zwischen gültigen Codes (bei Fehltrennung `false`).

- `note` ist leer (`""`), außer bei einem Grenzfall (≤ 1 Satz Begründung).

Die Eingabe ist ein JSON-Array von Objekten mit den Feldern `gs\_id`, `context\_before`, `sentence`, `context\_after`. Kodiere stets nur `sentence`; `context\_before`/`context\_after` dienen ausschließlich dem Verständnis (Regel 4 und 8) und der Erkennung von Segmentierungsbrüchen.

Beispiele

****Eingabe:****

```

[

{ "gs\_id": "B01", "context\_before": "", "sentence": "Wir entlasten Unternehmen von Bürokratie und stärken den Wettbewerb.", "context\_after": "" },

{ "gs\_id": "B02", "context\_before": "", "sentence": "Wasser und Energie gehören in öffentliche Hand.", "context\_after": "" },

{ "gs\_id": "B03", "context\_before": "", "sentence": "Wir stärken die Tarifautonomie und das Streikrecht der Gewerkschaften.", "context\_after": "" },

{ "gs\_id": "B04", "context\_before": "", "sentence": "Wir heben die Renten für langjährig Versicherte deutlich an.", "context\_after": "" },

{ "gs\_id": "B05", "context\_before": "", "sentence": "Menschen mit Behinderungen erhalten einen verbesserten Anspruch auf Eingliederungshilfe.", "context\_after": "" },

{ "gs\_id": "B06", "context\_before": "", "sentence": "Unsere Bäuerinnen und Bauern brauchen faire Preise und verlässliche Förderung.", "context\_after": "" },

{ "gs\_id": "B07", "context\_before": "", "sentence": "Wir wollen das Grundgesetz ändern, um den Klimaschutz als Staatsziel zu verankern.", "context\_after": "" },

{ "gs\_id": "B08", "context\_before": "In der Aussprache ging es um die Reform der Schuldenbremse.", "sentence": "Diese Grundgesetzänderung muss in diesem Hohen Haus eine Mehrheit finden.", "context\_after": "Wir appellieren an alle Fraktionen." },

{"gs\_id":"B09","context\_before":"","sentence":"Unser Land braucht endlich wieder eine handlungsfähige, stabile Regierung.","context\_after":""},

{"gs\_id":"B10","context\_before":"Wir investieren massiv in den sozialen Wohnungsbau.","sentence":"Ich möchte Ihnen das an zwei Beispielen verdeutlichen.","context\_after":"Erstens bauen wir 100.000 neue Sozialwohnungen pro Jahr."},

{"gs\_id":"B11","context\_before":"Sie verteilen hier Zahlen ohne Grundlage.","sentence":"Meine sehr verehrten Damen und Herren!","context\_after":"Lassen Sie uns zur Sache kommen."},

{"gs\_id":"B12","context\_before":"","sentence":"Unser Sozialstaat lässt immer mehr Menschen im Stich.","context\_after":""},

{"gs\_id":"B13","context\_before":"","sentence":"Die Sozialausgaben sind nicht mehr finanzierbar und müssen gesenkt werden.","context\_after":""},

{"gs\_id":"B14","context\_before":"","sentence":"Wer bei uns Schutz sucht, verdient ein faires Asylverfahren und eine echte Chance auf Integration.","context\_after":""},

{"gs\_id":"B15","context\_before":"","sentence":"Wir brauchen mehr Polizei und härtere Strafen gegen Kriminalität.","context\_after":""},

{"gs\_id":"B16","context\_before":"","sentence":"Nur ein geeintes und handlungsfähiges Europa kann sich behaupten.","context\_after":""},

{"gs\_id":"B17","context\_before":"","sentence":"Die Bundeswehr muss ihren Bündnisverpflichtungen nachkommen können.","context\_after":""},

{"gs\_id":"B18","context\_before":"","sentence":"Der Ausbau erneuerbarer Energien ist der Schlüssel zum Klimaschutz.","context\_after":""},

{"gs\_id":"B19","context\_before":"Die neue Förderregelung soll zum 1.", "sentence":"Januar 2025 in Kraft treten.", "context\_after":"Damit schaffen wir Planungssicherheit für Investoren."},

{"gs\_id":"B20","context\_before":"Wir verweisen ausdrücklich auf die EU-Verordnung Nr.", "sentence":"2019/1234 zur Lebensmittelsicherheit.", "context\_after":"Diese gilt unmittelbar in allen Mitgliedstaaten."}

]

...

**\*\*Ausgabe:\*\***

```

[

{"gs_id":"B01","marpor":"401","fehltrennung":false,"flag_unsure":false,"confidence":"medium","note":"Wettbewerb/Markt; 402 (Buerokratieabbau f. Unternehmen) denkbar"},

{"gs_id":"B02","marpor":"413","fehltrennung":false,"flag_unsure":false,"confidence":"high","note":""},

{"gs_id":"B03","marpor":"701","fehltrennung":false,"flag_unsure":true,"confidence":"medium","note":"Gewerkschaftsunterstuetzung; 405 Korporatismus denkbar"},

{"gs_id":"B04","marpor":"504","fehltrennung":false,"flag_unsure":false,"confidence":"high","note":"spezifische Sozialpolitik schlaegt 706"},

{"gs_id":"B05","marpor":"504","fehltrennung":false,"flag_unsure":false,"confidence":"medium","note":"spezifische Leistung schlaegt 705"},

{"gs_id":"B06","marpor":"703","fehltrennung":false,"flag_unsure":false,"confidence":"high","note":"Agrar-Sonderregel"},

{"gs_id":"B07","marpor":"501","fehltrennung":false,"flag_unsure":false,"confidence":"high","note":"Ziel schlaegt Mittel: Umwelt, nicht Verfassung"},

{"gs_id":"B08","marpor":"204","fehltrennung":false,"flag_unsure":false,"confidence":"medium","note":"Verfassungsaenderung selbst, kein uebergeordnetes Sachziel"},

{"gs_id":"B09","marpor":"305","fehltrennung":false,"flag_unsure":false,"confidence":"medium","note":"Staatsautoritaet, kein spezifischer Code"},

{"gs_id":"B10","marpor":"504","fehltrennung":false,"flag_unsure":false,"confidence":"medium","note":"Verbindungssatz erbt Wohnungsbau-Kontext, nicht 000"},

{"gs_id":"B11","marpor":"000","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":"wohlgeformte Anrede ohne Policy -> 000, NICHT Fehltrennung (Einheit intakt)"},

{"gs_id":"B12","marpor":"504","fehltrennung":false,"flag_unsure":false,"confidence":"medium",
"note":"Kritik am Ist-Zustand = pro Sozialstaat"}},

{"gs_id":"B13","marpor":"505","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":"explizit weniger -> Begrenzung"}},

{"gs_id":"B14","marpor":"607","fehltrennung":false,"flag_unsure":true,"confidence":"medium",
"note":"Integration/Vielfalt; 503 denkbar"}},

{"gs_id":"B15","marpor":"605","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":""}},

{"gs_id":"B16","marpor":"108","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":""}},

{"gs_id":"B17","marpor":"104","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":""}},

{"gs_id":"B18","marpor":"501","fehltrennung":false,"flag_unsure":false,"confidence":"high",
"note":""}},

{"gs_id":"B19","marpor":"000","fehltrennung":true,"flag_unsure":false,"confidence":"high",
note":"Datum '1. Januar' ueber Satzgrenze zerrissen -> Segmentierungsbruch"}},

```
{ "gs_id": "B20", "marpor": "000", "fehltrennung": true, "flag_unsure": false, "confidence": "high", "note": "Aktenzeichen 'Nr. 2019/1234' zerrissen, verwaister Fragmentsatz" }
```

```
]
```

```
'''
```

> Hinweis zum Kontrast B11 vs. B19/B20: alle drei ergeben `marpor = "000"`, aber B11 ist eine **intakte** Anrede (`fehltrennung = false`, echtes 000), während B19/B20 **zerrissene** Einheiten sind (`fehltrennung = true`). Downstream wird über das Flag entschieden, ob die Zeile als „kein Policy-Gehalt“ behalten oder als Datenfehler ausgeschlossen wird.

Appendix E: Code-to-Bucket Crosswalk

This appendix documents the complete assignment of the 56 MARPOR categories to the two aggregation schemes defined in Section 4.7.1. Table E.1 reports scheme A, which partitions all 56 codes into nine substantively defined buckets; every code is placed in exactly one bucket, so no probability mass is lost or duplicated in the aggregation of Section 4.7.2, and the row sums of the resulting indicator matrix equal one by construction. Table E.2 reports scheme B, which splits four domains into directional sub-buckets for the directional analysis. The remaining 32 codes fall into a residual category that drops out when the directional buckets are renormalised. Three of them — 406, 415 and 502 — lie inside the four domains under Aggregation A but carry no direction on the relevant axis, so scheme B is not a refinement of scheme A. Code 305 in particular maps to no directional bucket, which is why the 305 exclusion of Section 4.6.2 is immaterial on the B level. Category labels follow Handbook 4 (Volkens et al.); the assignment is defined once in the analysis configuration (00_config.R) and read by every downstream script.

Table E.1: Scheme A — partition of the 56 MARPOR categories into nine buckets.

Code	MARPOR category (Handbook 4)	Bucket (scheme A)
101	Foreign Special Relationships: Positive	Foreign Policy & Defence
102	Foreign Special Relationships: Negative	Foreign Policy & Defence
103	Anti-Imperialism	Foreign Policy & Defence

104	Military: Positive	Foreign Policy & Defence
105	Military: Negative	Foreign Policy & Defence
106	Peace	Foreign Policy & Defence
107	Internationalism: Positive	Foreign Policy & Defence
109	Internationalism: Negative	Foreign Policy & Defence
108	European Community/Union: Positive	European Integration
110	European Community/Union: Negative	European Integration
201	Freedom and Human Rights	Democracy & Political System
202	Democracy	Democracy & Political System
203	Constitutionalism: Positive	Democracy & Political System
204	Constitutionalism: Negative	Democracy & Political System
301	Decentralization	Democracy & Political System
302	Centralization	Democracy & Political System
303	Governmental and Administrative Efficiency	Democracy & Political System
304	Political Corruption	Democracy & Political System
305	Political Authority	Democracy & Political System
401	Free Market Economy	Economy
402	Incentives	Economy
403	Market Regulation	Economy
404	Economic Planning	Economy
405	Corporatism/Mixed Economy	Economy
406	Protectionism: Positive	Economy
407	Protectionism: Negative	Economy

408	Economic Goals	Economy
409	Keynesian Demand Management	Economy
410	Economic Growth	Economy
411	Technology and Infrastructure	Economy
412	Controlled Economy	Economy
413	Nationalisation	Economy
414	Economic Orthodoxy	Economy
415	Marxist Analysis	Economy
416	Anti-Growth Economy: Positive	Environment
501	Environmental Protection	Environment
502	Culture: Positive	Welfare & Social Policy
503	Equality: Positive	Welfare & Social Policy
504	Welfare State Expansion	Welfare & Social Policy
505	Welfare State Limitation	Welfare & Social Policy
506	Education Expansion	Welfare & Social Policy
507	Education Limitation	Welfare & Social Policy
603	Traditional Morality: Positive	Law & Order and National Identity
604	Traditional Morality: Negative	Law & Order and National Identity
605	Law and Order: Positive	Law & Order and National Identity
606	Civic Mindedness: Positive	Law & Order and National Identity

601	National Way of Life: Positive	Migration
602	National Way of Life: Negative	Migration
607	Multiculturalism: Positive	Migration
608	Multiculturalism: Negative	Migration
701	Labour Groups: Positive	Social Groups
702	Labour Groups: Negative	Social Groups
703	Agriculture and Farmers	Social Groups
704	Middle Class and Professional Groups	Social Groups
705	Underprivileged Minority Groups	Social Groups
706	Non-economic Demographic Groups	Social Groups

Note: Exact partition: each of the 56 codes appears exactly once. Codes are listed in MARPOR numerical order within buckets.

Table E.2: Scheme B — directional sub-buckets within four domains.

Domain	Directional (scheme B)	bucket MARPOR codes
Economy	Market liberalism	401 (Free Market Economy); 402 (Incentives); 407 (Protectionism: Negative); 414 (Economic Orthodoxy)
Economy	State intervention	403 (Market Regulation); 404 (Economic Planning); 405 (Corporatism/Mixed Economy); 409 (Keynesian Demand Management); 412 (Controlled Economy); 413 (Nationalisation)
Economy	General economy (residual)	408 (Economic Goals); 410 (Economic Growth); 411 (Technology and Infrastructure)
Welfare	Welfare expansion	503 (Equality: Positive); 504 (Welfare State Expansion); 506 (Education Expansion)

Welfare	Welfare retrenchment	505 (Welfare State Limitation); 507 (Education Limitation)
Migration	Migration restrictive	601 (National Way of Life: Positive); 608 (Multiculturalism: Negative)
Migration	Migration liberal	602 (National Way of Life: Negative); 607 (Multiculturalism: Positive)
Europe	Pro-EU	108 (European Community/Union: Positive)
Europe	Contra-EU	110 (European Community/Union: Negative)

Note: Scheme B covers 24 of the 56 codes. All remaining codes form a residual category (Andere) that is dropped under within-domain renormalisation. The bucket labels in the analysis code carry German names (Marktliberalismus, Staatsintervention, Wirtschaft Allgemein, Sozialstaat Ausbau, Sozialstaat Begrenzung, Migration restriktiv, Migration liberal, Pro-EU, Contra-EU); the manuscript uses the English forms shown here throughout.

Appendix F: Supplementary Results and Robustness Tables

This appendix collects the full-resolution results and the robustness evidence summarised in Chapter 5. Section F.1 documents the redundancy test of the two corrections (Section 4.6.2). Sections F.2 to F.10 report, in order: every H1a cell with its bootstrap interval (F.2); the six-band decomposition of the code-305 correction (F.3); the nearest-manifesto benchmark (F.4) and the between-party reference scale (F.5); the temperature and threshold sweeps (F.6, F.7); the migration re-split (F.8); the supplementary directional figures (F.9); and the LP-13 diagnostic (F.10). Unless stated otherwise every quantity uses the primary specification — full filter, code 305 mass subtracted (Aggregation A) or retained (Aggregation B), native decoding temperature — and every interval is a 1,000-draw non-parametric bootstrap (per-cell seeds derived from the base seed 20260507). The intervals are reported as lower bounds on total uncertainty and are not read as significance tests.

F.1 Redundancy Test of the Two Corrections

This appendix documents the comparison objects and derived quantities of the redundancy test reported in Section 4.6.2. The measured quantity is the mean per-cell contribution of the Democracy & Political System bucket to the manifesto–speech Jensen–Shannon divergence,

computed under Aggregation A with native temperature across the 38 valid party–LP cells. The four objects vary the two intervention axes: the filter stage (unfiltered, procedural filter, procedural and null-class filter) and the treatment of code 305 (included, excluded). Source: `intervention_000_vs_305_redundancy.csv`, produced by the robustness script (Part 3).

F.1.1 Comparison objects

Object	Specification (filter × code 305)	Role	Description	Mean DPS JSD contribution
D_raw	unfiltered × incl	Inflated baseline	No procedural filter, no null-class filter, code 305 included.	0.0387
B_old	filtered × excl	Old primary	Procedural filter with code 305 excluded, before the null-class filter was introduced.	0.0086
A_new	filtered_000 × incl	Filter only	Procedural and null-class filter applied, code 305 kept. Isolates what the null-class filter achieves on its own.	0.0236
C_dbl	filtered_000 × excl	Declared primary	Both corrections applied: null-class filter and symmetric exclusion of code 305.	0.0062

F.1.2 Derived quantities and pre-stated criteria

The exclusion would have been judged redundant if both pre-stated criteria had been met: an additional removal share below 15% on top of the null-class filter, and agreement of the filter-only specification with the old corrected primary to within 15% of the correction magnitude. The second criterion asks whether the null-class filter on its own already lands where the earlier, 305-corrected primary landed; if it did, the exclusion would add nothing. The criteria were declared before the results were known (Section 4.2).

Quantity	Definition	Value
Removed by the null-class filter alone	$D_raw - A_new$	0.0151
Removed by the 305-exclusion on top of the filter	$A_new - C_dbl$	0.0174
Additional share removed by the exclusion	$(A_new - C_dbl) / (D_raw - A_new)$	115.2% (redundancy threshold: below 15%)

Quantity	Definition	Value
Convergence of the filter-only specification and the old primary	$ A_{\text{new}} - B_{\text{old}} / D_{\text{raw}} - B_{\text{old}} $	49.8% (threshold: below 15%)
Verdict	Both criteria	Not redundant. The exclusion remains the primary correction.

Note. Contributions are taken from `intervention_000_vs_305_redundancy.csv` (rounded to four decimals there); the two shares are computed from these rounded values. C_{dbl} below A_{new} by construction does not indicate over-correction as long as the reduction reflects the removal of 305 mass on both sides; the per-bucket decomposition underlying this table is reported in `intervention_per_bucket_long.csv`.

F.2 Per-cell salience divergence (H1a) with bootstrap intervals

Table F.2.1 lists the salience-level (Aggregation A) manifesto–speech Jensen–Shannon divergence for each of the 38 party $\times$ legislative-period cells that enter H1a, together with its 95 % bootstrap interval and the number of speech sentences on which the cell rests. The cells are ordered by party and legislative period. The FDP is absent from LP 18 and the smaller parties enter only once they hold parliamentary seats, which is why the panel is unbalanced. The single largest value, the CDU/CSU cell at LP 13 (0.1125), and the pattern by which LP 13 is every party's maximum are examined in F.10.

Party	LP	JSD	95 % low	95 % high	Sentences (n)
CDU/CSU	13	0.1125	0.1116	0.1134	186,508
CDU/CSU	14	0.0377	0.0371	0.0383	160,735
CDU/CSU	15	0.0241	0.0235	0.0246	144,595
CDU/CSU	16	0.0235	0.0231	0.0240	183,016
CDU/CSU	17	0.0189	0.0185	0.0192	279,947
CDU/CSU	18	0.0268	0.0264	0.0273	255,663
CDU/CSU	19	0.0426	0.0421	0.0431	229,032
CDU/CSU	20	0.0195	0.0191	0.0200	158,802

Party	LP	JSD	95 % low	95 % high	Sentences (n)
SPD	13	0.0596	0.0588	0.0605	141,711
SPD	14	0.0383	0.0378	0.0390	210,661
SPD	15	0.0150	0.0146	0.0154	154,903
SPD	16	0.0215	0.0210	0.0219	189,120
SPD	17	0.0202	0.0197	0.0206	176,714
SPD	18	0.0284	0.0278	0.0289	173,434
SPD	19	0.0236	0.0231	0.0242	153,494
SPD	20	0.0334	0.0328	0.0340	174,767
FDP	13	0.0402	0.0394	0.0412	79,052
FDP	14	0.0114	0.0108	0.0120	63,566
FDP	15	0.0220	0.0210	0.0229	45,480
FDP	16	0.0134	0.0129	0.0140	90,192
FDP	17	0.0125	0.0121	0.0129	135,368
FDP	19	0.0292	0.0284	0.0301	80,049
FDP	20	0.0213	0.0207	0.0220	86,007
Grüne	13	0.0743	0.0729	0.0757	62,126
Grüne	14	0.0244	0.0237	0.0252	92,079
Grüne	15	0.0324	0.0315	0.0334	63,320
Grüne	16	0.0137	0.0131	0.0142	83,434
Grüne	17	0.0161	0.0156	0.0166	105,587
Grüne	18	0.0306	0.0299	0.0314	101,791
Grüne	19	0.0234	0.0227	0.0242	80,115
Grüne	20	0.0231	0.0225	0.0237	107,695
Linke	16	0.0214	0.0207	0.0221	83,410
Linke	17	0.0159	0.0153	0.0164	110,325

Party	LP	JSD	95 % low	95 % high	Sentences (n)
Linke	18	0.0402	0.0393	0.0411	95,911
Linke	19	0.0331	0.0322	0.0340	75,403
Linke	20	0.0219	0.0210	0.0230	41,562
AfD	19	0.0338	0.0331	0.0347	91,207
AfD	20	0.0273	0.0265	0.0281	76,761

Table F.2.1. Per-cell salience divergence (H1a), primary specification. JSD between each party's legislative-period speech distribution and its nearest manifesto over the nine salience buckets, with 95 % bootstrap interval (1000 draws) and speech sentence count. Aggregation A, full filter, code 305 subtracted, native temperature.

F.3 The code-305 correction: six specification bands (robustness arm 3)

The mass carried by code 305 (political authority / competence claims) is a genre feature of parliamentary speech rather than a policy position, so the primary specification subtracts it on the salience side. Table F.3.1 crosses the three speech-filter stages (no filter, procedural filter, full filter) with the two code-305 treatments (retained, subtracted) and reports, for each of the six bands, the mean per-cell JSD, the mean 95 % interval width, and the share of the remaining divergence carried by the Democracy & Political System bucket. The subtraction lowers the mean divergence at every filter stage, by 53.4 % without a filter, 49.7 % under the procedural filter and 41.8 % under the full filter. That the effect shrinks as the filter tightens is the complementarity of the two corrections seen from this side, since the null-class filter already removes part of the 305 excess and leaves less for the exclusion to take. In the primary band the Democracy & Political System bucket still carries 20.9 % of the divergence, the largest share of any bucket (Table 5.1).

Speech filter	Code 305	Mean JSD	Mean CI width	DPS share
no filter	incl	0.0715	0.0017	54.0%
no filter	excl	0.0333	0.0013	28.6%
procedural filter	incl	0.0641	0.0017	52.1%
procedural filter	excl	0.0322	0.0013	26.7%
full filter	incl	0.0510	0.0016	46.3%
full filter	excl	0.0297	0.0013	20.9%

Table F.3.1. Mean per-cell divergence across the six filter $\times$ code-305 bands. Mean JSD and mean 95 % bootstrap-interval width over the 38 H1a cells; "DPS share" is the share of the mean divergence carried by the Democracy & Political System bucket. Aggregation A. The primary specification is the final row (full filter, excl.).

Figures F.3.1 to F.3.4 unpack the correction. Figure F.3.1 shows the per-code contribution to the raw divergence, isolating code 305 as the dominant single contributor; Figure F.3.2 shows the asymmetry of code 305 between the manifesto and speech sides that motivates subtracting rather than down-weighting it. Figures F.3.3 and F.3.4 place the code-305 subtraction on the intervention ladder, first as its effect on the divergence-per-share and then across all nine buckets, confirming that no other bucket is disturbed by the correction. Figure F.3.5 applies the same diagnostic to the other correction. Measured on the unnormalised bucket masses, the null-class filter removes a quarter of the Democracy & Political System bucket against an overall rate of 14.5% across the analysis sample, and no other bucket lies above that rate. In the divergence decomposition the contribution of every bucket falls except one: Foreign Policy & Defence gains 54.8%, although the filter removes less of its mass than of almost any other bucket. The mass figures resolve the apparent contradiction. Because the removal falls disproportionately on a single bucket, every remaining bucket gains share as the denominator shrinks, Foreign Policy & Defence by 10.5%, and since it is one of the two buckets that speech already over-weights relative to the manifesto (Section 5.3.1), a higher speech share widens the gap rather than closing it. The rise therefore records renormalisation acting on the second half of the genre signature, not a loss of substantive content.

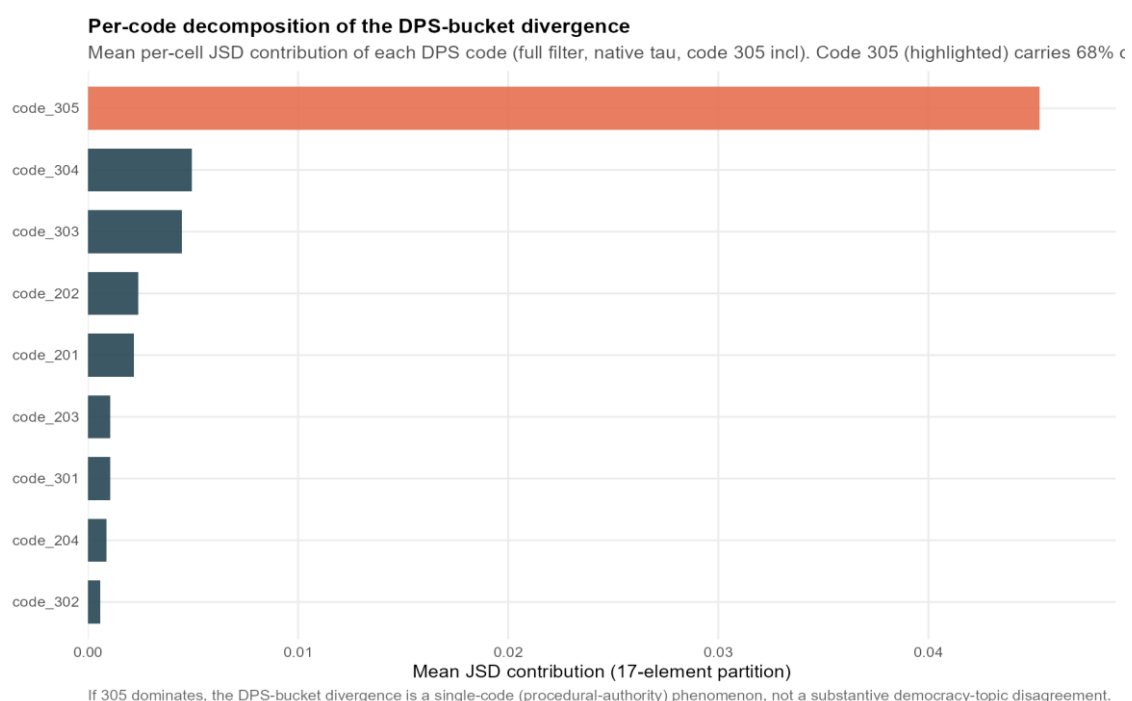


Figure F.3.1. Divergence-per-share by MARPOR code. Contribution of each code to the raw manifesto–speech divergence, ranked; code 305 is the single largest contributor on the speech side.

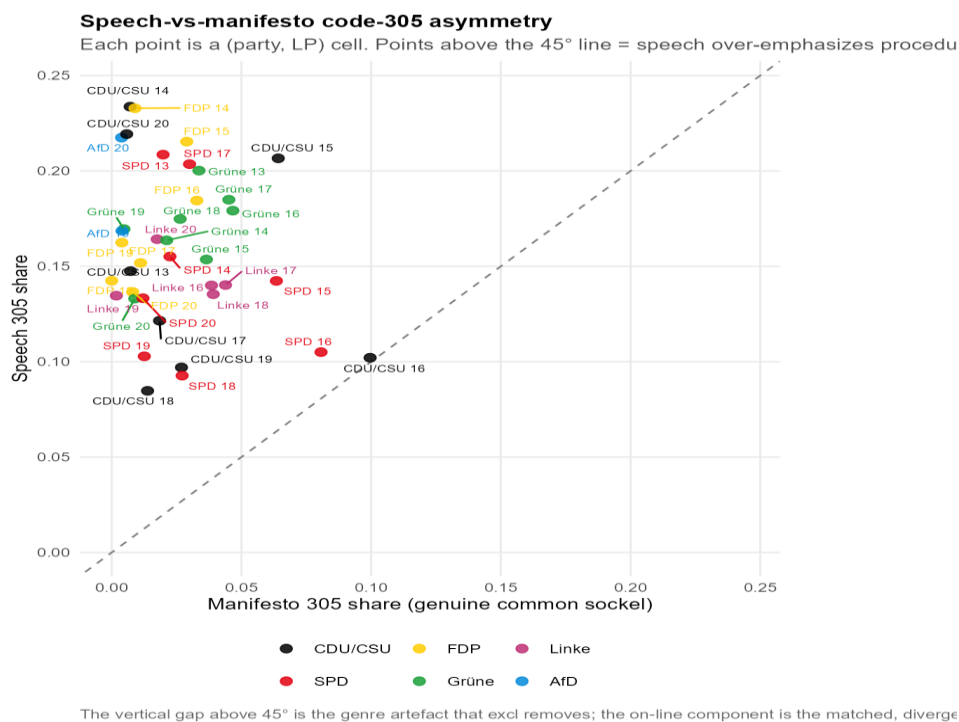


Figure F.3.2. Code-305 asymmetry between sides. Manifesto versus speech share of code 305 per cell; the systematic speech-side excess is the genre signal removed by the primary specification.

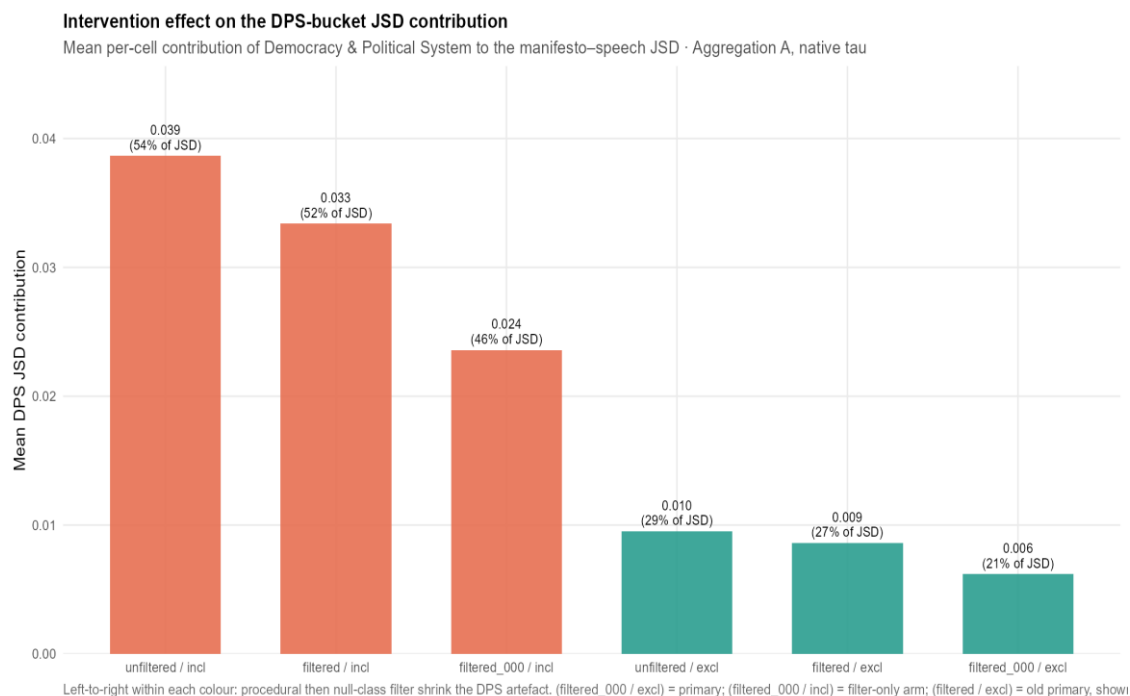


Figure F.3.3. Divergence-per-share along the intervention ladder. Effect of each successive correction (filtering, then code-305 subtraction) on the mean divergence-per-share.

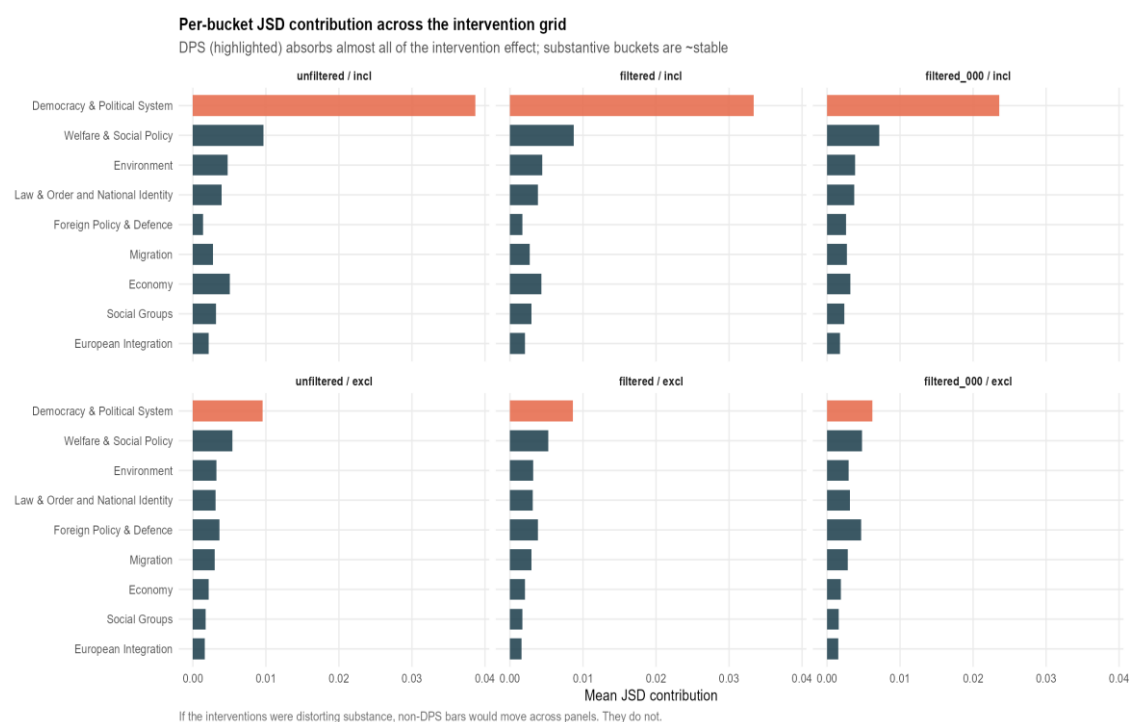


Figure F.3.4. Bucket-wise effect of the code-305 subtraction. Change in each of the nine salience buckets' mean share under the correction; only the code-305-bearing bucket moves materially.

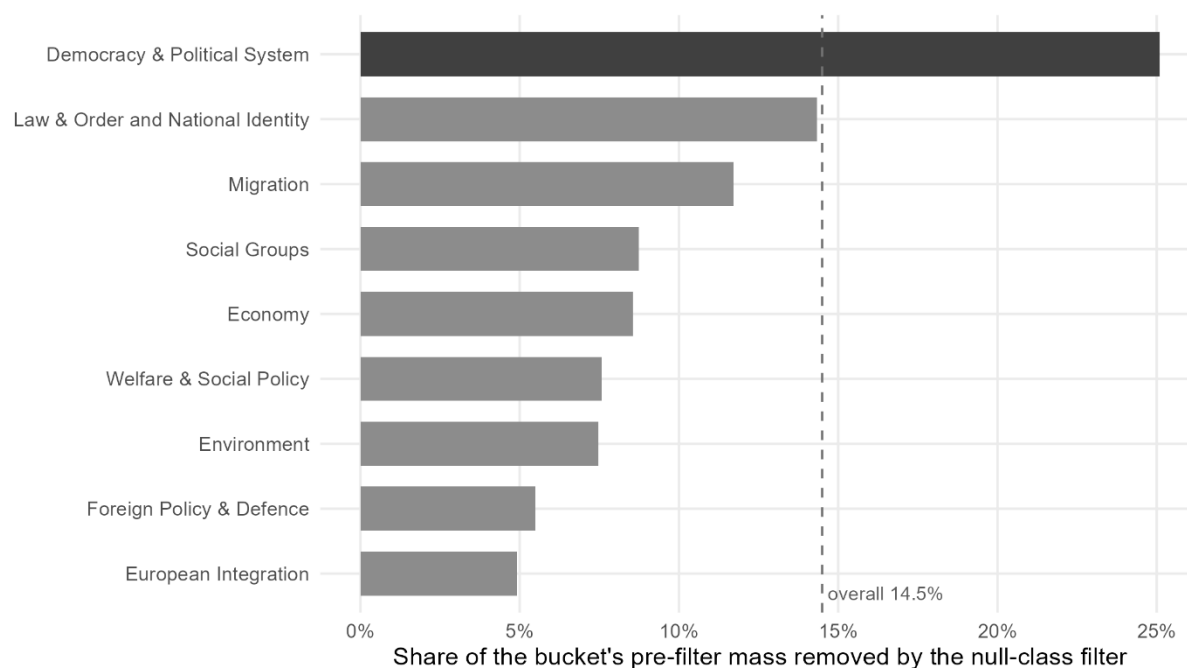


Figure F.3.5. Bucket-wise effect of the null-class filter. Share of each bucket's unnormalised mass removed when the null-class filter is added to the procedural filter, across the 38 cells of the analysis sample, against the overall removal rate of 14.5% as a reference line; code 305 is retained throughout so that the filter alone is at work. The Democracy & Political System bucket loses 25.1% of its mass, 10.6 percentage points above the overall rate and the only bucket above it. The rate is computed on the already procedurally filtered corpus, which is why it sits below the 19.34% the filter flags on the full corpus (Section 4.6.1).

F.4 Nearest-manifesto benchmark (identification of a party's own programme)

The nearest-manifesto benchmark asks, for each speech cell, whether the party's own manifesto is the closest of all parties' manifestos in the same period. Table F.4.1 reports, per party, how often a party's own manifesto is nearest, that hit rate, and the mean rank of that manifesto (1 = nearest). Across all 38 cells a party's own manifesto is nearest in 16 (42.1 %, against a 21.1 % chance baseline, the mean of one over the field size in each cell, with fields ranging from four to six parties) and its mean rank is 1.87. The FDP and AfD identify their own programme in every period; the Linke never does, reflecting its position near the field centroid. The full per-cell test behind these aggregates is reported in Table F.4.3.

Party	Own nearest	Hit rate	Mean rank
CDU/CSU	2 / 8	25.0%	2.50
SPD	1 / 8	12.5%	2.38
FDP	7 / 7	100.0%	1.00
Grüne	4 / 8	50.0%	1.62
Linke	0 / 5	0.0%	2.00
AfD	2 / 2	100.0%	1.00
Total	16 / 38	42.1%	1.87

Table F.4.1. Nearest-manifesto identification by party. Cells in which the party's own manifesto is the closest of all parties' manifestos, the corresponding hit rate, and the mean rank of that manifesto (1 = nearest). Aggregation A, primary specification. Chance baseline 21.1 %.

The complementary question, taken up as Figure 5.3 in the main text, is which single programme the chamber's agenda sits closest to across all parties. Table F.4.2 gives the supporting quantity: the mean divergence from every one of the 38 speech cells to each party's manifesto in the same period, both over all pairs and with each party's distance to its own speech removed, alongside the number of cells in which that manifesto is the nearest of all. The FDP programme is the closest to the pool of parliamentary speech by a clear margin, and stays first when own-party pairs are excluded, so the result is neither self-similarity nor an effect of the FDP appearing often as a candidate. The mechanism is one of salience, not position: the chamber's agenda is comparatively heavy on governance and economy and light on welfare and migration, which is the shape of the FDP's programmatic agenda, while the welfare-centred Linke and the migration-centred AfD sit farthest. This is discussed in the conclusion; it is a statement about which topics are emphasised, not about the positions taken on them.

Manifesto	Mean JSD (all)	Mean JSD (excl. own)	Times nearest
FDP	0.0257	0.0267	18
SPD	0.0379	0.0400	6
Grüne	0.0396	0.0422	7
CDU/CSU	0.0463	0.0484	5
Linke	0.0524	0.0564	0
AfD	0.0638	0.0705	2

Table F.4.2. *Agenda centre of gravity. For each party's manifesto: the mean Jensen–Shannon divergence from each of the 38 speech cells to that manifesto in the same period (n = 38 for every party except the AfD, n = 12) (lower = the programme the chamber's agenda most resembles), the same excluding each party's distance to its own speech, and the number of cells in which that manifesto is the nearest of all. Manifesto candidates are restricted to parties in parliament that period. Aggregation A, primary specification. Salience, not position.*

Party	LP	Candidate manifestos	Nearest manifesto	Rank of own manifesto
CDU/CSU	13	4	FDP	4
CDU/CSU	14	4	FDP	3
CDU/CSU	15	4	SPD	3
CDU/CSU	16	5	FDP	3
CDU/CSU	17	5	FDP	2
CDU/CSU	18	4	CDU/CSU	1
CDU/CSU	19	6	SPD	3
CDU/CSU	20	6	CDU/CSU	1
SPD	13	4	FDP	2
SPD	14	4	FDP	3
SPD	15	4	FDP	2
SPD	16	5	Grüne	3
SPD	17	5	FDP	2
SPD	18	4	CDU/CSU	2
SPD	19	6	SPD	1

Party	LP	Candidate manifestos	Nearest manifesto	Rank of own manifesto
SPD	20	6	CDU/CSU	4
FDP	13	4	FDP	1
FDP	14	4	FDP	1
FDP	15	4	FDP	1
FDP	16	5	FDP	1
FDP	17	5	FDP	1
FDP	19	6	FDP	1
FDP	20	6	FDP	1
Grüne	13	4	FDP	3
Grüne	14	4	FDP	2
Grüne	15	4	SPD	2
Grüne	16	5	Grüne	1
Grüne	17	5	Grüne	1
Grüne	18	4	Grüne	1
Grüne	19	6	Grüne	1
Grüne	20	6	CDU/CSU	2
Linke	16	5	Grüne	2
Linke	17	5	Grüne	2
Linke	18	4	SPD	2
Linke	19	6	SPD	2
Linke	20	6	FDP	2
AfD	19	6	AfD	1
AfD	20	6	AfD	1

Table F.4.3. Nearest-manifesto test per cell, primary specification (scheme A, full filter, code 305 excluded, native temperature).

Note: 38 valid cells (Section 4.8). Rank 1 denotes the nearest manifesto; ties take the minimum rank. A party's own manifesto ranks first in 16 of 38 cells (42.1%), and its mean rank is 1.87, and the rank distribution is 16/13/7/2 for ranks 1 to 4; the rank is never worse than 4. The derived chance baselines follow from the candidate counts: sixteen cells offer four candidates, ten cells five, and twelve cells six, which yields a random hit probability of $\text{mean}(1/n) = 8/38 = 21.1\%$ and an expected own-manifesto rank of $\text{mean}((n+1)/2) = 2.95$ (Section 4.8.1).

F.5 Pairwise between-party reference scale

The reference scale that calibrates H1a is built from between-party divergences on each side. Table F.5.1 reports the distribution of pairwise JSDs among speeches (the within-genre floor) and among manifestos (the cross-programme ceiling). The speech floor (mean 0.0074) and manifesto ceiling (mean 0.0520) bracket the observed H1a mean of 0.0297, which sits at 4.01 times the floor and 0.57 times the ceiling, that is, 49.9 % of the way along the floor-to-ceiling interval. The manifesto–speech divergence is therefore intermediate between the two reference regimes rather than small.

Side	Mean	Median	Q25	Q75	Min	Max	n
Speech (floor)	0.0074	0.0063	0.0034	0.0093	0.0011	0.0303	74
Manifesto (ceiling)	0.0520	0.0420	0.0281	0.0662	0.0082	0.1760	74

Table F.5.1. *Between-party pairwise divergence, by side. Distribution of pairwise JSDs among all party speeches (floor) and among all party manifestos (ceiling), pooled across periods. Aggregation A, primary specification. n = number of party pairs × periods.*

F.6 Decoding-temperature sweep (robustness arm 1)

The primary specification decodes the classifier at its native temperature. Table F.6.1 re-runs H1a and H1b at a sharp ($\tau = 0.5$), native ($\tau = 1.0$) and flat ($\tau = 2.0$) temperature and reports the Spearman rank correlations between the three resulting cell orderings, alongside the mean divergence level at each temperature. The cross-temperature rank correlations remain high for H1a (0.579 to 0.970) and for H1b (0.706 to 0.956); the sharp-to-native correlation, the pair that brackets the native specification, is 0.970 for H1a and 0.956 for H1b. Temperature shifts the level of the divergence but preserves the ordering of cells that carries the substantive claims.

Hypothesis	sharp–native	sharp–flat	native–flat
H1a (A)	0.970	0.579	0.729
H1b (B)	0.956	0.706	0.843

Table F.6.1a. *Rank stability across decoding temperature. Spearman ρ between cell orderings at sharp ($\tau = 0.5$), native ($\tau = 1.0$) and flat ($\tau = 2.0$) temperatures, for H1a (Aggregation A) and H1b (Aggregation B).*

Hypothesis	sharp (0.5)	native (1.0)	flat (2.0)
H1a (A)	0.0283	0.0297	0.0368
H1b (B)	0.0444	0.0506	0.0852

Table F.6.1b. Mean divergence level across decoding temperature. Mean per-cell JSD at each temperature; the native column reproduces the primary specification.

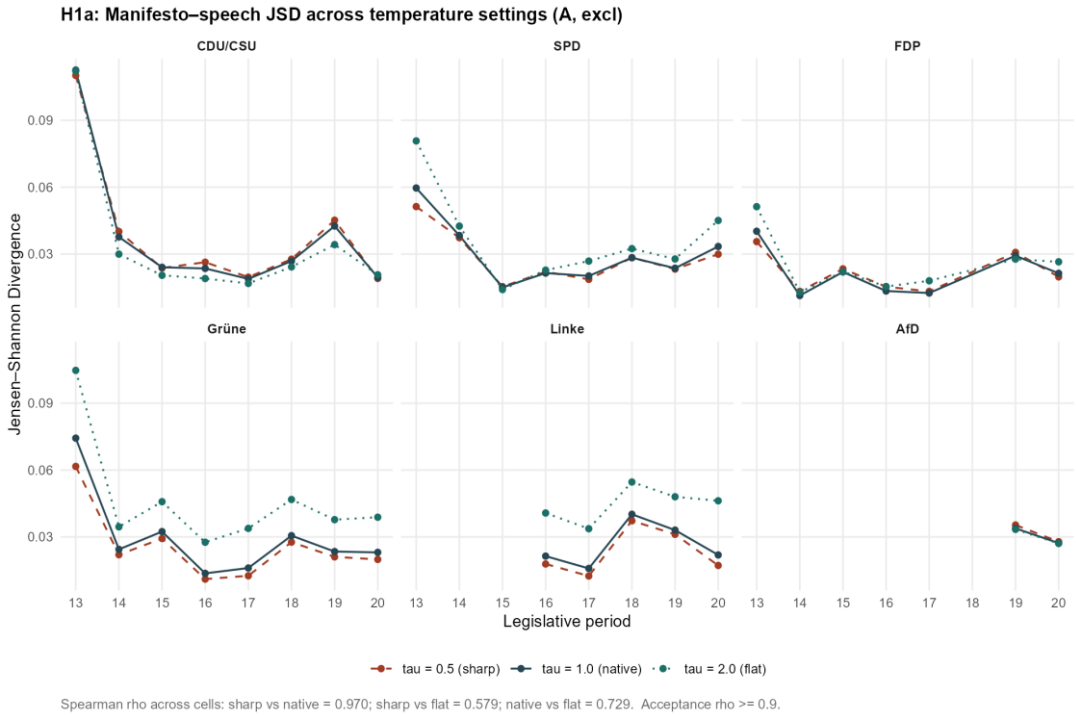


Figure F.6.1. H1a divergence across decoding temperature. Per-cell JSD at each temperature; cell orderings are preserved as the level shifts.

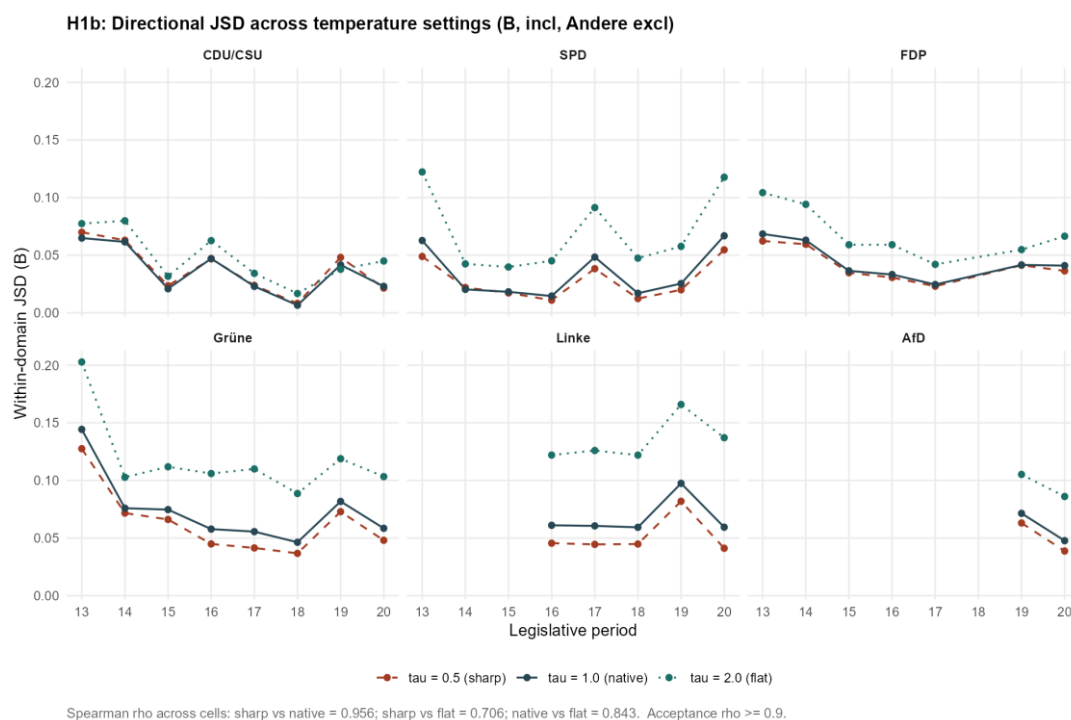


Figure F.6.2. H1b divergence across decoding temperature. As Figure F.6.1, for the directional specification (Aggregation B).

F.7 Soft-assignment versus hard-threshold classification (robustness arm 2)

The primary specification uses the classifier's soft label distribution. Table F.7.1 compares it against two hard-threshold variants that assign each sentence to a single bucket when the top label exceeds 0.4 or 0.5. The rank correlations between the soft ordering and each hard ordering stay high (H1a 0.929 to 0.943; H1b 0.772 to 0.861), and the two hard variants agree almost perfectly with each other (0.996 for H1a, 0.964 for H1b). The mean level rises modestly under hard assignment, as expected when soft mass is collapsed onto single buckets, without reordering the cells.

Hypothesis	soft-hard0.4	soft-hard0.5	hard0.4-hard0.5
H1a (A)	0.943	0.929	0.996
H1b (B)	0.861	0.772	0.964

Table F.7.1a. Rank stability under hard-threshold assignment. Spearman ρ between the soft-assignment ordering and each hard-threshold ordering (top label > 0.4 , > 0.5), and between the two hard variants. H1a and H1b.

Hypothesis	soft ($\tau = 1.0$)	hard > 0.4	hard > 0.5
H1a (A)	0.0510	0.0577	0.0602
H1b (B)	0.0506	0.0411	0.0409

Table F.7.1b. Mean divergence level under hard-threshold assignment. Mean per-cell JSD for the soft specification and each hard-threshold variant.

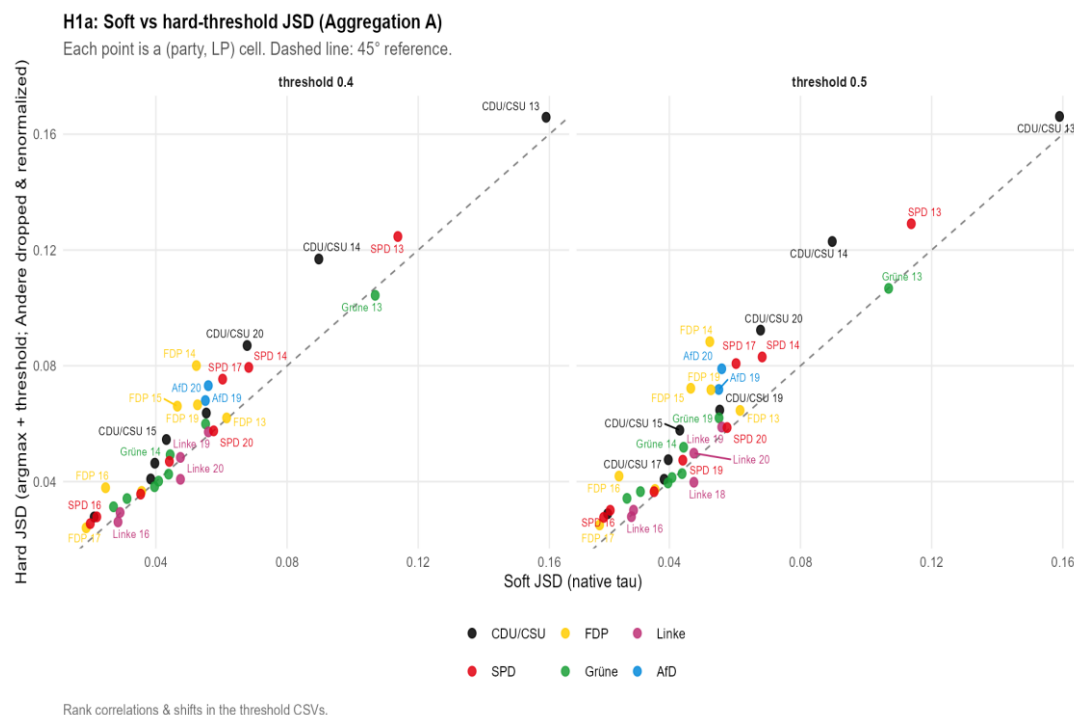


Figure F.7.1. H1a: soft versus hard-threshold divergence. Per-cell JSD under soft assignment against the hard-threshold variants; points track the identity line up to a level shift.

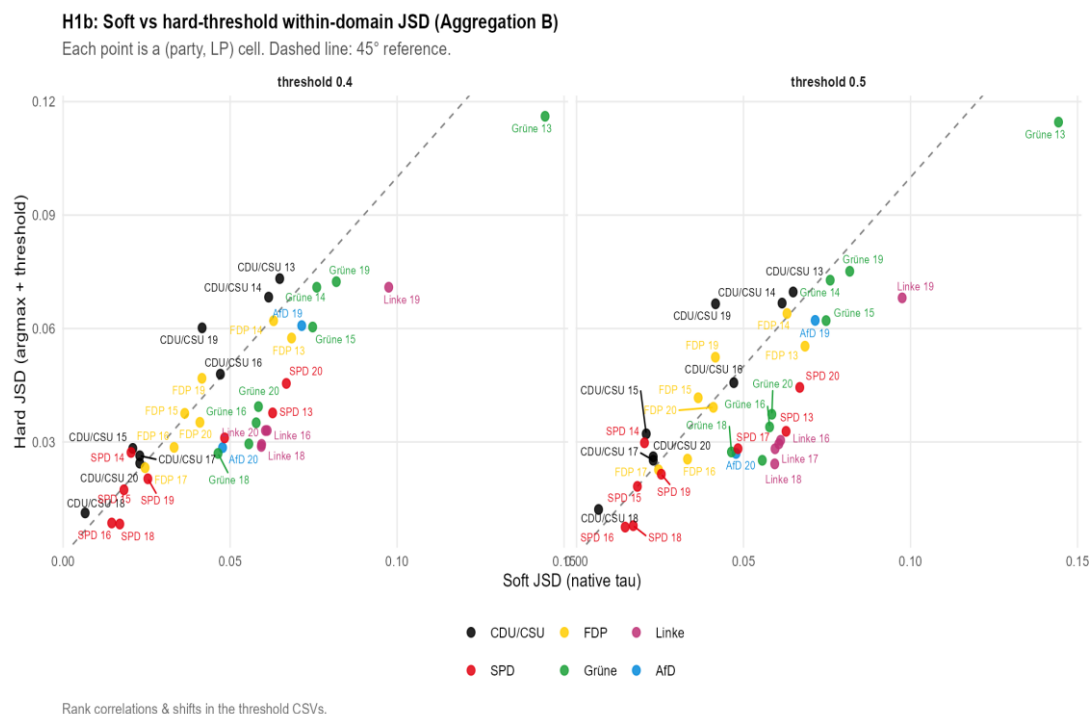


Figure F.7.2. H1b: soft versus hard-threshold divergence. As Figure F.7.1, for the directional specification.

F.8 Migration re-split (H1b, multiculturalism-only axis 608 vs 607; robustness arm 5)

The directional migration bucket pools the MARPOR national-way-of-life and multiculturalism codes. Because the split between border/immigration control and cultural integration is contested, Table F.8.1 re-runs the H1b migration cell two ways: the baseline restrictive/liberal split (601+608 restrictive, 602+607 liberal) and a narrower split that treats only multiculturalism as the axis (608 restrictive, 607 liberal). The mean divergence is essentially unchanged (0.1392 to 0.1344, a difference of -0.0062), and the restrictive share on the speech side falls from 68.4 % to 48.6 %, which localises the composition finding to the multiculturalism codes without overturning it.

Split	Cells (n)	Mean JSD	Restr. share (speech)
Baseline (601+608 / 602+607)	37	0.1392	68.4%
Multiculturalism only (608 / 607)	36	0.1344	48.6%

Table F.8.1. Migration re-split sensitivity (H1b). Mean migration-cell JSD and restrictive speech share under the baseline split (601+608 / 602+607) and the multiculturalism-only split (608 / 607). Δ mean JSD = -0.0062 , computed as the matched mean change across the 36 cells defined under both splits; the difference of the two column means is -0.0048 because they rest on 37 and 36 cells respectively. Aggregation B, primary specification.

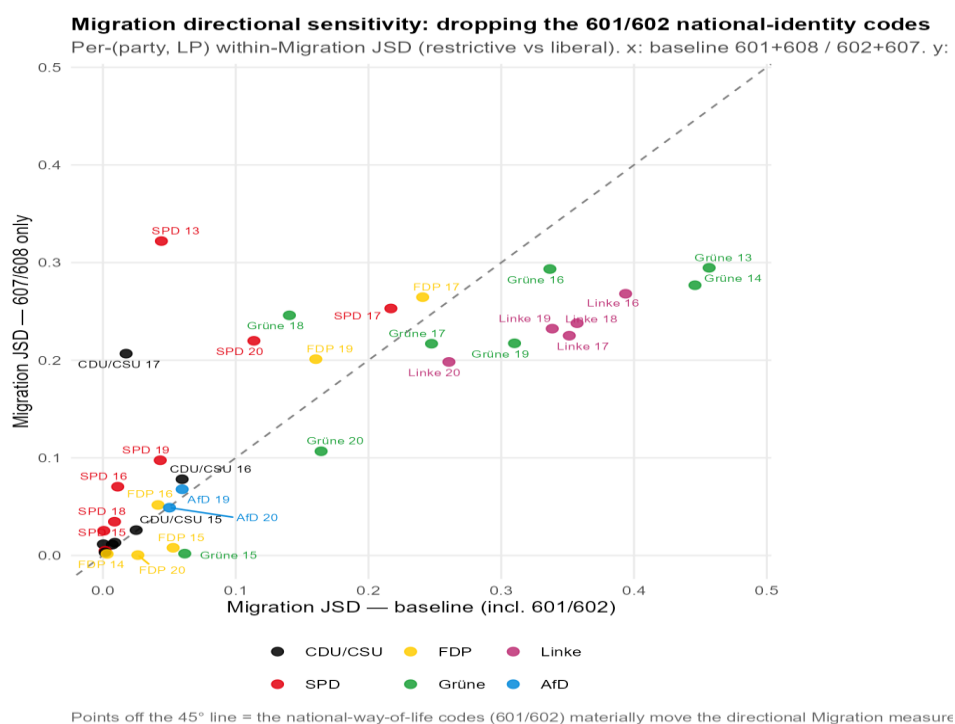


Figure F.8.1. Migration re-split sensitivity, per cell. Migration-cell JSD under the baseline and multiculturalism-only splits; the cell ordering is preserved and the level barely moves.

F.9 Supplementary directional figures

Four further figures support the directional analysis in Chapter 5. Figure F.9.1 is the H2b moderation scatter with adjusted scaling so that all dots-party pairs are readable, plotting each cell's speech-side against its manifesto-side polarisation score so that the slope comparison of H2b can be read cell by cell: clouds flatter than the 45-degree line indicate moderation, and a cloud displaced from the line indicates a level shift of the channel. Figure F.9.2 shows the H3 mean directional positions, the manifesto and speech means on each directional sub-scale over time. Figure F.9.3 reproduces the system-level polarisation series of Figure 5.9 without the excl-AfD overlay; the underlying values are identical, and the figure is provided so the all-party series can be read uncluttered. Figure F.9.4 disaggregates the migration domain to the party level and puts the weighted mean beside the weighted dispersion, which separates a common shift of the whole chamber from a widening between the parties.

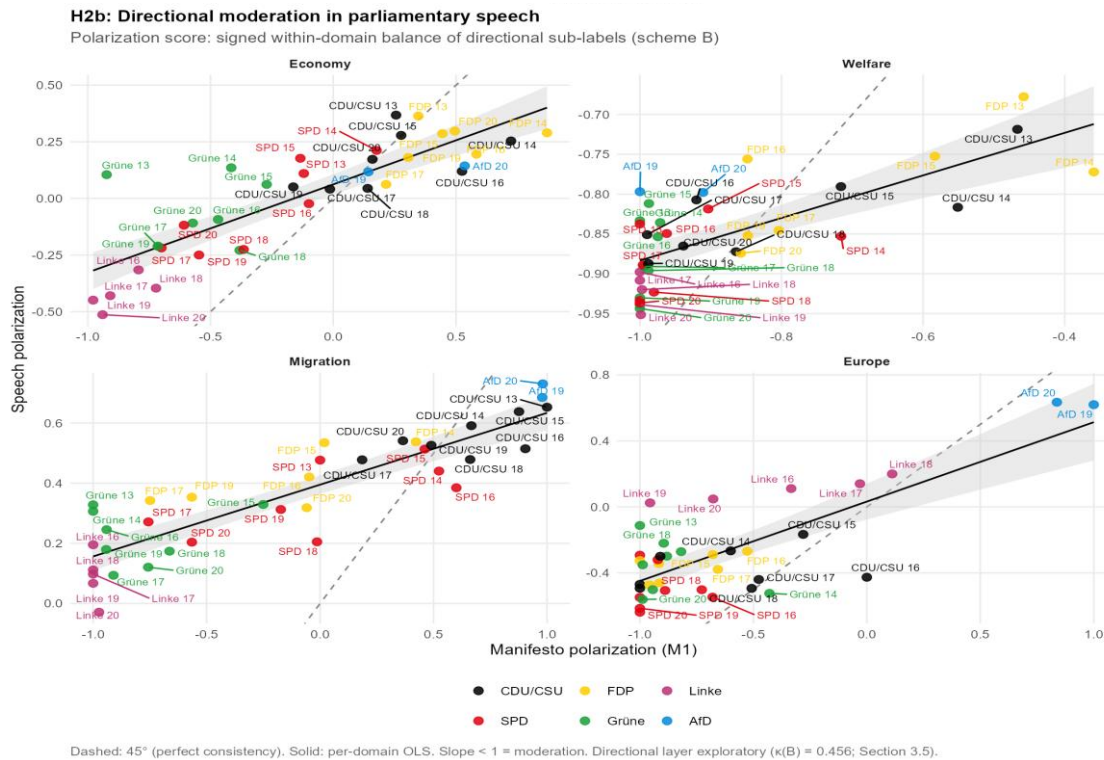


Figure F.9.1. H2b: speech versus manifesto polarisation scores. Each point is a party $\times$ period cell; axes are the signed within-domain directional balance in each channel, dashed line = 45° (perfect consistency), solid line = per-domain OLS. Aggregation B, primary specification. Scales are adjusted to make the annotation of the dots readable.

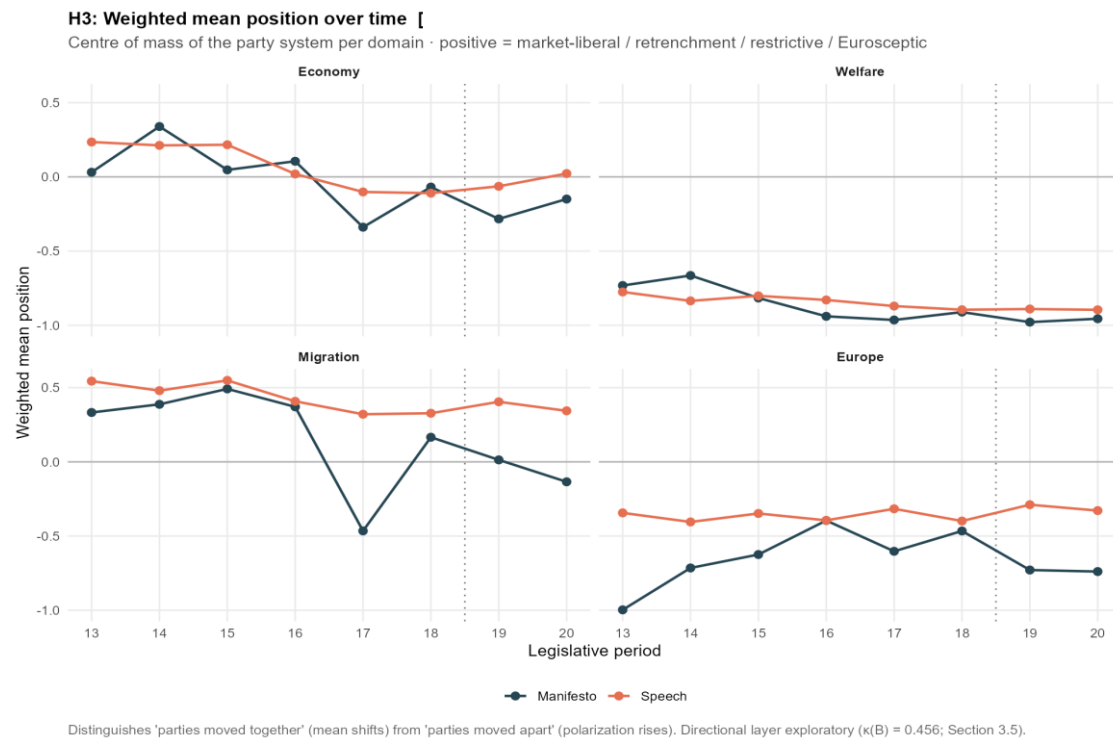


Figure F.9.2. H3: mean directional positions over time. Weighted mean position per legislative period, domain and channel, manifesto weighted by vote share and speech by seat share. The mean is the centre of the dispersion reported in Section 5.4 and separates a common shift of the whole chamber from a widening between parties. Aggregation B, primary specification. Descriptive series.

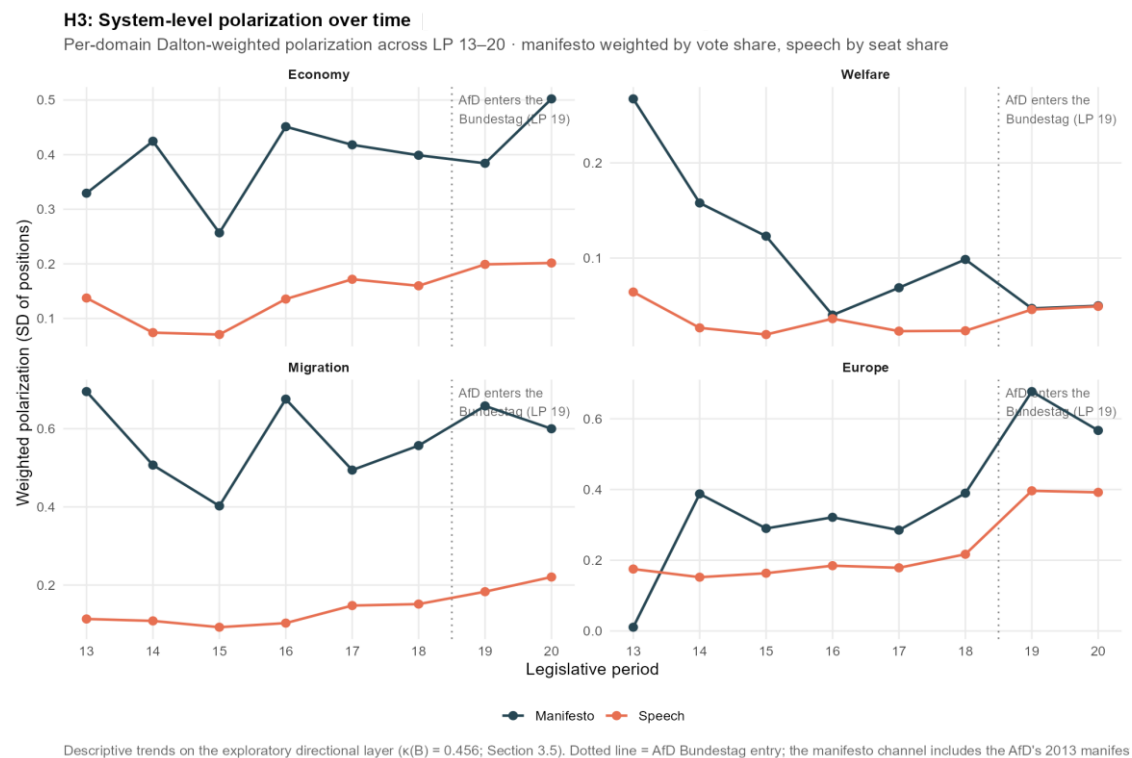


Figure F.9.3. System-level polarisation over time, all parties. Dalton-weighted SD of party positions per period, domain and channel (manifesto weighted by vote share, speech by seat share); identical series to the in-text figure, shown without the excl-AfD overlay. Aggregation B, primary specification.

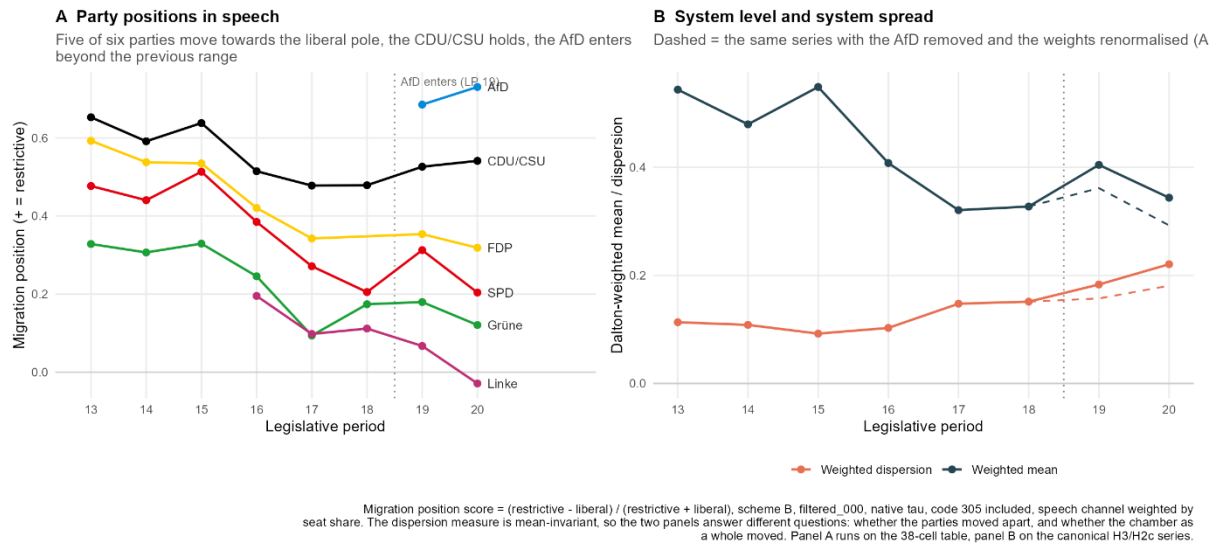


Figure F.9.4. Migration: party positions and system level over time. Panel A gives the speech-side position score of each party by legislative period, positive values denoting the restrictive pole. Panel B gives the Dalton-weighted mean and the Dalton-weighted dispersion of the same scores, the dashed lines repeating both series with the AfD removed and the remaining weights renormalised (arm 6). The FDP line breaks at LP 18, where the party held no seats, and the Linke enters the analysis sample at LP 16 (Section 4.8). Aggregation B, primary specification. Descriptive series.

F.10 The LP-13 elevation: a manifesto-side, window-boundary diagnostic

Section 5.2 reports that the manifesto–speech divergence peaks at LP 13, where all four cells (CDU/CSU, SPD, FDP, Grüne) are the maximum of their party and the CDU/CSU cell of 0.1125 is the largest value in the chapter. This subsection establishes that the elevation is a property of the manifesto input and of where the observation window opens, not an anomalous 1994 election and not a segmentation, classification, or mapping artefact. Three checks converge.

First, the divergence is localised to the manifesto side by a cross-swap. Holding the LP-13 speech distribution fixed and replacing the 1994 manifesto with the party's typical in-window manifesto, the mean of its other 1994–2021 programmes, removes most of the divergence, whereas holding the manifesto fixed and replacing the speech leaves it unchanged (Table F.10.1). For the CDU/CSU the manifesto swap reduces the divergence from 0.1125 to 0.0150, while the speech swap leaves it at 0.1214. The speeches at LP 13 are therefore ordinary; the 1994 manifestos carry the divergence.

Cell (LP 13)	Actual JSD	Manifesto → typical	Speech → typical
CDU/CSU	0.1125	0.0150	0.1214
SPD	0.0596	0.0207	0.0706

Cell (LP 13)	Actual JSD	Manifesto → typical	Speech → typical
FDP	0.0402	0.0175	0.0390
Grüne	0.0743	0.0171	0.0791

Table F.10.1. Cross-swap localisation of the LP-13 divergence. Each LP-13 cell's JSD (Actual), recomputed with the 1994 manifesto replaced by the party's typical in-window manifesto (Manifesto → typical) and with the LP-13 speech replaced by the party's typical speech (Speech → typical). Aggregation A, primary specification.

Second, the 1994 manifesto is not itself an aberrant document. Across each party's full manifesto record in the Manifesto Project main dataset, the change from the 1990 to the 1994 programme is unremarkable, and far larger shifts occurred in earlier decades (Table F.10.2, Figure F.10.1). For the CDU/CSU the 1994 change equals its long-run average and ranks sixth of twenty transitions; for the FDP it is one of the smallest on record, eighteenth of twenty. The Green record before 1994 is not captured by the party identifier used here, so the comparison is reported for the three parties with a long history; the 1994 Green change of 0.0369 is modest in absolute terms.

Party	1994 change	Pre-1994 mean	Pre-1994 max (year)	Rank of 1994 (1 = largest)
CDU/CSU	0.0619	0.0616	0.1579 (1957)	6 of 20
SPD	0.0427	0.0548	0.1015 (1976)	8 of 20
FDP	0.0112	0.0547	0.1079 (1972)	18 of 20
Grüne	0.0369	—	—	1 of 9

Table F.10.2. The 1994 manifesto change in historical perspective. JSD between each party's 1994 manifesto and its immediately preceding programme (1994 change), against the mean and maximum of that party's pre-1994 election-to-election changes and the rank of the 1994 change among all its transitions (1 = largest). Aggregation A.

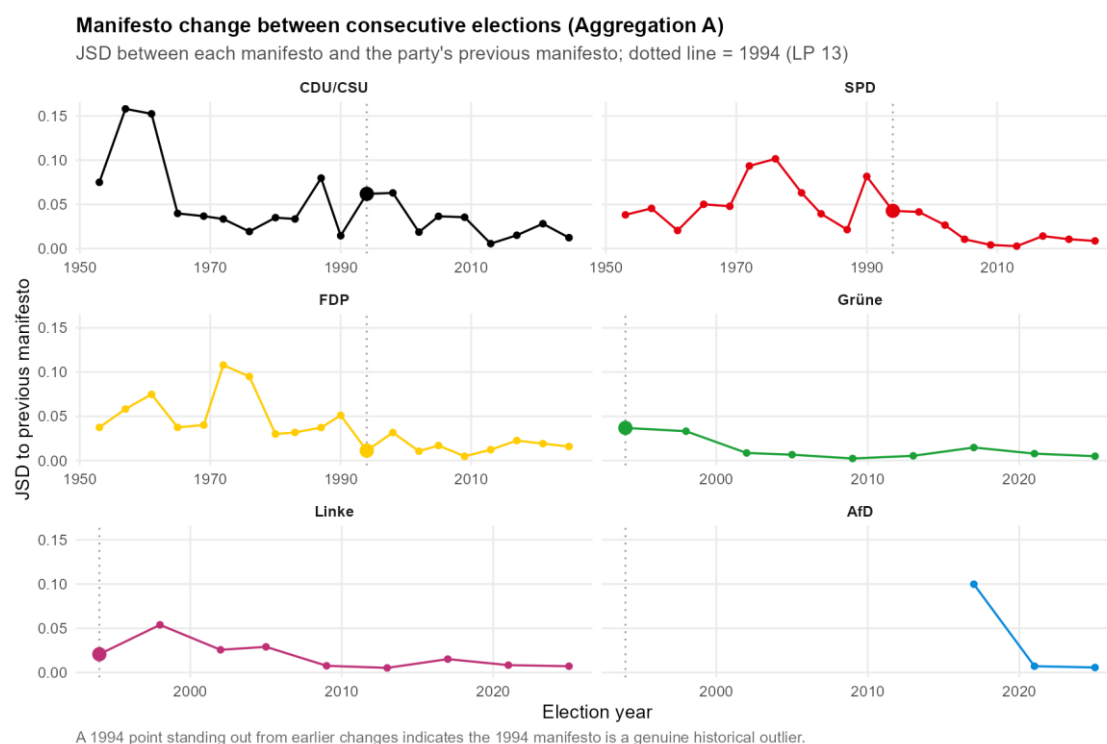


Figure F.10.1. Manifesto change between consecutive elections, by party. JSD between each manifesto and the party's previous programme; the dotted line marks 1994. The 1994 change does not stand out from the party's earlier transitions.

Third, the composition series explains why the 1994 programmes stand apart within the analysis window without being anomalous. Manifesto compositions drift steadily from one election to the next (Figure F.10.2), and for the three long-observed parties the election-to-election change is visibly smaller after the mid-1990s than before: the parties settle into the more similar profiles of 1998 to 2021. The 1994 manifesto is simply the earliest in the window, and therefore the most distant from that later cluster, which the genre-stable plenary speech resembles more closely than it resembles the older programme.



Figure F.10.2. Manifesto topic composition over time, by party. Nine-bucket manifesto share per election; the dotted line marks 1994. Compositions drift steadily, and the post-1994 profiles form a tighter cluster.

Taken together, the LP-13 elevation is a boundary effect of where the observation window opens, carried entirely by the manifesto side, and it requires no substantive account of the 1994 election. Chapter 5 reports it as such and does not read it as evidence about any single party or period.

Appendix G: Software, Runbook and Integrity Checks

This appendix documents the computational environment, the order in which the analysis scripts run, the integrity checks referenced in Section 4.7.4, and the training details of the non-codable filter of Section 4.6.1. All specification constants, including the primary specification, the temperature values, the bootstrap draws and the seeds, are declared once in the configuration script and read by every downstream script, which rules out divergent hard-coded settings by construction.

G.1 Computational Environment

Table G.1: Software, models and data versions.

Component	Version and details
Operating system	Windows 11

Hardware	NVIDIA RTX 5070 Laptop GPU (classification, filter training and inference)
R	key packages: data.table, arrow, manifestoR, polmineR, here
Python	torch (CUDA 12.8 build), transformers (≥ 4.40), pandas, pyarrow, numpy
Classifier	manifesto-project/manifestoberta-xlm-roberta-56policy-topics-context-2024-1-1 (Section 4.4)
Filter model	deepset/gbert-large, fine-tuned as described in G.4
Labelling model	Claude Sonnet (claude-sonnet-4-6, accessed via the batch API; Section 4.5.2)
Corpora and data	GermaParl2 v2.3.0-rc1 (doi:10.5281/zenodo.15495748); MARPOR Main Dataset, version 2025-1 (Lehmann et al., 2025)

G.2 Pipeline Runbook

Table G.2: Analysis scripts in execution order.

Script	Purpose
00_config.R	Single source of truth: paths, the 56-code list, both aggregation schemes, specification axes, primary-specification constants, seeds, JSD and bootstrap functions. Every downstream script sources this file and never hard-codes a specification.
01	Speech extraction from the CWB corpus: speaker roles, minimum length, sentence segmentation (Section 4.3.1).
04_classify_manifestoberta.py	GPU classification of all sentences into the 56 categories with context windows (Section 4.4); writes the chunked per-sentence posterior files.
05_pull_marpor.R	Manifesto data via manifestoR; per-manifesto code distributions.
06_gold_realign_validate.R	Gold-standard alignment and validation metrics (Section 4.5.1), including the carried triple comparison.
09	Silver-standard agreement analysis between ManifestoBERTa and the Sonnet labels (Sections 4.5.2 and 4.5.3).
10_train_000.py	Trains the non-codable filter (G.4); writes model_000_gbert/ and the cross-validated metrics.
11_apply_000.py	Applies the filter to all chunks; adds p_000 and is_000 to every sentence, non-destructively and resumably (Section 4.6.1).

12_aggregate.R	Single streaming pass over the corpus: all 18 specifications per scheme, manifesto distributions, bootstrap intervals, cell diagnostics (Section 4.7).
13_dps_speclock.R	DPS deep dive: 305 asymmetry, redundancy test, migration re-split arm, independent recomputation of the primary JSD (Sections 4.6.2 and 4.9).
17_robustness.R	Pre-committed robustness arms and acceptance criteria (Section 4.9).
19_migration_position_trajectories.R	Party-level read-out of the migration domain: speech positions per party against the Dalton-weighted mean and dispersion (Appendix F.9).
gold_000_check.py	Post-hoc check of the filter against the gold labels (Section 4.6.1).

Note: Script names follow the repository; scripts 01 and 09 are listed by their pipeline position. Intermediate outputs are written to Data/ and cache/ under the project root.

G.3 Integrity Checks

Three layers of checks guard the aggregation. First, the code-to-bucket indicator matrices are verified at load time: every one of the 56 codes is placed in exactly one bucket of scheme A, so the row sums of the indicator matrix equal one by construction and no probability mass can be lost or duplicated. Second, every per-cell distribution of every specification is renormalised and checked to sum to one within a numerical tolerance of 10^{-6} , on the speech side and the manifesto side alike. Third, the primary JSD values of the aggregation script are recomputed independently in the specification-lock script from the stored per-sentence bucket masses; the two computation paths agree to within the declared numerical tolerance of 10^{-6} (Section 4.7.4). In addition, the filter application of Section 4.6.1 writes atomically and resumably, so an interrupted run cannot leave partially written chunks, and the null-class probability is stored per sentence, so every threshold decision remains reversible without re-inference.

G.4 Non-Codable Filter: Training Details

Table G.3: Training and evaluation of the gBERT-000 filter.

Item	Value
Base model	deepset/gbert-large (Chan et al., 2020)
Task	Binary classification, codable versus non-codable (000)
Labels	Claude Sonnet silver labels, samples A and B combined; 19,821 usable sentences, 4,414 labelled 000 (22.3%)
Input	Sentence with surrounding context (use_context = true)

Evaluation	Stratified five-fold cross-validation, out-of-fold predictions; the production model is a final refit on all labelled data
Epochs per fold	3
Out-of-fold metrics	PR-AUC 0.845, ROC-AUC 0.931; at threshold 0.5: precision 0.801, recall 0.716, F1 0.756
Operating point	$\text{is_000} := \text{p_000} \geq 0.5$; p_000 is stored per sentence, so the threshold can be varied downstream without re-inference
Threshold justification	DPS share in the evaluation sample: 40.0% unfiltered, 32.2% at threshold 0.5, oracle 31.7% (Section 4.6.1)

Note: Metrics from model_000_text_metrics.json; see Section 4.6.1 for the gold-sample check of the filter.